

A MERGED RESEARCH REPORT

Surreal Scalars in Theoretical Physics

Black-Hole Singularities, Asymptotic Structure,
and the Limits of Replacing the Scalar Field

September 2026

Abstract

Three independently written manuscripts asked whether Conway's surreal field $\mathbf{No}$, or its algebraic closure $\mathbf{No}[i]$, can replace $\mathbb{R}$ and $\mathbb{C}$ in physics, and in particular whether admitting infinite scalars removes a black-hole singularity. This report is their union. All three reach the same verdict, and the verdict is negative.

Surreal and surcomplex scalars make the hierarchy of quantities near a singular regime *exactly representable*. They do not make the spacetime regular. For Schwarzschild with $m \neq 0$, the Kretschmann invariant is exactly $48m^2/r^6$. Under a formal ramification $r = s^q w(s)$, where $q \geq 1$ is an integer and $w \in \mathbb{R}[[s]]$ has $w(0) \neq 0$, the pole order becomes exactly $6q$ and the leading coefficient is $48m^2 w(0)^{-6}$, so no radial reparametrization of this form removes the blow-up; the proper time from rest at R to $r = 0$ is $(\pi/2)\sqrt{R^3/2M}$, finite — and infinitesimal, hence still finite, for infinitesimal R ; the tidal component $2M(2M - r)/r^4$ diverges; and the exact identity $r^6 \mathcal{K} = 48m^2$ has no field-valued solution at $r = 0$ in any characteristic-zero field. Evaluating at a nonzero infinitesimal radius names another point of a punctured domain. It does not supply the missing endpoint, a continuation law, or an observable rule.

Because the subject is empirical and the danger is overclaim, the report is built around a firewall. Every substantive statement carries one of three visible tags: **[E]** an *exact symbolic identity*, which is all that the three accompanying verification programs establish; **[C]** a *conditional mathematical theorem*, whose hypotheses are strong and are established for no physical model; or **[A]** an *assessment*, which is an argument and not a result. Imported literature carries **[I]**. Flattening these tiers would manufacture exactly the overclaim all three manuscripts were written to refuse.

What survives on the positive side is substantial and precisely delimited: exact curvature and infall identities, a small-angular-momentum crossover with an exact quadrature, inner and outer Hahn support criteria for a regular core that are sharp at $\alpha = 2/3$, formal and coefficientwise Einstein reduction theorems with counterexamples showing their hypotheses cannot be dropped, finite-dimensional surcomplex Hermitian and quantum-shadow theorems with their conditioning boundary, and an explicit incompatibility between the canonical integer-part global phase and a normalized scalar derivation. None of it is a singularity resolution, and none of it is claimed to be.

Status. A merged exposition assembled from three independent manuscripts, all pinned to VladimirReshetnikov/Surreal at commit 39f2be6667ade51bca2b45daa47e289d69c09764. Prepared with AI assistance; not refereed; no proof-assistant verification of any statement in this report. The three companion programs check $45 + 50 + 37 = 132$ finite symbolic identities and nothing else. No priority claim is made for any proposition, computation or benchmark. Section D is the consolidated register of what is *not* claimed; it is part of the result, not a disclaimer appended to it. The single sentence this report exists to refuse, in any form, is that surreal numbers resolve the black-hole singularity.

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Part I

The reading contract

1 What this report is

1.1 Three manuscripts, one verdict

This report is the union of three independently written manuscripts on the same commission: assess whether surreal and surcomplex scalars can replace the real and complex numbers in theoretical physics, with black-hole singularities as the test case. They were prepared separately, they are pinned to the same repository snapshot, and they reach the same conclusion by the same worked examples. Each of the three independently rebuilds the connection and the full Riemann tensor of the general static spherical metric in a computer algebra system and re-derives the same Schwarzschild quantities from the metric alone. Two of the three print the same Planck-curvature radius $r_Q = 48^{1/6}(m\ell_P^2)^{1/3}$ with the same disclaimer that it is dimensional power counting and not a derived threshold.

All three carry the following spine:

- (i) what a black-hole singularity actually is — coordinate divergence, curvature divergence, causal geodesic incompleteness and inextendibility at a named regularity are four different defects;
- (ii) Schwarzschild evaluated at an infinitesimal radius, with the exact proper-time infall law and the exact tidal Jacobi exponents;
- (iii) a conditional standard-part or coefficientwise reduction of the Einstein equations, with the observation that its hypotheses fail exactly where a singularity lives;
- (iv) a regular-core comparison in which the changed *dynamics*, not the changed scalar field, does the regularizing work;
- (v) finite-dimensional surcomplex quantum algebra, together with an explicit list of what does not extend;
- (vi) quantum fields, regulators, and transseries or renormalization examples showing that field enlargement does not fix a prescription;
- (vii) a comparison with other non-Archimedean and singular methods;
- (viii) a staged development program for the repository;
- (ix) a claim-status ledger and a curvature appendix.

And all three end at the same assessment: a strong exact asymptotic coefficient language, but extending the scalar field alone does not remove a singularity, define division by zero, fix ultraviolet dynamics, or supply an observable rule.

Duplication between the three is very high — on the order of two thirds. The curvature and infall computations, the Planck power counting with its identical r_Q , the regular-core example, the finite-dimensional quantum section, the transseries and regulator material, the framework comparison and the research program are common to all three, and each ships its own independent symbolic re-derivation of the same connection and curvature (45, 50 and 37 checks respectively). Whatever is shared is printed *once*. Where the same conclusion is reached by genuinely different arguments, both arguments are printed and marked as such. Nothing that any one manuscript

reached and the others did not has been dropped, and every honest limitation any of them states survives, in Section D.

1.2 What each member contributed

Member 12 — scalar-extension assessment (the base). A 34-page critical assessment with 25 references. It is the most complete of the three and supplies: the pole-persistence proposition invariant under formal ramification; the boundary-layer warning about differentiating standard parts; the exact inner and outer Hahn support criteria for the regular core, sharp at $\alpha = 2/3$; the formal Einstein-tensor reduction over $C^\infty(U, \mathbb{R})[[h]]$; the global-phase section deriving Exp_{EK} with kernel $2\pi i \mathbf{Oz}$ together with its proved incompatibility with a normalized Berarducci–Mantova derivation, and the real-time corollary $\text{Exp}_{\text{EK}}(i\omega t) = 1$; a units-and-falsifiability section; the explicit refutation-by-counterexample of the published surreal-singularity proposal; and audit, notation and claim-status appendices.

Member 13 — no-go obstructions. A deliberately negative-leaning assessment with a 50-check suite. It supplies the three-way taxonomy of replacement proposals kept rigorously separate throughout; the algebraic endpoint obstruction for $r^6 \mathcal{K} = 48m^2$ in any characteristic-zero field; anisotropy, weak singularities and scale invariants beyond one pole, with the Kasner coefficients; the double-null identities and the Borel-residue example; the non-multiplicativity and reparametrization dependence of finite-part extraction; the infinite-period proposition for a global phase homomorphism; the bounded-frame valuation invariance; the dominant-balance principle; the argument that the ultraviolet question is dynamical and not arithmetic; a measurement, recoverability and empirical-novelty section; and a five-stage development program with explicit failure conditions the system should *report* rather than paper over.

Member 14 — black-hole singularity limits. A 31-page assessment with a 37-check suite. It supplies the small-angular-momentum crossover with its exact dimensionless quadrature and uniform-on-bounded-intervals limit; the order-theoretic proof that a leading pole stays unlimited in *any* ordered extension fixing $\mathbb{R}$; the section arguing that the classical theory loses control at a positive real curvature scale before any literal infinitesimal radius; the noncanonical-subtraction example and the reading of standard part as a measurement rule; the remark that the regular-core r^5 term becomes $|x|^5$ along a Cartesian line; Kerr, near-extremal Reissner–Nordström and oscillatory regimes; the survey of published surreal-physics proposals; the beyond-all-orders factorial-series ambiguity; and a research program with explicit success criteria.

1.3 The merge rule

A merge is a union, not a selection. Concretely, in this report: a result every member proves is printed once; a result proved by genuinely different routes keeps both proofs, marked; anything one member reached that the others did not survives; every honest limitation survives; and nothing is strengthened beyond what a source proves. The last clause is the operative one for a physics report, and it is what Section 2 exists to enforce.

2 The claim-status firewall

2.1 Why this section comes before any physics

This is the only report in its collection whose subject is not mathematics. Its entire value is the firewall between what is proved and what is proposed. The characteristic failure mode of surreal-physics writing is not a wrong calculation; it is a correct calculation presented at the wrong epistemic grade, so that a formal identity is read as a statement about general relativity, or an evaluation at an infinitesimal radius is read as a physical regime. Both misreadings are available from perfectly true sentences. The only defence is to tag the grade of every statement and never let a tag be inferred from context.

Convention 2.1 (The three claim tiers, and the provenance marker). Every substantive statement in this report carries exactly one of the following tags, in its theorem header or at the head of its paragraph.

[E] Exact symbolic identity.

A finite algebraic or differential identity in ordinary real or complex symbols, checked exactly — not numerically, not to a tolerance — by at least one of the three accompanying programs, or derived here from such an identity by finite algebra. *This tier is the entire content of what the verification programs establish.* It includes the curvature components, the Schwarzschild and Kasner invariants, the infall and Jacobi exponents, the crossover integrand factor, the regular-core central limits, the finite-part shifts, the regulator integrals, and r_Q as an algebraic consequence of setting $\ell_P^4 \mathcal{K}$ to order one.

[C] Conditional mathematical theorem.

A theorem with an ordinary mathematical proof in the text, under hypotheses stated in full. The hypotheses are strong, and *they are established for no physical model.* This tier includes the formal and coefficientwise Einstein reductions, the quantum-shadow results, the finite Hermitian spectral theorem, the fine-topology collapse, the pole-persistence and endpoint obstructions, the Hahn support criteria, and the phase and period propositions. Nothing in this tier is machine-checked. A **[C]** statement about a formal coefficient ring is not a statement about general relativity, and this report never lets one be read as the other.

[A] Assessment.

An argument rather than a result: a judgement about what a proposal does or does not accomplish, a modelling choice declared as such, a power-counting estimate, a research proposal, or a negative claim scoped as a *missing implication* rather than an impossibility theorem. Assessments are the reason this report exists; they are also the tier on which nothing may be built.

Separately, **[I]** marks a result *imported* from the published literature, used with its own hypotheses intact and neither reproved nor strengthened here. **[I]** is a provenance marker, not a fourth claim tier: an imported result is evidence about the literature, not a claim of this report.

Warning 2.2 (The two inferences this report refuses). First: the Einstein reduction theorems of Section 12 are theorems about *formal coefficient equations under stated conditions*. They are not theorems about general relativity, and in particular they say nothing whatever about a metric whose leading term degenerates — which is exactly the situation at a singularity. Second: “Schwarzschild at an infinitesimal radius” is a *mathematical extension* of a classical formula, not a physical regime; Section 8 is placed immediately after the computation, rather than a dozen sections later, precisely so that this cannot be missed.

2.2 The ledger

Table 1 is the report’s claim-status ledger. It is duplicated in the accompanying `README.md` so that the contract is stated before anyone opens the article.

Table 1: Claim-status ledger. **[E]** exact symbolic identity; **[C]** conditional theorem proved in the text; **[A]** assessment; **[I]** imported.

Claim	Tier	Essential scope, and what is <i>not</i> claimed
General static-spherical $\mathcal{K}[f]$, Scal , G^t_t	[E]	Derived on a nondegenerate patch; independently rebuilt from the metric by all three programs. A rational identity in f and r , nothing more.
$\mathcal{K} = 48m^2/r^6$, $\mathcal{K}(2m) = 3/(4m^4)$, vacuum $\text{Ric} = 0$	[E]	Classical general relativity, recomputed and symbolically checked here. Not new physics.
$m^4\mathcal{K} = 48\varepsilon^{-6}$ at $r = m\varepsilon$	[E]	An exact Hahn-field value. <i>Not</i> a statement that a curvature observable has become a limited physical magnitude.
Infall $r^3 = \frac{9}{2}mu^2$, $\mathcal{K} = 64/(27u^4)$, tidal $4/(9u^2)$, $-2/(9u^2)$, Jacobi exponents $4/3, -\frac{1}{3}, \frac{2}{3}, \frac{1}{3}$	[E]	Exact for the $E = 1$ radial timelike geodesic in the stated frame convention. Generic rates only; special initial data change them.
Proper time from rest at R to $r = 0$ equals $(\pi/2)\sqrt{R^3/2M}$	[E]	Finite for finite R, M ; infinitesimal, hence still finite, for infinitesimal R . Infall terminates.
Crossover quadrature and its integrand factor; Φ endpoint asymptotics	[E]	Exact identity plus a limit uniform only on bounded positive intervals.
Hayward $\mathcal{K}(0) = 24/\ell^4$, $\rho(0) = 3/(8\pi\ell^2)$, $r_c = (2m\ell^2)^{1/3}$	[E]	For fixed positive <i>real</i> ℓ . C^2 -level curvature regularity for this ansatz only; not C^∞ , not geodesic completeness, not stability, not a microphysical Lagrangian.
Planck-curvature radius r_Q ; two-scale exponent $4\beta - 6\alpha$	[E]/[A]	The algebra is exact [E] ; its reading as a threshold is [A] dimensional power counting, not a derived universal radius and not a prediction of any quantum-gravity theory.
Finite-part shifts FP_ε versus FP_θ ; $a \mapsto a - c$	[E]	Exact. Shows a prescription dependence; proves no renormalization theorem.
Pole persistence under scalar extension and ramification	[C]	Hahn-field valuation statement; and, by a second independent route, an order-field statement invariant under any ordered extension fixing $\mathbb{R}$. Deliberately <i>not</i> about real limits.

Claim	Tier	Essential scope, and what is <i>not</i> claimed
No field-valued Schwarzschild endpoint	[C]	Conditional on preserving the same invariant identity at the endpoint; not a no-go for a modified metric, a different equation, nonclassical observables, or a geometry with no zero-area-radius event.
Set-sized collapse of the full fine topology	[C]	Set-sized subsets only. Explicitly <i>not</i> a claim that the proper class is discrete, and <i>not</i> applicable to the intrinsic valuation topology of a set-sized Hahn field.
Formal Einstein reduction over $C^\infty(U)[[h]]$	[C]	Requires a nondegenerate smooth leading metric on one common domain. Singular perturbations are outside it; a counterexample shows the hypotheses cannot be dropped. Says nothing about general relativity at a singularity.
Coefficientwise Hahn reduction and the shadow corollary	[C]	Strictly positive global support, common smooth domain, Neumann support lemma. Five named escape routes are left open.
Inner/outer strong-summability criteria, sharp at $\alpha = 2/3$	[C]	A criterion on Hahn-field valuation data, not on real convergence. It detects misuse of an expansion; it invents no solution.
Finite Hermitian spectral theorem; finite quantum shadow	[C]	Ordinary finite dimensions and step counts; standard-part readout adopted as a modelling rule. Conditioning covered only when the outcome probability has nonzero standard part. Not an infinite-dimensional spectral theorem.
Exp_{EK} with kernel $2\pi i \mathbf{Oz}$; phase/derivation incompatibility; $\text{Exp}_{\text{EK}}(i\omega t) = 1$; infinite periods	[C]	Statements about one specific commutative construction and one specific normalized derivation. They do not assert that no global surcomplex exponential exists — one is constructed here.
Surreal derivations, transseries embeddings, resurgent integration	[I]	Imported with their stated function-class restrictions. No general Einstein PDE transfer is claimed or available.
Penrose incompleteness, Raychaudhuri focusing, C^0 Cauchy-horizon stability, effective-field-theory corrections, quantum probes	[I]	Imported with hypotheses intact. Used to say what a singularity theorem does and does not conclude.

Claim	Tier	Essential scope, and what is <i>not</i> claimed
The conservative architecture: ordinary spacetime + controlled Hahn or transseries coefficients + explicit real-shadow map	[A]	Called the most defensible starting point. <i>Not</i> a derived necessity.
Standard part as the apparatus readout rule	[A]	An intelligible correspondence rule and an additional modelling assumption. Not forced by the field axioms; undefined on an unlimited curvature.
The staged program, benchmarks, projects and proposed module names	[A]	Proposals, not descriptions of existing modules and not results.
A physical resolution of black-hole singularities	—	Not supplied by scalar extension, and not claimed anywhere in this report.

3 The question that must actually be answered

3.1 Four different projects, three different replacement proposals

[A] Replacing the real numbers can mean four importantly different things. An *algebraic extension* lets a metric, a matrix or a curvature expression take values in a larger field while keeping the same formulas. An *asymptotic extension* interprets a small parameter as an infinitesimal and encodes a family of approximations in a Hahn series or a transseries. A *nonstandard reformulation* enlarges a structured mathematical universe, retaining specified transfer and internality principles and recovering ordinary quantities by a standard-part construction. A *new physical theory* changes the allowed states, dynamics, geometry or observational rules.

These projects are not equivalent. A useful reformulation may make proofs or computations easier without changing a single prediction. An algebraic extension can make an expression meaningful at a new nonzero argument without making it meaningful at its old singular argument. A genuinely new theory may use surreal numbers, but its physical content comes from the whole theory, not from the name of its field.

[A] Cutting the same question the other way gives three concrete replacement proposals, which this report keeps rigorously separate throughout.

(P1) Algebraic scalar extension.

Replace real or complex coefficients in finite algebraic expressions by elements of an ordered extension field or its complexification. Matrix identities, polynomial equations and rational curvature formulas remain meaningful wherever their denominators are nonzero. Mathematically straightforward; not yet a theory of time evolution.

(P2) Fields with non-Archimedean coefficients on ordinary spacetime.

Keep an ordinary real spacetime domain and let fields have Hahn-series or transseries coefficients, with coefficientwise spacetime derivatives and explicit support conditions. This is a workable formal deformation theory and is the principal constructive framework developed below.

(P3) A genuinely non-Archimedean spacetime.

Replace coordinate domains, curves, causal structure, integration, observables and possibly probability by new objects over a surreal field. A much larger proposal. A choice of field alone determines none of its topology, differential geometry, global existence theorems or empirical interpretation.

These proposals interact, but a result in one does not silently become a theorem in another. In particular, an expression evaluated at an infinitesimal radius is not automatically a solution on a larger spacetime containing the missing endpoint. Section numbers below state which proposal is in force.

3.2 An operational hierarchy of success

[A] It is useful to distinguish five increasingly demanding achievements.

- (1) A divergent quantity has a well-defined generalized numerical value.
- (2) An asymptotic solution has controlled algebraic and differential meaning.
- (3) A specified class of observers has well-defined observables through a previously singular regime.
- (4) The spacetime or its replacement has a suitable global evolution and a completeness or continuation theorem.
- (5) The model recovers tested physics and makes a discriminating, stable, empirically meaningful prediction.

Surreal arithmetic can directly assist the first two. The remaining steps need additional mathematics and physics. Failure to keep these levels separate is the main source of exaggerated claims about singularity resolution.

The same distinction can be phrased as a standard of evidence. A useful *representation* faithfully encodes specified equations and scale relations, with computable support or remainder certificates. An *analytic model* additionally explains which formal expressions correspond to actual solutions, with which domains and boundary data. A *physical singularity resolution* requires still more: an explicit dynamical law, a well-defined continuation or replacement of the singular regime, a stated regularity and measurement criterion, and physically interpretable predictions.

3.3 What a proposed resolution must deliver: a four-question test

[A] Concretely, require an explicit answer to four questions.

- (Q1) What states or events replace the classical endpoint?
- (Q2) What equations govern propagation there?
- (Q3) Which observables remain meaningful, and in what sense?
- (Q4) Does the replacement determine a continuation uniquely from admissible data, or does it introduce new boundary conditions?

The answers may involve nonclassical geometry or quantum states rather than a smooth metric; that is allowed, and no hypothesis of this report excludes it. What is not enough is to retain every classical formula, replace the word “divergent” by “surreal-valued”, and regard the missing choices as settled.

3.4 How strong is the affirmative case?

[I] There is substantial mathematics behind the affirmative case. Berarducci and Mantova constructed compatible differential structures on the surreal numbers and related them to transseries [11, 12]; Aschenbrenner, van den Dries and van der Hoeven placed the resulting differential field in a universal H -field framework with transseries and Hardy-field embeddings [13]; van den Dries and Ehrlich established surreal exponentiation and its elementary-extension properties [10]; Ehrlich and Kaplan constructed surreal ordered exponential fields together with trigonometric and surcomplex exponential structures [14, 15]; and Costin and Ehrlich developed extension and integration operators for a substantial, explicitly delimited class of resurgent functions, along with obstructions to substantially more unrestricted constructive extensions [16]. These are serious tools for asymptotic analysis, not informal manipulations of infinity. Their availability makes it reasonable to investigate reduced gravitational equations and multiscale models by surreal methods.

[A] The more ambitious statement — that physical spacetime or quantum amplitudes fundamentally use **No** or **No**[i] rather than $\mathbb{R}$ or $\mathbb{C}$ — does not follow. No predictive black-hole completion is established by the repository or by any surreal-physics proposal examined here. That is an assessment of the sources inspected, not a theorem excluding all future surreal-valued theories.

3.5 What the repository provides, and what it does not

All three manuscripts inspected `VladimirReshetnikov/Surreal` at the pinned snapshot

39f2be6667ade51bca2b45daa47e289d69c09764.

[A] Between them they read the root `README.md`, the documentation catalogue `docs/README.md`, the surcomplex analysis report’s `README` and the trigonometry report’s `README`. The catalogue describes fifteen research packages — six on surreal numbers, seven on surcomplex mathematics, two on foundations and computation — and labels them AI-assisted drafts rather than refereed or machine-checked foundations [1, 2, 3, 4].

The distinction between the reports and the formalized prerequisites matters. The root `README` describes implemented Lean prerequisites covering generic finite algebra and geometry, polynomial operations and jets, complexification, and a Hahn summability and standard-part layer with compatibility between complexification and Hahn coefficients. It states explicitly that these prerequisites do not yet construct the surreal field or establish its normal-form bridge to Hahn series, and do not establish all claims in the reports. It would therefore be wrong both to call the repository unformalized and to treat a successful Lean build as verification of its analytic reports. No manuscript ran that build, and neither did this merge.

Three documented features are used below. The analysis report separates strong Hahn summation from ordinary topological convergence and distinguishes coefficient rings and function classes — fixed-domain coefficient functions, common-domain germs, individually convergent germs with no uniform domain, and purely formal coefficients. The trigonometry report separates canonical finite-angle constructions from the additional choices involved in a global phase, and explicitly declines to select a scalar derivation. The documentation recommends set-sized coefficient workspaces over unrestricted proper-class constructions.

A concrete instance of the common-domain warning, taken from the catalogue, is the series

$$F(z) = \sum_{n \geq 1} \frac{t^{n\eta}}{1 - nz}, \quad \eta > 0. \quad (3.1)$$

Every coefficient is a holomorphic germ at zero, but the pole at $1/n$ prevents the family from sharing an ordinary disk of holomorphy. Treating this as a field on one fixed neighbourhood introduces a false hypothesis. The physics analogue is to claim a uniform spacetime domain from coefficientwise local solutions; Section 5.3 makes the requirement explicit.

[A] None of these ingredients is already a relativistic spacetime, an Einstein initial-value theory, a quantum state space with infinite-dimensional spectral theory, or a rule for measuring surreal-valued observables. A suitable extension of the repository would therefore have to add a *physical semantics* and a *differential-geometric layer*, not merely more arithmetic identities. The missing pieces are coefficient-function algebras, tensor differentiation, physical interpretation, analytic realization, and an account of evolution. Section 21 turns these into concrete development targets. Proposed physics work should build on the verified algebraic interfaces where applicable, and should not cite the repository as an already formalized theory of surreal general relativity.

4 The scalar systems and their observational shadows

4.1 Surreal numbers are a field, not a collection of error symbols

[I] Conway's surreal numbers form a proper-class ordered real-closed field containing $\mathbb{R}$ and the ordinals. Every individual surreal number has a set-sized normal form, although the collection of all of them is not a set. Adjoining i with $i^2 = -1$ gives the algebraically closed

$$\mathbf{No}[i] = \{a + ib : a, b \in \mathbf{No}\}, \quad \overline{a + ib} = a - ib, \quad |a + ib| = \sqrt{a^2 + b^2} \in \mathbf{No}_{\geq 0}. \quad (4.1)$$

The order lives in $\mathbf{No}$, not in $\mathbf{No}[i]$. Algebraic closure holds for ordinary finite-degree polynomials and is not analytic completeness; the modulus is surreal-valued and is not an ordinary real-valued Banach norm. The multiplicative and triangle identities are algebraic consequences of real closedness. These facts, the normal-form description and the exponential structure are in [7, 8, 11, 14, 9].

An infinite positive element ω satisfies $\omega > n$ for every ordinary natural number n , and $\varepsilon = \omega^{-1}$ is positive and smaller than every positive real. Nevertheless

$$\varepsilon \neq 0, \quad \varepsilon^{-1} = \omega, \quad 0^{-1} \text{ does not exist.} \quad (4.2)$$

[C] No field extension can change the last statement while retaining the field axioms: if $0x = 1$ then $0 = 1$. Likewise there is no largest positive surreal and no smallest positive infinitesimal, since $x + 1 > x$ and $0 < \varepsilon/2 < \varepsilon$.

[A] An infinitesimal is therefore not automatically a minimal length, a shortest time step, or an ultraviolet cutoff. Its presence adds finer scales rather than removing them. Choosing a surreal infinitesimal does not create a minimum length. Surreal numbers are also commutative and have no nonzero nilpotents: anticommuting Grassmann variables, noncommuting observables and operator algebras still require their own constructions. Nothing in this report is available over a noncommutative base without reproof, and the surquaternionic material in this repository's `docs/surquaternions/` family — where conjugation is an anti-automorphism, the modulus need not be multiplicative, and the value group need not be central — must be treated as a separate subject. Every result below uses commutativity somewhere: valuation additivity, the Neumann series for a metric inverse, st as a ring homomorphism, the Rayleigh-quotient step in the spectral theorem, Hahn convolution.

Definition 4.1 (Limited, infinitesimal, unlimited). In an ordered field $F \supseteq \mathbb{R}$, an element x is *limited* if $|x| < M$ for some real $M > 0$; *infinitesimal* if $|x| < r$ for every real $r > 0$; and *unlimited* if it is not limited.

This vocabulary is used deliberately in place of “finite” and “infinite” to block a specific slide: from “curvature is now a field element” to “curvature is a finite physical magnitude relative to ordinary units.” An infinite surreal is an individual field element; it is not thereby a limited observable. The two phrasings “finite element” and “limited element” denote the same set $\mathcal{O}_F$ below and are used interchangeably only where no physical reading is in play.

4.2 Set-sized Hahn workspaces

For concrete calculations, fix a divisible ordered abelian group Γ that is a *set* and work in

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)) = F_\Gamma[i]. \quad (4.3)$$

An element is a formal sum $a = \sum_{\gamma \in S} a_\gamma t^\gamma$ with $S \subseteq \Gamma$ well ordered. The convention is that t^γ is *small* for $\gamma > 0$, so the valuation of a nonzero series is $v(a) = \min \text{supp}(a)$, with $v(0) = +\infty$ as a bookkeeping symbol and not a newly adjoined field element. Then

$$v(ab) = v(a) + v(b), \quad v(a + b) \geq \min\{v(a), v(b)\}. \quad (4.4)$$

[1] For divisible Γ , F_Γ is real closed and K_Γ is algebraically closed. If Γ is realized as an additive subgroup of $\mathbf{No}$, the normal-form realization sends t^γ to $\omega^{-\gamma}$, consistently with this convention [11, 14, 7, 8]. The notation $\omega^{-\gamma}$ denotes Conway’s monomial map; for a general surreal exponent it must not be silently identified with a Gonshor exponential expression, and exponential or logarithmic closure is extra structure, not a property of every selected workspace [10, 14].

[A] Every set of surreal coefficients uses only a set of normal-form exponents, so the divisible additive subgroup they generate is a set-sized workspace containing that data. This removes the proper-class obstacle from individual calculations. It proves *no* closure under a chosen exponential, a scalar derivation, composition, integration or infinite summation; those obligations must be discharged separately. It also does not turn a proper-class-indexed field theory into a set-sized one by changing notation. The precise statement of the localization principle, its three clauses and the size obstructions that any candidate foundation must respect belong to the companion merged report *Foundations for Surreal and Surcomplex Mathematics*, held in `docs/foundations-and-computational/foundations/` of this repository, which proves that every *set* of surcomplex numbers lies inside one divisible set-sized algebraically closed workspace K_Γ , that any set of such workspaces lies inside a larger one, and that a polynomial splitting in one acquires no new roots in a larger surcomplex extension. This report imports that discipline and does not re-argue set-theoretic legality; readers weighing proposal (P3) of Section 3.1 should consult its size obstructions rather than this report.

For the calculations below, $\Gamma = \mathbb{Q}$ suffices for the Schwarzschild Puiseux exponents, $\Gamma = \mathbb{R}$ for arbitrary Kasner powers, and beyond-all-orders exponential scales require an enlarged group or a named transseries subfield — see Section 17.3, where $e^{-1/\varepsilon} \notin \mathbb{R}((\varepsilon^\mathbb{Q}))$ is proved by an exponent argument.

4.3 Strong Hahn summability is a support condition, not convergence

Definition 4.2 (Strong Hahn summability). A family $(a_j)_{j \in J}$ of Hahn series is *strongly summable* when the union of its supports is well ordered *and* each exponent lies in the support of only finitely many members. Addition is then coefficientwise.

The second clause is a finitary-multiplicity condition and is not implied by the first. It does the work in both directions.

Example 4.3 ([E] What is and is not strongly summable). For an infinitesimal ε of positive valuation,

$$\sum_{n \geq 0} \varepsilon^n = (1 - \varepsilon)^{-1}, \quad \text{with exact finite remainder} \quad (1 - \varepsilon)^{-1} - \sum_{n=0}^N \varepsilon^n = \frac{\varepsilon^{N+1}}{1 - \varepsilon}, \quad (4.5)$$

an exact algebraic identity, *not* a limit of partial sums in any topology. The family $\sum_{n \geq 0} n! \varepsilon^{n+1}$ is likewise strongly summable — its support lies in the positive integers and each exponent occurs once — although the corresponding real power series has zero radius of convergence. By contrast the ordinary convergent real series $\sum_{n \geq 0} 2^{-n}$, and the family of real constants $1/n!$, are *not* strongly summable when viewed as families of constant Hahn series: exponent 0 occurs infinitely often. So e is first defined by ordinary real analysis in the coefficient field and then embedded; it is not obtained by ignoring the support rule. Finally, a countable family each of whose members equals the same positive infinitesimal ε is not strongly summable, which is why countably additive non-Archimedean probability is not free (Section 15.4).

4.4 Finite elements, infinitesimals, and standard part

For an ordered field $F \supseteq \mathbb{R}$ write

$$\mathcal{O}_F = \{x \in F : |x| < n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m}_F = \{x \in F : |x| < 1/n \text{ for every } n \geq 1\}.$$

Proposition 4.4 ([C] Standard part). *Each $x \in \mathcal{O}_F$ has a unique decomposition $x = r + \delta$ with $r \in \mathbb{R}$ and $\delta \in \mathfrak{m}_F$. The map $\text{st}(x) = r$ is an order-preserving ring homomorphism with kernel $\mathfrak{m}_F$, inducing $\mathcal{O}_F/\mathfrak{m}_F \cong \mathbb{R}$. For $K = F[i]$ it extends componentwise to a homomorphism $\mathcal{O}_F + i\mathcal{O}_F \rightarrow \mathbb{C}$ commuting with conjugation.*

Proof. The set $A = \{a \in \mathbb{R} : a \leq x\}$ is nonempty and bounded above; let $r = \sup_{\mathbb{R}} A$. For every positive real h the supremum property gives $r - h < x < r + h$, with harmless weak inequalities at an endpoint, so $x - r$ is infinitesimal. Two distinct reals cannot differ by an infinitesimal, giving uniqueness. Sums of infinitesimals and products of an infinitesimal with a limited element are infinitesimal; expanding $(r + \delta)(s + \eta)$ proves the homomorphism assertion. Order preservation and the complex assertion are immediate. The proof uses completeness of $\mathbb{R}$ and nothing about F beyond its order. $\square$

For the Hahn field (4.3), $\mathcal{O}$ consists of the series of nonnegative valuation, $\mathfrak{m}$ of those of strictly positive valuation, and st extracts the coefficient of t^0 . Both descriptions are used below; they agree on $\mathcal{O}$.

Proposition 4.5 ([C] Two elementary obstructions). *No field extension makes division by zero valid. Moreover st cannot be extended to a unital ring homomorphism from an entire non-Archimedean field to $\mathbb{R}$ or $\mathbb{C}$ annihilating the nonzero infinitesimals.*

Proof. In a field $0b = 0$ for every b , so $0b = 1$ is impossible. For the second assertion pick a nonzero infinitesimal ε ; a multiplicative extension would give $1 = \text{st}(\varepsilon\varepsilon^{-1}) = \text{st}(\varepsilon)\text{st}(\varepsilon^{-1}) = 0$. $\square$

So $1/\varepsilon$ is meaningful and unlimited while $1/0$ remains undefined; and [A] “read the observable by taking the standard part” is a rule for *limited* values only, unless another measurement prescription is supplied. Infinite curvature has no finite real standard part merely because it has acquired a

surreal name. An observational rule based on standard part is a possible modelling choice, not a consequence of the field axioms; it discards infinitesimal corrections by design, and a theory that instead measures them must specify what apparatus, outcomes and probability rule do so. Section 19 returns to this.

4.5 Dimensionful constants are not infinitesimals by being small

[A] A physical length is a numerical value *and* a unit. The statements $r/m = \varepsilon$ and $\ell_P/m = \varepsilon^\beta$ are well-defined scale comparisons; the statement “the radius is ω^{-1} ”, with no unit or reference length, is not. Multiplication by a fixed positive real conversion factor preserves infinitesimality, but a complete model must also say which dimensional constants and dimensionless ratios are held fixed in its asymptotic family. Throughout this report a *dimensionless ratio*, never a dimensionful quantity, is what may be declared infinitesimal.

The Planck length is a positive real length in conventional physics; its small numerical value in metres does not make it infinitesimal. If m and ℓ_P are fixed nonzero real lengths and ε is a genuine infinitesimal then $m\varepsilon \ll \ell_P$ — not merely “somewhat below an astrophysical radius”. A formal parameter representing a real-family limit $\ell_P/m \rightarrow 0$ is useful, but identifying that parameter with a physical constant is a separate interpretive step. This distinction becomes load-bearing in Section 8.

[A] There is also no physical reason, from the field alone, to prefer ω^{-1} to ω^{-2} or to $\exp(-\omega)$. Their radically different orders become useful when dictated by an equation or a matching condition. Choosing one as a fundamental cutoff without such a principle adds unexplained structure.

4.6 Transfer: what one may and may not infer

[I] Real-closed-field algebra has first-order transfer in the language of ordered fields, and richer elementary embeddings preserving restricted analytic and exponential structure exist in the settings studied by van den Dries, Ehrlich and Kaplan [10, 14, 9]. It would be incorrect to say that surreals admit no transfer.

[A] What does not follow from a bare field embedding is a transfer principle for all real functions, all subsets, Lebesgue measure, Hilbert spaces, or theorems about solutions of nonlinear partial differential equations. The language, the expanded structure and the admissible quantifiers matter. A nonstandard superstructure carrying internal sets and transferred integration supplies additional data that an ordered field alone does not specify; Loeb’s construction of an ordinary measure from nonstandard measure data is a useful illustration of how much surrounding structure such a result needs [39]. Thus “the formula still makes sense” is a much weaker statement than “the existence, uniqueness and stability theorem still holds”, and it would be equally misleading to say that everything transfers or that nothing does. The question is always which language, structure and hypotheses have actually been transported.

5 Topology and calculus are part of the physical proposal

5.1 The full fine topology is not ordinary spacetime topology

Use the positive surreal-valued modulus to define *fine balls* $B(a, r) = \{z \in \mathbf{No}[i] : |z - a| < r\}$, allowing *every* positive surreal radius r ; equivalently, take order balls on $\mathbf{No}$ and modulus balls on $\mathbf{No}[i]$ as the default neighbourhoods on the full proper class. All three manuscripts prove the

following, by the same Conway-cut argument; it is printed once, with the union of their statements and of their caveats.

Proposition 5.1 ([C] Set-sized collapse in the full fine topology). *Every set $A \subseteq \mathbf{No}[i]$ is closed and discrete in the full fine topology. Every convergent set-indexed net is eventually equal to its limit. Every continuous map from an ordinary connected real interval into this topology is constant. In particular the usual topology of $\mathbb{R}$ is not the subspace topology inherited from the full fine space.*

Proof. Fix $x \in \mathbf{No}[i]$. The nonzero distances $|x - a|$ for $a \in A \setminus \{x\}$ form a set of positive surreals, so the Conway cut $\{0 \mid |x - a|\}$ produces a positive δ smaller than every member of that set. The ball $B(x, \delta)$ therefore meets A only at x if $x \in A$, and misses A otherwise; an empty distance set causes no difficulty, since any positive radius serves. This gives discreteness and, applied to points outside A , closedness. The range of a set-indexed net together with its proposed limit is a set, so a ball isolating the limit forces eventual equality. Finally the image of a connected real interval is a set, hence discrete, and the continuous image of a connected space is connected, so the image is a single point. $\square$

Warning 5.2 (The three things Theorem 5.1 does not say). (a) It does *not* say the proper class $\mathbf{No}[i]$ is a discrete space. A singleton is not an open ball, and the proof cannot choose one radius below a *proper class* of positive distances. The restriction to set-sized subsets is essential and is stated as a hypothesis, not a convenience.

- (b) It does *not* apply to the *intrinsic* valuation or order topology of a set-sized Hahn field. In $\mathbb{R}((t^{\mathbb{Q}}))$ one has $t^n \rightarrow 0$ in its own valuation topology; after embedding into $\mathbf{No}$ a smaller scale such as $\omega^{-\omega}$ blocks that convergence. The two ambient topologies must never be identified. Even in a non-Archimedean ordered field, the embedded real sequence $1/n$ does not approach zero in the full order topology, because one fixed positive infinitesimal is below all of its terms.
- (c) It does *not* rule out coarser topologies or coefficientwise smooth structures. Four alternatives change the model in four different ways: the valuation topology on a set-sized field; a coefficientwise topology; the standard-part topology on $\mathcal{O}$, defined by inverse images of ordinary open sets under st , which cannot separate two points with the same standard part and whose natural separated quotient is ordinary space; and a specialized notion of surreal analytic function. None may be silently substituted for another in a proof about causal curves or limits.

[A] The constructive consequence adopted below is the fourth option in Section 3.1: retain ordinary real spacetime as the parameter domain and attach formal coefficients to its fields. That is proposal (P2), and it is chosen precisely so that no claim depends on ordinary real histories being continuous in the full fine topology.

5.2 Three derivatives, and a fourth object, that must not be conflated

This is the report's central hygiene rule. All three manuscripts state a version of it; their union is the following table, which is referred to throughout and never restated.

Table 2: The derivative and exponential ledger: domain, law, and what fails. No row may be used for another row's object.

Object	Domain and normalization	Law it obeys, and what fails
Coefficientwise spacetime derivative ∂_μ	Series $\sum_\gamma a_\gamma(x)t^\gamma$ with all $a_\gamma \in C^\infty(U)$ on <i>one common</i> ordinary domain $U \subseteq \mathbb{R}^4$; $\partial_\mu t^\gamma = 0$	Commutates with the other ∂_ν , Leibniz, commutes with st and with π_0 by construction. This is the derivative used for every metric perturbation below. It is <i>not</i> a derivative in the asymptotic parameter.
Parameter derivative $d/d\varepsilon$	$d(\varepsilon^q)/d\varepsilon = q\varepsilon^{q-1}$	Ordinary calculus in the parameter. Distinct from both neighbours; holding t constant while claiming to differentiate in r after setting $t = r/m$ is an error, since then $\partial_r = m^{-1}\partial_t$.
Scalar derivation D on No	Berarducci–Mantova derivation, normalized $D\omega = 1$, $D _{\mathbb{R}} = 0$ [11, 13]	Leibniz and strong additivity on No ; Liouville closed. Then $D(\omega^{-1}) = -\omega^{-2} = -\varepsilon^2$, neither 1 nor 0. It differentiates <i>elements read as asymptotic functions</i> ; it is not a physical time derivative, and one scalar derivation does not supply four commuting coordinate derivatives or a connection. Choosing it by analogy with time evolution can make a supposedly constant regulator time-dependent.
Derivative of a surreal <i>function</i>	Costin–Ehrlich extensions and integration on specified resurgent classes [16]	Genuine calculus on its stated class, with proved obstructions to broader constructive extension. A fourth object, not the scalar derivation.
Fine difference-quotient derivative	A chosen topology on a non-Archimedean field	Definable, but does not automatically preserve mean-value, compactness or existence theorems.
Real surreal exponential on No	$\exp : \mathbf{No} \rightarrow \mathbf{No}_{>0}$ [10]	Order isomorphism onto the positives; growth and decay only. Defines no oscillatory phase.
Infinitesimal Hahn exponential E	$E(h) = \sum_{n \geq 0} h^n/n!$ for $h \in \mathfrak{m}_{\mathbb{C}}$	Strongly summable; a homomorphism $(\mathfrak{m}_{\mathbb{C}}, +) \rightarrow \text{units}$, injective, and satisfies the chain rule for a compatible derivation. Defined only on infinitesimals.

Object	Domain and normalization	Law it obeys, and what fails
Finite-angle phase cis	$\text{cis}(\theta) = e^{i \text{st } \theta} E(i(\theta - \text{st } \theta))$ for $\theta \in \mathcal{O}$	Group law with kernel $2\pi\mathbb{Z}$; the ordinary factor comes from complex analysis, the infinitesimal tail from strong summation. Says nothing about infinite angles.
Global phase Exp_{EK}	Ehrlich–Kaplan integer-part construction with $\mathbf{Oz}$ as the canonical initial integer part [14]	A group homomorphism with kernel $2\pi i \mathbf{Oz}$, extending the ordinary complex exponential — but <i>no</i> chain rule for any derivation with $D\omega = 1$ (Theorem 16.2), and $\text{Exp}_{\text{EK}}(i\omega t) = 1$ for every real t (Theorem 16.3).
Quaternionic radial exponential	Surquaternionic manuscripts, docs/surquaternions/	<i>Not</i> an additive-to-multiplicative homomorphism for noncommuting arguments, and not an everywhere Hahn-summed power series. Named here only so that Exp_{EK} ’s homomorphism property is not carried across; no result of this report may be lifted to a noncommutative base without reproof.

Warning 5.3 (The rows are different objects, not competing accounts). Exp_{EK} *is* a group homomorphism; the quaternionic radial exponential is *not*; and a componentwise derivation can admit *no* global commuting exponential chain rule. All three statements are true, of three different objects. Flattening them into one symbol “the surreal exponential” manufactures a contradiction between correct theorems. Likewise $D(\omega^{-1}) = -\varepsilon^2$ contradicts nothing: ∂_μ , $d/d\varepsilon$ and D answer different questions, and $D(t) = 0$ for every real scalar t does not mean that a real-coordinate physical field has zero derivative.

5.3 Common domains and support conditions are essential

An expression $A(x) = \sum_\gamma a_\gamma(x) t^\gamma$ is used below only when the coefficient functions live on a specified common ordinary domain and the support rules justify the operations being performed. Having an analytic germ for each coefficient is strictly weaker than having a common neighbourhood: (3.1) is the witness. A coefficient domain that shrinks to nothing cannot support an advertised common spacetime neighbourhood.

[A] Likewise, a Taylor expansion does not determine an arbitrary smooth function: e^{-1/x^2} for $x > 0$, continued by zero for $x \leq 0$, has zero Taylor series at the origin and is not identically zero. A rule that extends all smooth functions by formal Taylor coefficients necessarily loses such information. A physical proposal must choose a function class or a richer extension rule rather than assuming that every classical field extends canonically.

5.4 Integration: substantial results, specified scope

[I] Costin and Ehrlich construct extensions of a specified class of resurgent real functions to surreal arguments, together with compatible antidifferentiation and integration; their class includes

important Borel-summable examples and solutions of nonresonant meromorphic systems, and changes of variables allow finite singular locations to be treated. They also prove obstructions to substantially more unrestricted constructive extensions [16].

[A] This is a positive starting point for physics, and a reduced gravitational ODE might fall inside such a class — in which case a surreal function extension would be far more informative than assigning a formal value to a divergent term. Membership, the relevant sector or branch and the chosen integration operator would all have to be checked. These results are not a general Einstein PDE existence theorem, and they concern derivations on *functions*, a different object from the scalar derivation of Table 2.

In the coefficientwise framework, integration over a fixed ordinary domain is instead defined by integrating each coefficient, provided all coefficient integrals exist and the resulting family remains admissible:

$$\int_U \left(\sum_{\gamma} L_{\gamma}(x) t^{\gamma} \right) d^4x := \sum_{\gamma} \left(\int_U L_{\gamma}(x) d^4x \right) t^{\gamma}, \quad (5.1)$$

whose support is contained in the original well-ordered support. This is not the full fine-topological limit of Riemann sums, and it is the only integration used below.

5.5 Coordinate and frame discipline

Proposition 5.4 (**[C]** Bounded-frame valuation invariance). *For a finite tuple V over a Hahn field put $v_*(V) = \min_j v(V_j)$. If a matrix A and its inverse A^{-1} both have limited entries, then $v_*(AV) = v_*(V)$.*

Proof. Limited entries give $v_*(AV) \geq v_*(V)$ by (4.4); applying the same inequality to $A^{-1}(AV)$ gives the reverse. $\square$

[A] An infinite boost, or a transformation of infinitesimal determinant, need not preserve this criterion, so component valuations are *not* unrestricted coordinate invariants. Curvature scalars, areal radius, proper time and tidal eigenvalues in a specified observer frame are the better primary diagnostics, and a general theory must say which formal coordinate changes preserve its coefficient algebra and what counts as a physically comparable observer.

5.6 Valuations classify balances, not solutions

Lemma 5.5 (**[C]** Dominant balance: the minimum is attained twice). *Suppose a dimensionless equation is a finite sum $\sum_{j=1}^N c_j \prod_{k=1}^d X_k^{a_{jk}} = 0$ with nonzero coefficients and integer exponents where the denominators exist. If the valuations of the terms had a unique smallest value, the corresponding leading coefficient could not cancel. Hence any solution requires the minimum valuation to be attained at least twice.*

[A] This gives a concrete dominant-balance algorithm: enumerate competing leading orders, solve the resulting leading-coefficient equations, then determine corrections. It can guide asymptotic Einstein or matter equations once a gauge and ansatz are fixed. It is a *necessary* condition and explicitly not sufficient: leading coefficients may fail to solve, constraints may conflict, logarithmic terms may be forced, and an apparently valid recursion may have no controlled real realization. Valuation arithmetic is attractive when several corrections compete — a candidate term proportional to $\varepsilon^a x^{-b}$ changes order according to $a - b\alpha$ when $x = \varepsilon^\alpha$, and a collection of such inequalities can

locate balances and propose rescalings before a costly expansion is attempted — but the physical coefficients and admissible operators must still come from an effective action or another specified dynamical model.

Part II

Black holes: the worked examples, and what they settle

6 What a black-hole singularity is

6.1 Four notions that must remain separate

[A] A coordinate formula can diverge even when the underlying geometry is regular. A curvature tensor can diverge in a freely falling frame even if some scalar invariants remain bounded, and the absence of a particular divergent invariant is not a general regularity theorem. A causal geodesic can be incomplete with no established curvature blow-up. And a spacetime can admit a continuous metric extension but no extension with the differentiability needed for the classical field equations. These are four different defects requiring four different notions of resolution [19, 21, 20, 22, 23].

Artificially excising a regular point produces incompleteness with no physical curvature singularity; demanding maximality or appropriate inextendibility addresses that separate issue [20]. So the questions “does a bad chart merely need replacement”, “is the spacetime extendible, and with which differentiability”, “do curvature scalars or freely falling tidal components diverge”, and “can a chosen matter field or quantum observable continue even when classical geometry does not” are logically independent. Answering one settles none of the others.

[A] These distinctions survive any serious change of scalars. A metric must be nondegenerate to define its inverse. Curvature needs derivatives in a specified sense. A geodesic needs an affine parameter and a solution theory. An extension needs a larger space, compatible charts or their replacement, and agreement with the original geometry. Adding ω to the scalar vocabulary supplies none of these structures. Replacing coefficients by surreal numbers cannot supply the missing geometric definitions; without them, “the geodesic continues through an infinite value” has no precise geometric content.

[A] The phrase “where usual real analysis fails” therefore needs refinement. Real differential geometry successfully identifies incomplete geodesics, distinguishes coordinate artefacts from invariant pathology, and estimates diverging tidal fields. Real analysis often succeeds precisely at proving that a particular classical solution cannot be extended with a specified regularity. That is a diagnosis of the solution or the model, not a contradiction in the real numbers, and describing the problem as a failure of real arithmetic misidentifies what needs repair.

6.2 What the singularity theorems actually constrain

[I] One standard form of Penrose’s theorem assumes global hyperbolicity with a noncompact Cauchy surface, an appropriate compact trapped surface and the null convergence condition, and concludes future null geodesic incompleteness [17, 19, 21]. The conclusion is not that every possible singularity

is a point of infinite density, nor that every curvature scalar must diverge, nor that a computer has encountered a large number. Its proof combines local focusing with global geometry.

[I] The local null Raychaudhuri equation for a twist-free congruence in four dimensions gives, under the null convergence condition,

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^ak^b \leq -\frac{1}{2}\theta^2, \quad (6.1)$$

with the shear contraction on the positive screen space [18, 19]. An initially negative expansion is driven to focusing within affine parameter at most $2/|\theta_0|$, as long as the regular evolution in question exists. The equality model is

$$\theta(\lambda) = \frac{\theta_0}{1 + \frac{1}{2}\theta_0\lambda}, \quad \theta_0 < 0, \quad (6.2)$$

with a pole at $\lambda_* = -2/\theta_0$.

Example 6.1 ([E] Evaluating the focusing model at an infinitesimal). Evaluating (6.2) at $\lambda_* - \varepsilon$ replaces a large negative real expansion by an unlimited negative surreal expansion; at λ_* the denominator is still zero. Infinite numbers alone do not prevent focusing. The algebraic inequality (6.1) can be represented in an ordered extension field; the global causal and completeness assumptions cannot be replaced by field arithmetic.

[A] A caustic of a congruence is not itself necessarily a spacetime singularity: the global argument is what turns focusing, under further assumptions, into incompleteness. A new theory may evade a hypothesis by modifying the effective energy condition, causal structure, topology, differentiability, or the spacetime concept. Adopting an ordered extension of $\mathbb{R}$ does not identify which hypothesis changes and proves no replacement theorem. Conversely, showing that the classical proof cannot be copied into an unfamiliar topology — where, by Theorem 5.1, global compactness and causal-topological arguments are not preserved — does not exhibit a nonsingular physical spacetime.

6.3 Changing variables does not automatically change the physics

Example 6.2 ([E] A pole, a projective coordinate, and what each means). For $y' = y^2$ the solution is $y = (t_* - t)^{-1}$. At $t = t_* - \varepsilon$ the value is ε^{-1} ; at $t = t_*$ the identity $(t_* - t)y = 1$ has no field-valued solution at all. The projective coordinate $z = 1/y$ obeys $\dot{z} = -1$ and crosses zero smoothly, so a compactified state space continues the *projective* trajectory. But the inverse map fails at $z = 0$: if y is the measured stress, replacing it by z has not made the measured stress bounded. Which of y and z is the physical observable is additional information.

[A] This guards against two opposite errors. Not every divergence is a fundamental obstruction, because some are removable by changing the geometric description. But not every compactification is a physical resolution, because the regularity of the relevant observables must be checked.

[E] Reparametrizing time is likewise not a cure. If a timelike geodesic reaches the end of its domain at finite proper time τ_* , then $u = -\log(\tau_* - \tau)$ moves that endpoint to $u = +\infty$ without changing the elapsed proper time; u is not the proper-time parameter and in general is not affine. Incompleteness is not cured by relabelling the clock. A surreal value for u may encode an extraordinarily late stage of that reparametrization; the missing proper-time continuation is a separate problem. A compactification, a blow-up, a reciprocal field and a geodesic completion are four different operations.

7 Schwarzschild at an infinitesimal radius

7.1 Units and conventions

Signature $(-, +, +, +)$, geometrized units $G = c = 1$. The mass parameter $m = GM/c^2$ has dimensions of length; proper time is expressed in length units, so $\tau = c\tau_{\text{physical}}$ and $[\mathcal{K}] = \text{length}^{-4}$. Per Section 4.5, a dimensionless ratio is what may be declared infinitesimal.

7.2 The horizon is already regular over the reals

The Schwarzschild metric in its familiar static chart is

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2d\Omega^2, \quad f(r) = 1 - \frac{2m}{r}, \quad (7.1)$$

with $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. Introduce the advanced coordinate $v = t + r_*$ with $dr_*/dr = f^{-1}$; substitution gives

$$ds^2 = -f(r)dv^2 + 2dvdr + r^2d\Omega^2. \quad (7.2)$$

Computation 7.1 ([E] The horizon needs no new scalars). The (v, r) block of (7.2) has determinant -1 even at $r = 2m$, and away from the ordinary angular-coordinate poles the full determinant is $-r^4 \sin^2\theta$, nonzero there. Hence the divergence of g_{rr} at the horizon in (7.1) is removed by an ordinary *real* coordinate change. No generalized number is needed. All three companion programs check this determinant.

[E] Inside the horizon $g^{ab}(\partial_a r)(\partial_b r) = f(r) < 0$, so the gradient of the areal radius is timelike. [A] The future singular boundary is therefore not analogous to an obstacle at the centre of a room that a traveller could circle around, and allowing a complex-valued radius does not preserve the real Lorentzian problem.

7.3 The invariant calculation

Computation 7.2 ([E] General static-spherical curvature). For a metric of the form (7.1) with general f ,

$$\mathcal{K}[f] := R_{abcd}R^{abcd} = (f'')^2 + 4\left(\frac{f'}{r}\right)^2 + 4\left(\frac{1-f}{r^2}\right)^2, \quad (7.3)$$

and, contracting before squaring,

$$\text{Scal} = -f'' - \frac{4f'}{r} + \frac{2(1-f)}{r^2}, \quad G^t_t = G^r_r = \frac{rf' + f - 1}{r^2}; \quad (7.4)$$

for $f = 1 - 2M(r)/r$ the last becomes $G^t_t = -2M'(r)/r^2$. Section A derives these by an orthonormal coframe. *Each of the three companion programs rebuilds the Christoffel symbols, the full four-index Riemann tensor, the Ricci tensor, the scalar curvature and the mixed Einstein component from the metric alone, with f an unevaluated symbolic function, and contracts them to confirm (7.3)–(7.4). None of them substitutes into an assumed answer.*

Computation 7.3 ([E] Schwarzschild). Substituting $f = 1 - 2m/r$ into (7.3) gives

$$\mathcal{K} = \frac{48m^2}{r^6}, \quad \mathcal{K}(2m) = \frac{3}{4m^4}, \quad (7.5)$$

with the three contributions of (7.3) equal to $16m^2/r^6$ each. For $r > 0$ Schwarzschild is vacuum: all sixteen Ricci components and the Ricci scalar vanish, and $G^t_t = 0$. The divergent invariant is built from the full Riemann tensor, equivalently the Weyl tensor in this vacuum region.

Warning 7.4 ([A] The divergence is not an infinite matter density). Because $\text{Ric} = 0$ for $r > 0$, it is misleading to explain this example as an infinite ordinary matter density that could be replaced by a surreal density. The vanishing of the Ricci tensor in this vacuum solution does not contradict the nonzero, divergent Weyl curvature. Whatever a replacement theory does, it is not merely renaming a density.

Computation 7.5 ([E] Evaluation at an infinitesimal radius). Let $x = r/m$, so the dimensionless invariant is $m^4\mathcal{K} = 48x^{-6}$; with $x = \varepsilon > 0$ a positive infinitesimal,

$$m^4\mathcal{K} = 48\varepsilon^{-6}. \quad (7.6)$$

Equivalently, with an ordinary reference length L , $m = \mu L$ and $r = L\varepsilon$,

$$L^4\mathcal{K} = 48\mu^2\varepsilon^{-6}. \quad (7.7)$$

This is an exact and genuinely useful non-Archimedean description: it records both the power and the coefficient of the divergence, and the compared quantity is dimensionless. The value can be added, multiplied, compared and inverted; it dominates $m^{-4}\varepsilon^{-5}$ and is dominated by $m^{-4}\varepsilon^{-7}$, information that a bare symbol $+\infty$ destroys. The valuation of $m^4\mathcal{K}$ is exactly the blow-up exponent.

Warning 7.6 ([A] What Theorem 7.5 does and does not say). The value (7.6) is unlimited: larger than every positive real and with no finite standard part. It exceeds every real curvature bound in these units, and the physics of tidal exposure is not made finite by reclassifying it as a legitimate scalar. The radius here is the *areal* radius, defined by the area $4\pi r^2$ of the symmetry spheres, so no change of coordinates alters the scalar value at a fixed event; one may alter the notion of event or replace the spacetime model, but that is more than an algebraic relabelling. At $r = 0$ the defining expression still has a pole.

7.4 Poles persist: two independent routes

Two of the three manuscripts prove that the pole survives, by arguments with different hypotheses and different strengths. Both are printed.

Proposition 7.7 ([C] Route 1, valuation-theoretic: poles survive scalar extension and ramification). *Let $p \geq 1$ be an integer, let $u(r) \in \mathbb{R}[[r]]$ have nonzero constant term c , and put $f(r) = r^{-p}u(r)$. Evaluate the series at a positive infinitesimal ε in a Hahn field, using strong summation. Then*

$$\mathbf{v} f(\varepsilon) = -p \mathbf{v} \varepsilon, \quad f(\varepsilon) = c\varepsilon^{-p}(1 + \delta), \quad \delta \in \mathbf{m}_F,$$

so $f(\varepsilon)$ is unlimited. Moreover, under a formal change of radial variable $r = s^qw(s)$ with $q \geq 1$ an integer and $w \in \mathbb{R}[[s]]$ having $w_0 = w(0) \neq 0$, one has

$$f(s^qw(s)) = s^{-pq}\tilde{u}(s), \quad \tilde{u} \in \mathbb{R}[[s]], \quad \tilde{u}(0) = cw_0^{-p} \neq 0.$$

Thus the pole order is exactly pq , and its leading Laurent coefficient is cw_0^{-p} . The pole persists although its coefficient can change.

Proof. Strong summation of $u(\varepsilon)$ is valid by the positive-valuation power-series rule, and its constant term is c , so $u(\varepsilon) = c(1 + \delta)$ with $v\delta > 0$ unless $\delta = 0$. Multiplying by ε^{-p} gives the stated leading term and valuation. For the change of variable,

$$f(s^q w(s)) = s^{-pq} w(s)^{-p} u(s^q w(s)).$$

The substitution into u is defined as a formal power series because $s^q w(s)$ has positive order. Since $w_0 \neq 0$, the factor $w(s)^{-p}$ is a power-series unit with constant term w_0^{-p} , while $u(s^q w(s))$ has constant term c . Their product is the required $\tilde{u}(s)$, with nonzero constant term $c w_0^{-p}$. No term of higher s -order can cancel the resulting leading term at exponent $-pq$. $\square$

Proposition 7.8 ([C] Route 2, order-theoretic: a leading pole stays unlimited in every ordered extension). *Let $F \supseteq \mathbb{R}$ be an ordered field, $\varepsilon > 0$ infinitesimal, and $p > 0$ such that ε^p is defined and infinitesimal in F . If $a \in \mathbb{R} \setminus \{0\}$ and u is infinitesimal, then $I = a\varepsilon^{-p}(1 + u)$ is unlimited; and no ordered extension of F fixing $\mathbb{R}$ makes that same element limited.*

Proof. Since $|u| < 1/2$ we get $|I| > |a|\varepsilon^{-p}/2$, which exceeds every positive real bound because ε^p is infinitesimal. Every inequality with a real bound is preserved by an ordered extension. $\square$

Remark 7.9 (Why both routes are kept). They are not the same theorem. Theorem 7.7 is a statement about Hahn-field valuation data and is what yields *ramification invariance*: no radial reparametrization of the stated form removes the blow-up. Theorem 7.8 is a statement about the order alone and is what yields *extension invariance*: passing from a small ordered asymptotic field to all of **No** cannot turn the same curvature observable into a limited one. Neither implies the other, and the two together are the precise content of “a larger field does not help”. Subtracting terms, rescaling the measured observable by an infinitesimal, or changing the metric can change the conclusion — but each introduces additional structure or a new physical claim.

Computation 7.10 ([E] Ramification invariance, checked). For $m \neq 0$, Theorem 7.7 applies to (7.5) with $p = 6$ and $c = 48m^2$. In the checked special case $w(s) = 1 + s$,

$$\mathcal{K}(s^q(1 + s)) = 48m^2 s^{-6q}(1 + s)^{-6}.$$

Expansion for $q = 1, 2, 3$ gives pole order exactly $6q$ and leading coefficient $48m^2$. Its preservation in these tests comes from the normalization $w(0) = 1$. For a general unit w , the proposition instead gives $48m^2 w(0)^{-6}$; for example, the constant unit $w = 2$ gives $3m^2/4$. The pole remains in each case.

The finite verification reconstructed the Christoffel symbols and the Riemann tensor from the metric and contracted them to obtain $\mathcal{K} = 48M^2/r^6$ before making the normalized substitutions. Those three tests illustrate the general formal-series argument; they do not assert an unchanged coefficient for arbitrary $w(0) \neq 0$. *No formal radial reparametrization satisfying the proposition’s hypotheses removes the pole.*

Corollary 7.11 ([C] The real-analytic geometric consequence). *If a dimensionless scalar curvature invariant tends to infinity along a curve, there is no extension along that curve to an ordinary nondegenerate C^2 metric at an endpoint at which that invariant is continuous and finite: continuity would give a finite limit.*

Status. A restricted and useful obstruction. [A] It is *not* a claim that all conceivable generalized geometries have been ruled out, and it says nothing about a degenerate limiting metric, a weaker regularity class, or a geometry with no such endpoint event.

7.5 The algebraic endpoint obstruction

Two of the three manuscripts prove the following by the same one-line argument; it is printed once.

Proposition 7.12 ([C] No field-valued Schwarzschild endpoint). *On the Schwarzschild region,*

$$r^6 \mathcal{K} = 48m^2. \quad (7.8)$$

Let E be any characteristic-zero field containing a nonzero m . There is no assignment of a value $\mathcal{K}_0 \in E$ at $r = 0$ extending (7.8) to that endpoint. In particular, adjoining surreal or surcomplex values does not extend this invariant identity through $r = 0$: a regular extension preserving this relation at its endpoint is impossible in any field, not merely in $\mathbb{R}$.

Proof. At $r = 0$ the left side of (7.8) is zero for every field element $\mathcal{K}_0$, including one unlimited relative to $\mathbb{R}$. The right side is nonzero. $\square$

Status. Deliberately a precise, limited statement, and explicitly a *conditional* obstruction. [A] It does not forbid a modified metric, a different field equation, a nonclassical notion of observable, a quantum replacement of the endpoint, or a geometry that has no zero-areal-radius event at all. Identifying an endpoint by an adjoined compactification symbol would change the algebra and would still require a metric and an evolution law there. Setting $r = \varepsilon$ merely avoids $r = 0$; it does not provide the missing assignment.

7.6 An unlimited number is not an added regular event

[A] Assigning a nonzero infinitesimal radius describes another point in a punctured extension of the coordinate domain. It does not add the missing endpoint. There is also no “last positive radius” before zero, since $\varepsilon/2$ is smaller. A proposed rule that replaces zero by one chosen infinitesimal has introduced a *regulator*, and must explain the choice, its physical units, the resulting equations, and the invariance of observations under changing that choice.

[A] Allowing a metric to have unlimited coefficients might be part of a broader theory. In that case the theory must define regularity for those coefficients and the behaviour of clocks, rods and tidal observables there. It cannot infer regularity from the fact that a coefficient belongs to **No**.

8 The classical theory loses control before a literal infinitesimal scale

Warning 8.1 (Placement is deliberate). This section follows Section 7 immediately, and not a dozen sections later, because the preceding computation is the one a reader is most likely to misread as a physical regime. One of the three manuscripts makes the present argument its headline; another makes it inside an effective-field-theory section; the third defers it. The buried version invites exactly the misreading this report exists to prevent, so the argument is promoted here to the status the first manuscript gives it.

8.1 A positive real high-curvature scale is encountered first

[I] Low-energy quantum gravity can be treated as an effective field theory; its controlled use depends on suitable separation from the ultraviolet scale, not on an assumption that the classical equations

hold at arbitrarily high curvature [24, 25]. The coefficients, field content and state are physical inputs, not consequences of choosing **No**.

Computation 8.2 ([E] The Planck-curvature radius). With $\ell_P = \sqrt{\hbar G/c^3}$, the dimensionless indicator $\mathcal{Q} = \ell_P^4 \mathcal{K} = 48m^2 \ell_P^4 / r^6$ on Schwarzschild. Setting $\mathcal{Q} \sim 1$ gives

$$r_Q = 48^{1/6} (m \ell_P^2)^{1/3}. \quad (8.1)$$

For $m \gg \ell_P$,

$$\frac{r_Q}{\ell_P} = 48^{1/6} \left(\frac{m}{\ell_P}\right)^{1/3} \gg 1, \quad \frac{r_Q}{2m} = \frac{48^{1/6}}{2} \left(\frac{\ell_P}{m}\right)^{2/3} \ll 1. \quad (8.2)$$

Status. The algebra of (8.1)–(8.2) is exact and is checked by two of the three programs. **[A]** Its *reading* is dimensional power counting, not a derived universal threshold, not a sharply defined transition radius, and not an exact prediction of any quantum-gravity theory. Derivatives, matter scales, the quantum state and the coefficients of the effective action can introduce other relevant criteria and can invalidate a simple curvature criterion sooner.

[A] Two consequences follow, and they are the point of this section.

First, a macroscopic areal radius deep inside a large black hole can coexist with Planck-scale curvature: by (8.2) the radius at which $\mathcal{Q} \sim 1$ is far above the Planck length while far below the horizon. “Small radius” and “high curvature” cannot be identified without specifying the mass. A black-hole horizon, a Planck-curvature regime and a classical singular boundary are three distinct things, and low horizon curvature for a large black hole is entirely compatible with high interior curvature.

Second — and this is the load-bearing statement — fix ordinary positive real L and ℓ_P/L . A literal positive infinitesimal ε satisfies $\varepsilon < \ell_P/L$, and indeed lies below *every* positive real cutoff ratio. Evaluating the unchanged classical solution at $r = L\varepsilon$ therefore explores a regime beyond any such fixed cutoff. Equivalently, if $r = m\varepsilon$ with fixed real $m, \ell_P > 0$ then

$$\mathcal{Q} = 48 \left(\frac{\ell_P}{m}\right)^4 \varepsilon^{-6}, \quad (8.3)$$

which is unlimited. A surreal-valued computation can faithfully report that the expansion parameter is large; it cannot thereby justify neglecting the higher-order terms that made the classical approximation unreliable. The computation of Section 7 is therefore a *mathematical extension* of a classical formula, not a physical regime.

8.2 Two-parameter power counting becomes exact valuation arithmetic

The three manuscripts present two versions of the same device, with different parametrizations. Both are printed because their independent variables differ.

Computation 8.3 ([E] Version A: radius and Planck length scaled against the mass). Let $r/m = \varepsilon^\alpha$ and $\ell_P/m = \varepsilon^\beta$ with $\alpha, \beta > 0$ real. Then

$$\ell_P^4 \mathcal{K} = 48 \varepsilon^{4\beta - 6\alpha}, \quad (8.4)$$

which is infinitesimal if $2\beta > 3\alpha$, of order one if $2\beta = 3\alpha$, and unlimited if $2\beta < 3\alpha$.

Computation 8.4 ([E] Version B: mass scaled against the Planck length). Let $m = \ell_P \varepsilon^{-a}$ and $r = m \varepsilon^b$ with $a, b > 0$. Then

$$\mathcal{Q} = \ell_P^4 \mathcal{K} = 48 \varepsilon^{4a-6b}, \quad (8.5)$$

sub-Planckian for $b < 2a/3$, of order one at $b = 2a/3$, super-Planckian for $b > 2a/3$ — even though r/m is infinitesimal in *every* case.

[A] Version B does not contradict (8.3): it changes an explicit assumption, namely that the mass is held at a fixed real value. That is precisely the merit of the device. “Infinitesimally far inside relative to the mass scale” does not determine the curvature regime when the mass is simultaneously scaled, and a valuation representation makes the competing assumptions visible and mechanically checkable. [A] The ratio assignments describe a *family of models*; they do *not* assert that a fixed measured Planck length is literally an infinitesimal. Physical application still requires choosing a finite large mass and estimating how accurately the formal family describes it.

8.3 Hawking modes are a different question from the central singularity

[A] A local mode frequency is measured relative to an observer, for example through $-k_a u^a$. Large backward-traced frequencies in horizon calculations concern redshift, states, propagation and short-distance dynamics; they are not the same invariant issue as $\mathcal{K} \rightarrow \infty$ at $r = 0$. [I] Unruh and Schützhold derive conditions under which modified high-frequency dispersion still yields the Hawking effect to leading order, and also exhibit ways those conditions can fail [31]. [A] A surcomplex symbol for an unlimited phase or frequency selects none of the dispersion law, the state or the boundary condition used in such a calculation. It can be a bookkeeping variable inside a specified model; it is not a substitute for those inputs.

9 Proper-time infall and the physical tidal field

9.1 An exact radial geodesic

Computation 9.1 ([E] The $E = 1$ radial infall). For a radial timelike Schwarzschild geodesic with conserved specific energy $E = f \, dt/d\tau$, four-velocity normalization gives $(dr/d\tau)^2 = E^2 - f(r)$. Choose infall from rest at infinity, $E = 1$:

$$\left(\frac{dr}{d\tau}\right)^2 = \frac{2m}{r}, \quad \frac{dr}{d\tau} = -\sqrt{\frac{2m}{r}}. \quad (9.1)$$

With $u = \tau_* - \tau > 0$ the remaining proper time,

$$u = \frac{2r^{3/2}}{3\sqrt{2m}}, \quad r^3 = \frac{9}{2}mu^2, \quad r = \left(\frac{9m}{2}\right)^{1/3}u^{2/3}, \quad \frac{m}{r^3} = \frac{2}{9u^2}, \quad \mathcal{K} = \frac{64}{27u^4}. \quad (9.2)$$

The cancellation of the mass in the proper-time curvature is exact for this geodesic.

[A] The fractional power $r \propto u^{2/3}$ is exactly what a ramified Hahn or Puiseux workspace represents well, and taking $u = L\varepsilon$ produces exact infinitesimal distances and unlimited curvature while retaining the relation to the infaller’s clock. That is an improvement in *representation*. It is not an improvement in the boundedness of (9.2), and it does not make the remaining proper time to the classical endpoint infinite.

Computation 9.2 ([E] The infall terminates at finite proper time — and at infinitesimal proper time for infinitesimal radius). For radial infall from rest at areal radius R , the proper time to reach $r = 0$ is

$$\tau = \int_0^R \frac{dr}{\sqrt{2M(r^{-1} - R^{-1})}} = \frac{\sqrt{2}\pi R^{3/2}}{4\sqrt{M}} = \frac{\pi}{2} \sqrt{\frac{R^3}{2M}}, \quad (9.3)$$

finite for finite R, M . This was verified independently by evaluating the quadrature in closed form. Moreover, for *infinitesimal* R the value (9.3) has $\sqrt{\tau} = \frac{3}{2} \sqrt{R} > 0$, so it is infinitesimal — and hence still limited, indeed still *finite*. The infall terminates either way.

Warning 9.3 ([A] The one place a reader may hope for relief, and why there is none). A tempting thought is that at an infinitesimal radius the remaining proper time becomes “infinitely short” in a way that makes the endpoint unreachable. Theorem 9.2 refutes it exactly: the proper time is infinitesimal, and an infinitesimal is limited. It is not $+\infty$, it is not unreachable, and no ordered extension makes it so. The clock runs out.

9.2 Geodesic deviation resolves the scale hierarchy

Computation 9.4 ([E] Tidal equations and Jacobi exponents). In a parallel-propagated radial freely falling orthonormal frame, with the convention in which the radial relative acceleration is stretching,

$$\frac{d^2 \xi_r}{d\tau^2} = \frac{2m}{r^3} \xi_r = \frac{4}{9u^2} \xi_r, \quad \frac{d^2 \xi_\perp}{d\tau^2} = -\frac{m}{r^3} \xi_\perp = -\frac{2}{9u^2} \xi_\perp. \quad (9.4)$$

Since $d^2/d\tau^2 = d^2/du^2$, the ansatz $\xi = u^a$ gives the indicial equations $a(a-1) = \frac{4}{9}$ and $a(a-1) = -\frac{2}{9}$, hence

$$\xi_r = A u^{4/3} + B u^{-1/3}, \quad (9.5)$$

$$\xi_\perp = C u^{2/3} + D u^{1/3}. \quad (9.6)$$

For a generic diagonal family of three independent Jacobi separations the radial direction stretches like $u^{-1/3}$ while the two transverse directions shrink like $u^{1/3}$, so the associated volume scales like $u^{1/3}$ and tends to zero. Independently, the tidal component $R^r_{trt} = 2M(2M - r)/r^4$ diverges as $r \rightarrow 0$ and is unlimited at infinitesimal r .

Status. Exact for this geodesic family in the stated frame convention. [A] The displayed rates are *generic*: special initial data can remove a leading term and change them, though that only relocates the dominant power and never removes the divergence of the curvature itself. A radial boost preserves the principal tidal coefficients, which is why (9.4) holds in the freely falling frame and not only in the static one.

[A] The surreal description retains all these scales simultaneously: for $u/m = \varepsilon$ it records an unlimited tidal coefficient, an unlimited radial stretching mode and infinitesimal transverse separations, all at once, and it connects the singular regime to an observer’s clock and to neighbouring freely falling trajectories. That is more informative than “infinite curvature is now allowed”. Calling these legitimate numbers is mathematically coherent. Calling them bounded physical forces, or a nonsingular passage, is not. A surreal evaluation supplies no surviving detector, restores no nonzero volume and determines no propagation through the endpoint.

9.3 Why a formal bounce is not a geodesic extension

[A] One can write $r \propto |u|^{2/3}$ on both sides of $u = 0$, and the algebraic relation $r^3 = (9m/2)u^2$ admits a formal other branch. This does not produce a smooth Schwarzschild continuation through the singularity: the first derivative of r is unbounded there, the curvature still behaves as u^{-4} , and the original metric and geodesic equation are undefined at the joining event. The relation alone omits the differentiability, curvature, orientation and matching information a physical extension needs. Matching two formulas across an omitted point supplies neither the missing geometry nor a unique law for outgoing matter.

Computation 9.5 ([E] A smoother parameter is not a regular metric). Write $u = L\lambda^3$ with a fixed real length L , so $r \propto \lambda^2$ and the fractional power in the curve disappears. But $d\tau/d\lambda = -3L\lambda^2$ vanishes at zero, and the proper-time metric on the curve pulls back to

$$-d\tau^2 = -9L^2\lambda^4 d\lambda^2, \quad (9.7)$$

which degenerates there. This is not a nonsingular coordinate transformation at the endpoint.

Status. [A] Resolving a ramified formula, or blowing up an algebraic variety, does not resolve a nondegenerate Lorentzian metric. Algebraic desingularization remains a useful organizational tool provided the distinction stays explicit.

[A] There is no assertion here that a quantum-gravitational bounce is impossible. Such a bounce would need a different dynamical or geometric account of the transition. Surreal methods might describe its asymptotic incoming and outgoing regions; the transition law is precisely the physical information the classical formulas do not contain.

10 A constructive multiscale application: small angular momentum

The u^{-4} curvature law of Theorem 9.1 must not be universalized to every Schwarzschild geodesic. Its dependence on angular momentum is the clearest example in this report of something an ordered asymptotic field genuinely organizes.

Computation 10.1 ([E] Nonradial infall changes the proper-time exponent). Restrict by spherical symmetry to the equatorial plane and let $j = r^2 d\phi/d\tau$ be the conserved specific angular momentum. For finite E and $j \neq 0$,

$$\dot{r}^2 = E^2 - \left(1 - \frac{2m}{r}\right) \left(1 + \frac{j^2}{r^2}\right) = \frac{2mj^2}{r^3} + \frac{2m}{r} - \frac{j^2}{r^2} + E^2 - 1. \quad (10.1)$$

The first term dominates as $r \downarrow 0$ with j fixed and nonzero, and integrating the leading balance gives

$$r \sim \left[\frac{5}{2}\sqrt{2m}|j|s\right]^{2/5}, \quad \mathcal{K} \sim 48m^2 \left[\frac{5}{2}\sqrt{2m}|j|\right]^{-12/5} s^{-12/5}, \quad (10.2)$$

where s is the remaining proper time. The divergence remains; its power in remaining proper time changes from -4 to $-12/5$.

[A] This is not a change in the curvature as a function of areal radius. It is a change in how the worldline approaches the same invariantly characterized region.

Computation 10.2 ([E] An exact dimensionless crossover quadrature). Let $J = |j| > 0$, set $r = JR$, and define the dimensionless remaining clock $S = \sqrt{2m} J^{-3/2} s$. For the ingoing branch reaching $r = 0$, integrating the exact first integral (10.1) gives

$$S = \int_0^R \frac{u^{3/2} du}{\sqrt{1 + u^2 + \frac{J}{2m} [(E^2 - 1)u^3 - u]}}, \quad (10.3)$$

the radicand being positive in the sufficiently small interior region under consideration. *The exact integrand denominator factor in (10.3) is symbolically checked.*

Proposition 10.3 ([C] The small- J limit, uniform on bounded intervals). *As $J/m \rightarrow 0$ with E fixed and R ranging over any bounded positive interval, the reciprocal square root in (10.3) converges uniformly to that of $1 + u^2$. Consequently*

$$S \rightarrow \Phi(R) := \int_0^R \frac{u^{3/2}}{\sqrt{1 + u^2}} du, \quad (10.4)$$

with the elementary endpoint asymptotics

$$\Phi(R) \sim \frac{2}{5} R^{5/2} \quad (R \downarrow 0), \quad \Phi(R) \sim \frac{2}{3} R^{3/2} \quad (R \rightarrow \infty). \quad (10.5)$$

Status. The first asymptotic in (10.5) yields the angular-momentum-dominated law (10.2); the second recovers the radial law (9.2) in the overlap region $J \ll r \ll m$. [A] The convergence is uniform *only* on bounded positive R -intervals; the large- R behaviour of the limiting function is a matched-asymptotic statement and is *not* a claim of uniform convergence on an unbounded interval.

Computation 10.4 ([E] Exponent sector tests). With $m = \mu L$, $J = L\varepsilon^\beta$ and $s = L\varepsilon^\alpha$ for real $\alpha, \beta > 0$, the consistency conditions are

$$\alpha < \frac{3}{2}\beta : \quad r/L \text{ has leading exponent } \frac{2\alpha}{3}, \quad r \gg J, \quad (10.6)$$

$$\alpha > \frac{3}{2}\beta : \quad r/L \text{ has leading exponent } \frac{2(\alpha+\beta)}{5}, \quad r \ll J. \quad (10.7)$$

At equality neither monomial may be discarded, and Φ describes the transition. Real coefficients of order one can be restored without changing these exponent tests.

[A] This is what useful surreal-compatible physics mathematics looks like: retain several small scales, determine which terms dominate, avoid taking incompatible limits. Note carefully what supplies what. Ordinary real analysis supplies the actual crossover *function* Φ ; Hahn valuations supply a concise way to *select* its asymptotic regions. The construction offers no new gravitational law, and it is equally expressible in a suitable transseries or generalized-series field. Its value must be judged by clarity, support control and computational reliability — not by whether the ambient field contains a proper class of still larger numbers.

11 Beyond one pole: anisotropy, weak singularities, and other interiors

11.1 Kasner geometry as an anisotropic benchmark

Computation 11.1 ([E] Kasner curvature). For the vacuum Kasner metric, with reference scales absorbed into the coordinates,

$$ds^2 = -d\tau^2 + \sum_{j=1}^3 \tau^{2p_j} dx_j^2, \quad \sum_j p_j = 1, \quad \sum_j p_j^2 = 1, \quad (11.1)$$

a diagonal orthonormal frame gives curvature components with coefficients proportional to $p_j(p_j - 1)/\tau^2$ in the time-space planes and $p_j p_k/\tau^2$ in the spatial planes, whence

$$\mathcal{K} = \frac{4}{\tau^4} \left[\sum_j p_j^2 (p_j - 1)^2 + \sum_{j < k} p_j^2 p_k^2 \right]. \quad (11.2)$$

For $(p_1, p_2, p_3) = (-\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$ the coefficient is $64/27$ — the same coefficient met in the Schwarzschild proper-time infall (9.2). For each of the three permutations of $(1, 0, 0)$ it vanishes, corresponding to a flat coordinate example rather than a genuine curvature divergence. Both constraints, the $64/27$ coefficient and all three flat exceptional cases are symbolically checked.

[A] The value of a Hahn description here is not “curvature is infinite”. It simultaneously retains three anisotropic scales, their products and the orders of the tidal terms, and it makes *cancellation* visible: identical exponents alone do not prove a nonzero leading curvature coefficient, as the flat exceptional case demonstrates. **[I]** Rigorous singular asymptotics already exist in special Einstein–matter settings; Andersson and Rendall’s quiescent Einstein–scalar and stiff-fluid analysis is a concrete benchmark [26]. **[A]** A useful surreal project would reproduce a stated solution class and its error estimates, not replace those estimates by an appeal to infinitesimal notation.

Warning 11.2 (**[A]** One Kasner expansion is not a theorem about interiors). Generic oscillatory sequences of changing Kasner epochs raise additional issues, and a single well-ordered expansion must not be assumed to encode every possible interior evolution. **[I]** BKL-type analyses describe changing anisotropic regimes, often represented by billiard behaviour, with nontrivial scope and assumptions [27]. **[A]** Hardy fields and most asymptotic transseries constructions are designed for eventually ordered, nonoscillatory behaviour; oscillatory phases, repeated changes of dominant balance, or spatially localized features may require additional variables, sectors, or a different function category. A Schwarzschild Puiseux law licenses *no* inference about generic interiors. The real surreal exponential does not by itself define an unrestricted periodic function at every infinite argument.

11.2 Divergent curvature need not have the same integrated effect

Computation 11.3 ([E] An integrated tidal criterion). Consider the schematic tidal magnitude — *not* a universal formula for black-hole interiors —

$$|T(s)| \asymp \frac{A}{s^2 [\log(L_0/s)]^q}, \quad 0 < s < s_0 < L_0/e. \quad (11.3)$$

A natural absolute-integrability diagnostic for the displacement kernel of a Jacobi equation is the twice-integrated, s -weighted quantity

$$\int_0^{s_0} s|T(s)|ds \asymp A \int_{\log(L_0/s_0)}^{\infty} v^{-q} dv, \quad (11.4)$$

by $v = \log(L_0/s)$, which is finite exactly when $q > 1$. The weight s arises by integrating a positive forcing twice up to the endpoint and reversing the order of integration. Both the $q \neq 1$ primitive and the logarithmic boundary primitive are symbolically checked.

Status. Curvature can diverge while this weighted tidal integral converges, so two profiles both with leading power s^{-2} can differ in integrated effect because of logarithmic factors. **[A]** This is a diagnostic of an integral equation and a *model* criterion; it is explicitly not a theorem giving an extension of a spacetime or its matter fields, and not a substitute for all hypotheses in a geometric definition of singularity strength.

[A] A surreal or transseries representation preserves that distinction, which the undifferentiated statement “the tidal field is an infinite number” cannot. Integrability of the distortion kernel is information supplied by the calculus and the equations; it does not follow from making $T(\varepsilon)$ a named field element. **[I]** Weak mass-inflation-type singularities motivate exactly this examination; representative charged black-hole analyses separate diverging curvature from finite tidal distortion [30].

11.3 Charged and rotating interiors are not Schwarzschild copies

[I] Inner-horizon instability and mass inflation in charged or rotating models introduce blueshift, late-time tails and regularity questions different from the spacelike Schwarzschild singularity. Dafermos’s spherically symmetric Einstein–Maxwell–scalar analysis shows mass-inflation behaviour coexisting with a continuous metric extension while stronger regularity is obstructed [22]; Dafermos and Luk prove C^0 stability of the Kerr Cauchy horizon under prescribed interior-data hypotheses, obtaining continuous metric extensions across a nontrivial portion of that horizon [23].

[A] Neither is a theorem that generic black holes have a smooth, unique, physically harmless interior continuation. What they establish for present purposes is that even inside real mathematics, asking whether “the singularity is removed” without naming the regularity class is an incomplete question. Finite or continuous metric components do not imply bounded curvature, a C^2 extension, or uniqueness of the continued Einstein evolution. A C^0 extension does not by itself provide finite curvature or a classical C^2 solution beyond the boundary, and replacing those components by surreal numbers settles none of these distinctions.

[A] Exponential growth factors together with inverse powers or logarithms of an advanced-time parameter are natural candidates for transseries descriptions, and surreal exponentiation and logarithms can retain separated scales that a truncated power series misses. But a schematic expression involving $e^{\kappa v} v^{-p}$ is *not* a universal theorem about all Kerr interiors; the model, tail assumptions, gauge and quantities being estimated must all be specified. Computing a large generalized scalar is not a substitute for a nonlinear stability argument.

Computation 11.4 ([E] Near-extremal Reissner–Nordström bookkeeping). For $f(r) = 1 - 2m/r + q^2/r^2$ the horizon polynomial is $r^2 - 2mr + q^2$. In geometrized units, with $\delta = \sqrt{m^2 - q^2}/m$ for $0 < |q| < m$,

$$r_{\pm} = m(1 \pm \delta), \quad \kappa_{+} = \frac{r_{+} - r_{-}}{2r_{+}^2} = \frac{\delta}{m(1 + \delta)^2}. \quad (11.5)$$

A symbolic infinitesimal δ records the splitting of a double root and the small surface-gravity scale cleanly, and several parameters can keep this limit separate from a wave-frequency limit or a small perturbation amplitude.

Status. Algebraic identities and asymptotic bookkeeping. **[A]** Explicitly *not* a claim that extremality or a Cauchy horizon is physically harmless.

Part III

The positive theorems, all of them conditional

12 Formal and coefficientwise Einstein reduction

Warning 12.1 (Read Theorem 2.2 again before this section). Everything in this section is tier **[C]**. The theorems below are statements about *formal coefficient equations under stated hypotheses*. They are not theorems about general relativity, they assert nothing about a metric whose leading term degenerates, and they must never be quoted as showing that a non-Archimedean deformation of a singular spacetime is well behaved. Section 13 exhibits the counterexamples that prove their hypotheses cannot be dropped.

12.1 Two conservative geometric frameworks

The three manuscripts use two genuinely different coefficient algebras, and both are kept. In each, the set of spacetime events is *not* replaced: U is an ordinary coordinate domain in a smooth four-manifold, or in $\mathbb{R}^4$, and fields are formal families over it. This is proposal (P2) of Section 3.1.

Framework F1: formal power series in one parameter. Put

$$\mathcal{A}(U) = C^\infty(U, \mathbb{R})[[h]], \quad (12.1)$$

with h a formal parameter killed by the coordinate derivatives. Products are formal Cauchy products, with only finitely many contributions at each integer order. Reduction

$$\pi_0 : \mathcal{A}(U) \rightarrow C^\infty(U), \quad \sum_{n \geq 0} f_n h^n \mapsto f_0,$$

is a ring homomorphism and satisfies $\pi_0 \partial_\mu = \partial_\mu \pi_0$ *by definition*. Formal series can later be evaluated at a positive infinitesimal in a suitable Hahn workspace, coefficient by coefficient. That evaluation does not make the map $x \mapsto f(x; h)$ smooth in the full fine topology; the specified derivative, not an unstated topological identification, carries the construction.

Framework F2: a Hahn coefficient-function algebra. Put

$$\mathcal{A}_\Gamma(U) = \left\{ A(x) = \sum_{\gamma \in S} A_\gamma(x) t^\gamma : A_\gamma \in C^\infty(U), S \subseteq \Gamma \text{ well ordered} \right\}, \quad (12.2)$$

where S is the *global* coefficient support. Products use Hahn convolution. This is a ring, not a field: smooth coefficients have zeros. Write $\mathcal{A}_\Gamma^{\geq 0}(U)$ and $\mathcal{A}_\Gamma^{> 0}(U)$ for the nonnegative- and positive-support parts; extraction of the exponent-zero coefficient gives

$$\text{st} : \mathcal{A}_\Gamma^{\geq 0}(U) \rightarrow C^\infty(U), \quad \text{st } \partial_\mu = \partial_\mu \text{st}.$$

Restriction to smaller ordinary domains is allowed and does not enlarge support. *The distinction between (12.2) and a collection of coefficient germs with shrinking domains is essential* (Section 5.3); so is the distinction between $\mathcal{A}(U)$, where each order receives finitely many contributions automatically, and $C^\infty(U)((t^\Gamma))$, which needs controlled supports and summability under every operation employed.

[A] The notation deliberately separates two questions: whether a formal equation is *well defined*, and whether its formal solution represents an actual family of real solutions with controlled remainders. The first is algebraic. The second is analytic and generally much harder.

12.2 The reduction theorem in Framework F1

Theorem 12.2 (**[C]** Reduction of the Einstein tensor over $C^\infty(U, \mathbb{R})[[h]]$). *Let $g(h) = g_0 + hg_1 + h^2g_2 + \dots$ be a symmetric 4×4 matrix with entries in $\mathcal{A}(U)$, and assume g_0 is an ordinary smooth nondegenerate Lorentzian metric on U . Define the connection and curvature by the usual coordinate formulas, differentiating coefficients and holding h fixed. Then the inverse metric exists in $M_4(\mathcal{A}(U))$ and reduction modulo h satisfies*

$$\pi_0(g^{-1}) = g_0^{-1}, \tag{12.3}$$

$$\pi_0(\text{Ric}[g(h)]) = \text{Ric}[g_0], \tag{12.4}$$

$$\pi_0(G[g(h)]) = G[g_0]. \tag{12.5}$$

If $\Lambda(h)$ and $T(h)$ have the same coefficientwise interpretation and $G[g(h)] + \Lambda(h)g(h) = 8\pi T(h)$ holds coefficientwise on U , then $G[g_0] + \Lambda_0g_0 = 8\pi T_0$. In particular, for real constants Λ, κ , if $G[g] + \Lambda g = \kappa T$ then $G[g_0] + \Lambda_0g_0 = \kappa T_0$.

Proof. Write $g = g_0(I + H)$ with $H \in hM_4(\mathcal{A}(U))$. Then

$$g^{-1} = \left(\sum_{n \geq 0} (-H)^n \right) g_0^{-1}, \tag{12.6}$$

and at any specified power of h only finitely many terms contribute; this is a formal inverse, using no topological convergence, and it proves (12.3). The map π_0 preserves finite sums and products and commutes with ∂_μ , since the latter differentiates only coefficient functions. Apply it to

$$\Gamma^a_{bc} = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}), \quad R^a_{bcd} = \partial_c \Gamma^a_{db} - \partial_d \Gamma^a_{cb} + \Gamma^a_{ce} \Gamma^e_{db} - \Gamma^a_{de} \Gamma^e_{cb}.$$

Reduction commutes with the contractions defining the Ricci curvature and the scalar curvature, hence with $G_{ab} = R_{ab} - \frac{1}{2} \text{Scal}(g)g_{ab}$. Applying reduction to the formal Einstein equation gives the last assertions. $\square$

Status. **[C]**. Requires a nondegenerate smooth leading metric on *one common* domain. The same argument works for common-domain Hahn coefficient families when their supports justify multiplication and the inverse expansion, but those hypotheses must be *stated*, never inferred from pointwise values. **[A]** Self-contained; not a claim that the repository has formalized Einstein geometry.

12.3 The reduction theorem in Framework F2, and its shadow corollary

This is a genuinely different theorem: its coefficient algebra is a Hahn ring rather than a power-series ring, its perturbation has strictly positive *global* support rather than positive integer order, and its inverse needs a support lemma rather than an order count. Both are printed.

Theorem 12.3 ([C] Standard-part reduction over a Hahn coefficient algebra). *Assume a formal metric deformation*

$$g = g_0 + h, \quad h \in M_4(\mathcal{A}_F^{\geq 0}(U)), \quad (12.7)$$

on one common smooth domain U , with g_0 an ordinary smooth Lorentzian metric with $\det g_0$ nowhere zero, h globally well supported of strictly positive order, and $T \in M_4(\mathcal{A}_F^{\geq 0}(U))$. Then

$$g^{-1} = \sum_{n \geq 0} (-g_0^{-1}h)^n g_0^{-1} \quad (12.8)$$

is a valid matrix Hahn sum, and

$$\text{st}(g^{-1}) = g_0^{-1}, \quad \text{st}(G(g)) = G(g_0). \quad (12.9)$$

If $G_{\mu\nu}(g) + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$ holds coefficientwise with $\Lambda \in \mathbb{R}$, its ordinary shadow satisfies

$$G_{\mu\nu}(g_0) + \Lambda(g_0)_{\mu\nu} = 8\pi \text{st}(T_{\mu\nu}). \quad (12.10)$$

The same coefficient-zero compatibility holds for the coordinate Riemann and Ricci tensors and their finite scalar contractions.

Proof. Positive global support and the Neumann support lemma guarantee both a well-ordered resulting support and finitely many coefficient contributions at each exponent, so (12.8) is a legitimate element of $M_4(\mathcal{A}_F(U))$; only the $n = 0$ term has exponent zero, giving the first identity of (12.9). Standard part commutes with addition, with products, with real constants, and with the coefficient derivatives; applying it to the Christoffel formula yields the Christoffel formula for g_0 , and applying it to the finite curvature and contraction formulas gives the second identity. Reducing the formal Einstein equation coefficientwise gives (12.10). $\square$

Status. [C]. At each ordinary point the positive-order perturbation preserves the signature over the ordered field — put g_0 in a real Lorentz-diagonal frame and use finite elimination; the pivots remain infinitesimal perturbations of nonzero real pivots and keep their signs. *That statement is pointwise only*; no compactness and no uniform positive lower bound on U is inferred.

Corollary 12.4 ([C] Positive-order corrections cannot repair a singular shadow). *A metric satisfying the hypotheses of Theorem 12.3 cannot have a smooth, nondegenerate ordinary shadow through a point of zero areal radius while that shadow agrees with the Schwarzschild geometry on a punctured neighbourhood with $m \neq 0$.*

Proof. A smooth nondegenerate ordinary metric has finite continuous curvature scalars locally. On the punctured Schwarzschild region the scalar is $48m^2/r^6$ by Theorem 7.3, unbounded as the areal radius tends to zero. That contradicts such an extension. $\square$

Status. A *conditional* obstruction, not a universal theorem against non-Archimedean physics. [A] It leaves open, explicitly, five escape routes: a degenerate or nonexistent standard-part metric; negative-order fields; singular boundary layers; new field equations; and nonclassical observables or a different notion of spacetime. Any proposed evasion should identify precisely which hypothesis is being replaced, and how the missing physics is supplied.

12.4 Connection, Bianchi identities and gauge

[C] In either framework, with commuting coefficient derivatives, define

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\beta}(\partial_{\mu}g_{\beta\nu} + \partial_{\nu}g_{\beta\mu} - \partial_{\beta}g_{\mu\nu}), \quad (12.11)$$

and form the curvature, Ricci curvature and Einstein tensor by their usual finite contractions. The algebraic derivations of torsion-freeness, metric compatibility and the Bianchi identities — including the contracted Bianchi identity — use only these operations and their identities, so they hold coefficientwise. At first order one obtains the linearized Einstein operator together with the corresponding matter perturbation.

[A] Ordinary coordinate transformations of U act on the coefficient tensors in the usual way, and formal diffeomorphisms near the identity can be defined order by order using Lie derivatives. Neither operation changes the invariant content merely because the coefficients lie in a larger algebra. Conservation must either follow from the chosen matter equations or be imposed consistently with the Bianchi identity; arbitrarily choosing coefficients of T does not guarantee an admissible matter model. And this is a route to formalization, not an assertion that any of it is already present in the repository.

[A] A practical implementation should treat support conditions as part of every operation's input contract: inversion requires an invertible leading term, composition requires the relevant positive-order or domain hypotheses, and integration requires both a coefficientwise integral and an admissible combined support. It is unsafe to derive an identity in a formal ring, interpret it as an ordinary convergent sum, and then apply ordinary interchange-of-limit theorems without proving a bridge.

12.5 What the reduction theorems say about new physics

[A] A *regular* infinitesimal deformation of a nondegenerate classical metric obeys the classical Einstein equation at leading order. So a model cannot both keep this regular common-domain reduction and obtain an arbitrary different macroscopic leading equation without changing another assumption — the source, the action, or the scaling regime. Read positively, that is a precise consistency theorem: ordinary general relativity is recovered at leading order in a regular formal deformation.

[A] Read negatively — and this is the direction that matters here — the same theorem limits what the construction can claim. If the leading metric has a classical singular boundary, the theorem does not extend it through that boundary. Its hypotheses require a common smooth domain and an invertible leading metric. The theorems do *not* exclude singular perturbations: a leading metric may degenerate at a boundary, inverse coefficients may acquire negative valuation, resummed terms may become dominant, a new leading geometry may appear, or derivatives may promote a seemingly small correction to leading order, so that a rescaled inner problem differs substantially from the outer one. Those are exactly the situations a controlled asymptotic language is for. They are singular perturbation problems, not counterexamples to the reduction theorems, whose hypotheses have simply ceased to apply.

13 Why the hypotheses cannot be dropped

13.1 A boundary layer with uniformly small values and order-one curvature

Example 13.1 ([E] The family b_δ). For a positive ordinary real parameter δ put

$$b_\delta(x) = \frac{\delta^2}{1 + (x/\delta)^2} = \frac{\delta^4}{\delta^2 + x^2}. \quad (13.1)$$

For every real x one has $0 \leq b_\delta(x) \leq \delta^2$, so the family converges uniformly to zero, and its first derivatives are uniformly of order δ . Nevertheless

$$b''_\delta(0) = -2 \quad (13.2)$$

for every δ , while the limit function has second derivative zero. Both (13.2) and the derivative scaling are symbolically checked.

[A] Uniform smallness of a metric perturbation therefore does not by itself imply small curvature: second-derivative control is the missing ingredient, and this family has first-derivative control without it. Evaluating the same rational formula at an infinitesimal δ , every pointwise value has zero standard part, but a coefficientwise or appropriately extended second derivative at zero still yields -2 . There is no common formal expansion in nonnegative powers of δ with smooth coefficients valid across $x = 0$: away from zero, expansion in δ^2/x^2 produces coefficients singular at zero. So the example does not contradict Theorem 12.2; it demonstrates exactly why its hypotheses cannot be omitted, and why a *pointwise* standard part is an insufficient basis for differentiating a limiting metric.

13.2 A purely real counterexample to interchanging reduction and differentiation

Example 13.2 ([E] The family $\delta \sin(x/\delta)$). The real functions $f_\delta(x) = \delta \sin(x/\delta)$, $\delta > 0$, converge uniformly to zero as $\delta \downarrow 0$, yet $f'_\delta(0) = 1$ for every δ , while the derivative of the limit function is zero.

[A] This is a real-family counterexample: no value for $\sin \omega$ is needed to state it, and no non-Archimedean object appears in it. Its role is to identify precisely what Theorem 12.2 and Theorem 12.3 are buying. In those theorems reduction commutes with differentiation *by definition of the regular formal algebra*. That commutation does not hold for arbitrary families followed by a limiting or standard-part operation. Highly oscillatory and boundary-layer families require additional estimates before reduction and differentiation may be interchanged.

13.3 Position-dependent exponents break the coefficient algebra

[C] A substantial issue arises when an exponent varies with position: differentiating $t^{p(x)}$ introduces $p'(x) \log t$, so the constant-exponent coefficient algebra (12.2) is *not* closed under such expressions.

[A] Three responses are available, and one of them must be chosen: restrict the ansatz; enlarge the differential transseries class with a proof of closure; or use separate local expansions with a justified matching rule. This is exactly the kind of concrete obstruction a research prototype should expose rather than paper over.

13.4 Formal certificates versus analytic certificates

[A] A coefficient solution of a formal Einstein equation is not automatically an actual family of real solutions with that asymptotic expansion. For that one needs a realization or summation theorem, error bounds, a gauge choice, constraint propagation, and a well-posed problem in an appropriate function space. A recursion with nonzero denominators at every formal order need not have controlled real coefficients or a convergent expansion.

Two different outputs must therefore be distinguished by name.

A formal certificate

proves support admissibility and vanishing of the residual through a valuation threshold.

An analytic certificate

additionally proves approximation or summability and quantifies a real error.

[A] The first is valuable without the second, but it must not be advertised as an actual spacetime solution. A remainder of valuation greater than N bounds scale in the formal ordering only; a numerical bound at a finite real parameter needs control of the coefficients and a realization map. Conversely two real solutions differing by $e^{-A/\varepsilon}$ can have identical power-series data while remaining physically distinct in a process that amplifies that sector. A representation is most useful when it prevents such information from being discarded prematurely, and when it *rejects* a product or substitution whose support condition fails rather than assigning a symbolic infinity and proceeding.

13.5 Actions, conservation laws, and measures, with an honest scope

[C] On a fixed ordinary compact integration domain, a common-support family of smooth densities integrates coefficientwise by (5.1), and the support of the result is contained in the original well-ordered support. Ordinary integration by parts applies separately to every coefficient under its usual boundary assumptions. For the Einstein–Hilbert density, the inverse metric exists formally by (12.6) or (12.8), and since $\det g$ has a nonzero ordinary leading term, the branch of $\sqrt{-\det g}$ around a nondegenerate Lorentzian g_0 can be expanded by an admissible positive-order binomial series. Consequently a formal local action principle, Euler–Lagrange equations and coefficientwise conservation identities can be developed, and their local Euler–Lagrange expressions recover the formal equations of Section 12.

[A] That is a conservative extension of classical variational algebra, and nothing more. It does not define a nonperturbative path-integral measure, select boundary conditions at a singularity, or prove convergence of the expansion, and it does not pretend that a fine-topological Riemann integral over the whole surreal class has been constructed. A quantum functional integral is a different task: the field space, the measure or its replacement, the regularization, the gauge treatment and the limiting procedure all need definitions, and these are already substantial over $\mathbb{R}$ and $\mathbb{C}$. A proper-class sum over all surreal-valued histories is *not* licensed by Hahn summability. Formal sums, finite-dimensional integrals and rigorously constructed functional measures must remain explicitly different objects.

14 A regular-core comparison: where new physics actually enters

14.1 A real modification already removes the central pole

To separate a change of dynamics from a change of scalars, all three manuscripts use the static metric function of Hayward’s regular-core model [28]:

$$f_\ell(r) = 1 - \frac{2mr^2}{r^3 + 2m\ell^2}, \quad m > 0, \ell > 0, \quad (14.1)$$

inserted into the static spherical form (7.1).

Computation 14.1 ([E]) The central limits, for fixed real ℓ). At large radius the correction to Schwarzschild in f begins at order r^{-4} . At small radius

$$f_\ell(r) = 1 - \frac{r^2}{\ell^2} + \frac{r^5}{2m\ell^4} + O(r^8), \quad (14.2)$$

a de Sitter-like core, and substituting into (7.3) gives

$$\mathcal{K}(0) = \frac{24}{\ell^4}, \quad \text{Scal}(0) = \frac{12}{\ell^2}, \quad (14.3)$$

with the three contributions of (7.3) tending to $4/\ell^4$, $16/\ell^4$ and $4/\ell^4$ respectively. The effective mass function and, via $G^t_t = -2M'_\ell(r)/r^2$ and $G_{ab} = 8\pi T_{ab}$ with $T^t_t = -\rho$, the effective energy density are

$$M_\ell(r) = \frac{mr^3}{r^3 + 2m\ell^2}, \quad \rho(r) = \frac{M'_\ell(r)}{4\pi r^2} = \frac{3m^2\ell^2}{2\pi(r^3 + 2m\ell^2)^2}, \quad \rho(0) = \frac{3}{8\pi\ell^2}, \quad (14.4)$$

with $p_r = -\rho$, so at the centre the stress is of cosmological-constant form. The crossover radius, from balancing the denominator terms, is

$$r_c = (2m\ell^2)^{1/3}, \quad (14.5)$$

and for fixed $r > 0$ letting the real $\ell \rightarrow 0$ recovers Schwarzschild pointwise. All of (14.2)–(14.4), the expansion through r^5 , the leading outer correction, the mass-function identity, the Einstein source identity and the $\ell \rightarrow 0$ limit are symbolically checked.

Warning 14.2 ([A]) What does the regularizing, and what “regular” means here). The local de Sitter-type core and the finite limiting curvature are consequences of the *new metric function* and its effective source, not of any non-Archimedean arithmetic. Unlike Schwarzschild, the model is not vacuum in the core. Four qualifications are part of the statement.

- (a) “Regular core” here means finite curvature plus the corresponding classical C^2 -level metric regularity — *not* analytic or all-orders smoothness. In fact the r^5 term in (14.2) becomes $|x|^5$ along a Cartesian line through the centre, so this ansatz must not be advertised as a C^∞ Cartesian metric there. The vanishing spherical-coordinate determinant at the centre must be assessed in Cartesian coordinates; the expansion yields a nondegenerate C^2 local metric, and no claim about higher derivatives follows from the curvature limit alone.
- (b) Writing $T_{ab} = G_{ab}/8\pi$ is a consistent *effective* description. It supplies no microphysical matter Lagrangian and establishes no stability.

- (c) A finite-curvature core is not a proof of global geodesic completeness, unique evolution, or physically stable evaporation. The regularity class, inner-horizon behaviour, global causal assumptions and stress-energy interpretation must all be examined. [I] Classifications of geodesically complete black-hole scenarios make exactly this larger geometric issue explicit [29].
- (d) Keeping a finite real ℓ , possibly motivated by a physical ultraviolet scale, can give finite central invariants — but choosing the scale and the modified source remains a physical hypothesis. Surreal arithmetic does not determine that ℓ is the Planck length, or that a particular regular-core ansatz is the correct quantum-gravitational dynamics.

14.2 Making the core radius infinitesimal reverses the conclusion

Computation 14.3 ([E] Regulator nonuniformity). Set $\ell = m\varepsilon$ with ε a positive infinitesimal. For every fixed positive ordinary regulator, (14.3) is finite. As a non-Archimedean value, however,

$$m^4\mathcal{K}(0) = 24\varepsilon^{-4}, \quad (14.6)$$

which is unlimited; equivalently, with $\ell = L\delta$ and δ infinitesimal, $L^4\mathcal{K}(0) = 24\delta^{-4}$. The denominator at the centre is algebraically nonzero, and the central curvature is bounded by no real number.

Warning 14.4 ([A] Two statements that are not the same). *Regular for each fixed real regulator* and *uniformly finite in the regulator's infinitesimal limit* are different assertions, and the first does not imply the second. Equally, *a field-valued smooth expression exists* and *the curvature has a finite ordinary observable shadow* are different regularity criteria. Taking the regulator infinitesimal makes the nonuniformity explicit; it does not supply a uniformly finite core, and the scalar field has not supplied a finite observational curvature bound.

[A] The two real-parameter limits illustrate the same nonuniformity from the other side. At every fixed $r > 0$, letting real $\ell \rightarrow 0$ recovers Schwarzschild; near $r^3 \sim 2m\ell^2$ that outer approximation ceases to be uniform and the core must be resolved. A Hahn representation is well suited to organizing the outer and inner scales. It is the altered metric or action, not surreal arithmetic, that changes the central behaviour.

14.3 Hahn summability detects the correct boundary layer

This is the sharpest genuinely positive application in the report, and it is unique to one member. Write

$$x = \frac{r}{m}, \quad \varepsilon = \frac{\ell}{m}, \quad f(x, \varepsilon) = 1 - \frac{2x^2}{x^3 + 2\varepsilon^2}, \quad (14.7)$$

treat ε as a Hahn monomial of positive valuation, and take $x = \varepsilon^\alpha$ with $\alpha > 0$ real; the requisite real exponents lie in a divisible set-sized value group.

Proposition 14.5 ([C] Outer and inner strong-summability criteria). *For (14.7), the outer geometric expansion*

$$f = 1 - \frac{2}{x} \sum_{k \geq 0} \left(-\frac{2\varepsilon^2}{x^3} \right)^k \quad (14.8)$$

is strongly Hahn summable exactly when $\alpha < 2/3$, whereas the inner expansion

$$f = 1 - \frac{x^2}{\varepsilon^2} \sum_{k \geq 0} \left(-\frac{x^3}{2\varepsilon^2} \right)^k \quad (14.9)$$

is strongly Hahn summable exactly when $\alpha > 2/3$. At $\alpha = 2/3$ neither displayed geometric family is strongly summable. The exact transition-scale expression, with $x = \varepsilon^{2/3}X$, is

$$f = 1 - 2\varepsilon^{-2/3} \frac{X^2}{X^3 + 2}, \quad (14.10)$$

whose denominator is a unit, and which is therefore well defined, for positive limited X with positive real standard part.

Proof. The ratio in (14.8) has valuation $(2 - 3\alpha) v(\varepsilon)$, positive exactly for $\alpha < 2/3$; its powers then have increasing well-ordered support with finite multiplicity at each exponent, so the geometric identity applies. For $\alpha > 2/3$ the exponents strictly decrease and the family is not summable. The ratio in (14.9) has valuation $(3\alpha - 2) v(\varepsilon)$, giving the complementary criterion. At $\alpha = 2/3$ all geometric terms contribute at the same exponent, so infinitely many nonzero contributions land on that exponent and strong summability fails. Equation (14.10) follows by direct substitution in the original rational function, without summing either invalid family. $\square$

Status. [C]. A criterion on Hahn-field data — valuations of the expansion ratio — and *not* on real convergence. Its outer, inner and transition-scale rational rewrites are symbolically checked.

Computation 14.6 ([E] The exact finite remainder underlying the criterion). For every finite N ,

$$\frac{1}{1+A} = \sum_{k=0}^{N-1} (-A)^k + \frac{(-A)^N}{1+A}, \quad (14.11)$$

so if $v A > 0$ the remainder has valuation $N v A$. This identity was checked exactly for $N = 1, \dots, 8$. It provides a certificate of asymptotic order and is a natural target for existing Neumann-series and rational-jet infrastructure.

[A] The crossover is at $r^3 \sim 2m\ell^2$, matching (14.5); a formally small correction in the outer region becomes leading in the inner region, and the same sector test can be phrased in exponents: with $r/L = \varepsilon^a$, $\ell/L = \varepsilon^b$ and m/L ordinary and nonzero, comparing $3a$ with $2b$ identifies the outer and inner sectors, with both terms retained at equality. The gain is real and it is precisely delimited: *support conditions detect when an approximation has been used outside its asymptotic regime*. At the transition the failure is a failure of that expansion, not a singularity of the rational model, and no new physical solution has been invented by the series algebra.

Part IV

Quantum structures: what extends, and what does not

15 Finite-dimensional surcomplex quantum algebra

15.1 The algebraic part works well

Let $F \supseteq \mathbb{R}$ be a real-closed ordered field and $K = F[i]$. Define

$$\overline{a + ib} = a - ib, \quad |a + ib| = \sqrt{a^2 + b^2}, \quad \langle \psi, \phi \rangle = \sum_{j=1}^n \overline{\psi_j} \phi_j, \quad \|\psi\|^2 = \sum_{j=1}^n |\psi_j|^2. \quad (15.1)$$

[C] Then $\langle v, v \rangle > 0$ for every $v \neq 0$, since it is a finite sum of squares of real and imaginary parts, so (15.1) is a positive-definite Hermitian form on K^n . Positivity and Cauchy–Schwarz hold by ordinary finite algebra, unitary matrices preserve the norm, and a normalized vector has nonnegative candidate Born weights $p_j = |\psi_j|^2$ summing to one in F . Orthogonality, tensor products of finitely many systems and projective-state algebra all make sense. The same finite algebra supports Lorentzian quadratic forms, their linear isometry groups, Clifford algebras and finite spinor constructions over appropriate extensions.

[A] What has been preserved is an *algebraic* structure. Compactness of a group, spectral measures, the topology of a state space and continuum dynamics are separate issues.

Theorem 15.1 ([C] Finite Hermitian spectral theorem). *Let F be a real-closed ordered field containing $\mathbb{R}$, let $K = F[i]$, and let n be an ordinary finite integer. Every Hermitian $A = A^\dagger \in M_n(K)$ has an orthonormal eigenbasis, and all its eigenvalues belong to F .*

Proof. The characteristic polynomial splits over the algebraically closed K , so there is an eigenvector $v \neq 0$ with eigenvalue λ . Since $v^\dagger v > 0$,

$$\lambda v^\dagger v = v^\dagger A v = \overline{v^\dagger A v} = \bar{\lambda} v^\dagger v,$$

whence $\lambda = \bar{\lambda} \in F$. Normalize v using the positive square root of $v^\dagger v$, which exists because F is real closed. The orthogonal complement of v is invariant under A , because $\langle v, A w \rangle = \langle A v, w \rangle = \lambda \langle v, w \rangle$, and the restriction is Hermitian. Induction on the ordinary finite n completes the proof. $\square$

Status. All three manuscripts prove this, by the same argument; it is printed once. The proof requires *no* completeness of F . [A] It is a finite algebraic theorem and not an infinite-dimensional spectral theorem for unbounded operators. Finite quantum kinematics, finite spin matrices and finite unitary algebra therefore do not force an Archimedean scalar field — but that is a scalar-extension result, not a new quantum prediction. An ordered extension also permits infinitesimal probabilities, whose physical interpretation is an additional rule; and unlimited eigenvalues can be represented, but whether they represent physical energies requires an interpretation and a dynamics.

15.2 Finite observations have an ordinary shadow

The three manuscripts state three overlapping versions. Their union is the following, with the sharp conditioning boundary that only one of them makes explicit.

Theorem 15.2 ([C] Reduction of finite quantum protocols). *Let all dimensions and step counts be ordinary finite integers.*

- (a) *If $\psi \in K^n$ satisfies $\psi^\dagger \psi = 1$ then every component is limited, and $\psi_0 = \text{st } \psi \in \mathbb{C}^n$ satisfies $\psi_0^\dagger \psi_0 = 1$; moreover each $p_a = |\psi_a|^2$ is limited with*

$$\sum_a |\psi_{0,a}|^2 = 1, \quad \text{st}(p_a) = |\psi_{0,a}|^2. \quad (15.2)$$

- (b) *Every unitary $U \in M_n(K)$ has limited entries and $U_0 = \text{st } U$ is an ordinary complex unitary; every orthogonal projection $P = P^\dagger = P^2$ has limited entries and $P_0 = \text{st } P$ is an ordinary orthogonal projection.*

- (c) $\text{st}(\psi^\dagger P \psi) = \psi_0^\dagger P_0 \psi_0$ and $\text{st}(U \psi) = U_0 \psi_0$; finite observables with limited entries have their expectation values $\psi^\dagger A \psi$ reduced correctly. Consequently the real standard parts of joint probabilities in a fixed finite sequence of unitary evolutions and projective measurements agree with the corresponding ordinary complex calculation.
- (d) Conditioning on an outcome reduces correctly provided the outcome probability has nonzero standard part.

Proof. Each $|\psi_j|^2$ is nonnegative and at most the total sum 1, so each component is limited; apply Theorem 4.4 to the norm identity. The same argument applied to the columns of U bounds every entry by 1 and gives $U_0^\dagger U_0 = I$. For a projection and a coordinate vector e_j , $\|P e_j\|^2 = P_{jj}$ and $\|(I - P)e_j\|^2 = 1 - P_{jj}$ are both nonnegative, so $0 \leq P_{jj} \leq 1$ and every entry of the j th column has modulus at most 1. Reduction preserves the self-adjointness and idempotence identities, and it commutes with conjugation, with finite sums and with finite products, hence with the expressions in (15.2) and with the finite products defining a protocol; joint probabilities may be computed as squared norms of products of the relevant projectors and unitaries applied to ψ , with no normalization by a possibly infinitesimal intermediate probability. For (d), conditioning divides by a probability and takes a positive square root for normalization, and both commute with reduction when the denominator's standard part is positive. $\square$

Status. [C], and conditional on adopting real standard parts as measured probabilities (Section 19.2). [A] It is *not* a universal theorem about every non-Archimedean probability interpretation, and not a theorem that non-Archimedean models are experimentally indistinguishable. Two exclusions are part of the statement: an outcome of infinitesimal probability has zero ordinary shadow, so reduction cannot determine the state obtained by dividing by that infinitesimal; and infinite or hyperfinite experiments lie outside the theorem entirely.

Example 15.3 ([E] A rare outcome stays rare for finitely many trials). For the two-level state

$$\psi = \frac{(1, \varepsilon)}{\sqrt{1 + \varepsilon^2}}, \quad p_2 = \frac{\varepsilon^2}{1 + \varepsilon^2}, \quad \text{st}(p_2) = 0,$$

any fixed ordinary finite number of independent repetitions preserves that zero standard-part probability for observing the rare outcome. The normalization and the rare-outcome limit are symbolically checked.

Status. [A] Introducing an infinite repetition count, or an amplifier with infinite resources, changes the operational assumptions; it is not evidence that the original finite experiment had a different real shadow. Postselection on an infinitesimal-probability event lies outside Theorem 15.2.

15.3 Nonuniform amplification: the interchange genuinely fails

Example 15.4 ([E] $H = \varepsilon \sigma_z$ with $T = \varepsilon^{-1}$). In units with $\hbar = 1$, take $H = \varepsilon \sigma_z$ and evolution time $T = \varepsilon^{-1}$. Then $HT = \sigma_z$, so the finite-phase evolution is nontrivial although $\text{st} H = 0$. Reducing H before an unbounded duration loses the information.

Status. This is why Theorem 15.2 does not imply equivalence under arbitrary nonuniform limits: reduction and unbounded duration do not commute. [A] It demonstrates a *mathematical* failure of interchange. It expressly does not establish a feasible way to observe a literal infinitesimal probability or energy shift, and

a proposed observable effect must state how such a duration, coupling, postselection or amplification is physically realized. More generally $x \mapsto x/\varepsilon$ promotes an infinitesimal deviation to an order-one result; the operation is defined algebraically, but an experimental proposal must identify the physical gain, its energy or stability cost, and what fixes ε — without silently inserting an infinite resource.

15.4 Probability requires a summation theory, not just positive scalars

[C] For infinitely many outcomes, “the probabilities sum to one” needs a definition. A countable family assigning the same positive infinitesimal ε to each outcome is *not* strongly Hahn summable, since infinitely many summands contribute to the same exponent (Theorem 4.3). Ordinary finite additivity therefore cannot be promoted to a countably additive probability theory by declaring each term infinitesimal.

[I] Non-Archimedean probability theories exist and introduce explicit structures for infinite addition and conditioning [38]. **[A]** They are examples of additional mathematics a physical proposal would have to choose, not consequences of F being an ordered field. An exact infinitesimal probability is nonzero in the ordered field and has zero standard part; for a fixed ordinary finite number of trials the probability of seeing such an event remains infinitesimal in a finite independent-trial model. A new probability interpretation is possible, but it must specify how records relate to the field-valued weights. **[I]** Ordinary measure-theoretic or nonstandard constructions — Loeb’s among them — can be incorporated only after their assumptions and interpretation are supplied [39].

[A] Two precise notions of countable addition for Hahn-valued weights are worked out in the companion report held in `docs/surreal/hahn-valued-measures-and-probability/`. Under strong Hahn countable additivity each Hahn coefficient is carried by finitely many points (`meas:thm:atomic`); a coefficientwise notion that admits diffuse laws is in its Part V. Neither is a physical interpretation: that report does not say how records relate to field-valued weights.

15.5 What this does and does not say about dynamics

[A] Theorem 15.2 associates an ordinary unitary matrix to every chosen finite surcomplex unitary matrix. It does not prove that the resulting family $U_0(t)$ satisfies a specified Schrödinger equation, is strongly continuous, or has a particular Hamiltonian: those are differential and topological assertions. An observable can have unlimited eigenvalues even though all normalized-state components are limited, and a rule for reading such an observable cannot be obtained by taking the standard part of an unlimited number.

[A] The expression $U(t) = \exp(-iHt/\hbar)$ is likewise not defined by the existence of $K = F[i]$. A real exponential on F defines growth and decay; it does not choose values for oscillatory phases with unlimited arguments. A general complex exponential requires a compatible function theory, periodic structure, and branch or extension choices; parametrizing the unit circle algebraically or using finite angles does not by itself define a canonical value of $\sin \omega$. Three tractable subclasses exist: keep the unperturbed real-time propagator over $\mathbb{C}$ and solve a formal perturbation problem coefficientwise; exponentiate a positive-order formal matrix, for which each coefficient receives finitely many relevant terms; or work with an explicitly constructed analytic extension on a restricted domain. What is not legitimate is to infer a global unitary evolution for an unlimited Hamiltonian from Theorem 15.1 alone.

[A] The immediate lesson is favourable but modest: finite quantum algebra is not obstructed by real-closed non-Archimedean coefficients. To obtain new physics one must explain which dynamics or observational rule cannot already be represented by the ordinary shadow.

15.6 Infinite-dimensional quantum theory is not supplied by algebraic closure

[A] Ordinary Hilbert-space completeness, unbounded self-adjoint operators, spectral measures, strong continuity of time evolution and countably additive probability require more structure than the field operations. A space K^d with an F -valued norm is not automatically a Hilbert space in the usual real-normed sense, and a surreal-valued inner product is not an ordinary Hilbert norm. Increasing d to a surreal number is not a definition of a vector space of that dimension, let alone a measure-theoretic construction. The full fine topology is especially unsuitable for ordinary countable approximation processes, by Theorem 5.1. Finite Fourier identities and finite Hermitian spectral algebra do not establish infinite Fourier completeness. A generalized quantum theory must explicitly choose its state spaces, topology, operator domains and observables; a hyperfinite nonstandard model, a coefficientwise deformation of an ordinary Hilbert space, and a module over a Hahn field are three different options, and none inherits the useful features of the others merely because its scalars admit infinitesimals.

[I] There is also a distinct notion of *quantum* singularity. Horowitz and Marolf study scalar quantum probes of static singular spacetimes and relate unique quantum evolution to essential self-adjointness of the relevant spatial operator [33]. **[A]** Their setting establishes no universal resolution of the spacelike Schwarzschild singularity. It does illustrate what a meaningful question looks like: does the dynamics require extra boundary conditions? Changing scalars alone selects no operator domain and removes no ambiguity in a self-adjoint extension. For semiclassical gravity the demand is greater still — one must specify the quantum state, the renormalized stress tensor and the backreaction prescription, none of which is selected by the availability of unlimited surreal energies or surcomplex amplitudes.

16 Global phases exist, but are not automatically time evolution

16.1 Finite phases, and a genuine global extension

[A] It would be wrong to dismiss surcomplex quantum theory by saying that no global surcomplex exponential exists. One does.

[C] For a limited surreal angle $\theta = a + \delta$ with $a \in \mathbb{R}$ and δ infinitesimal, the ordinary real phase together with an infinitesimal Taylor expansion gives

$$\text{cis}(\theta) = e^{ia} \sum_{n \geq 0} \frac{(i\delta)^n}{n!} = e^{ia} E(i\delta), \quad (16.1)$$

where the ordinary factor e^{ia} is supplied by complex analysis and the infinitesimal tail is a strong Hahn sum. Every point of the surcomplex unit circle has a *limited* angle — for instance by the rational half-angle chart together with the ordinary choice of branch for its finite part. **[A]** This finite-angle fact assigns no physical temporal frequency to an arbitrary unlimited angle.

[I] Ehrlich and Kaplan construct a global exponential using an integer part of the ordered field; for initial surreal structures there is a canonical choice, the omnific integers $\mathbf{Oz}$, and the repository's trigonometry report includes that construction [14, 4]. Writing a general surreal as $b = b_{\text{PI}} + b_{\text{fin}}$, separating its purely infinite normal-form terms from its finite terms, the global normalization takes the form

$$\text{Sin}_{\text{EK}} b = \sin(b_{\text{fin}}), \quad \text{Cos}_{\text{EK}} b = \cos(b_{\text{fin}}), \quad (16.2)$$

because $b_{\text{PI}}/(2\pi)$ belongs to $\mathbf{Oz}$, so the purely infinite part is a period in the integer-part construction. The corresponding surcomplex exponential is

$$\text{Exp}_{\text{EK}}(a + ib) = \exp(a)(\text{Cos}_{\text{EK}} b + i \text{Sin}_{\text{EK}} b), \quad (16.3)$$

with kernel $2\pi i \mathbf{Oz}$. **[C]** It extends the ordinary complex exponential and *has* the group-homomorphism property.

Remark 16.1 (The construction uses more than the bare field). The normalization (16.2) uses the chosen integer part and, for canonicity, the surreal initial-subtree structure — strictly more than the ordered field plus local Taylor axioms. This report adopts the usual $\cos b + i \sin b$ convention in (16.3), not the reversed pair appearing in one displayed preprint formula; the convention is fixed by the value 1 at $b = 0$ together with the addition law. The published erratum to [14] concerns in-text bibliography reference numbering and not the withdrawal of the construction [15].

16.2 An explicit derivative incompatibility

The global group law is not enough to ensure compatibility with a selected scalar derivation.

Proposition 16.2 (**[C]** Canonical phase versus scalar differentiation). *Let D be a derivation on $\mathbf{No}$ that kills $\mathbb{R}$ and satisfies $D\omega = 1$ — in particular the normalized Berarducci–Mantova derivation. Then the functions in (16.2) do not satisfy the global scalar chain rule $D(\text{Sin}_{\text{EK}} x) = \text{Cos}_{\text{EK}}(x) D(x)$ for every $x \in \mathbf{No}$.*

Proof. Since ω is purely infinite, $\text{Sin}_{\text{EK}} \omega = 0$ and $\text{Cos}_{\text{EK}} \omega = 1$. Therefore $D(\text{Sin}_{\text{EK}} \omega) = D(0) = 0$ while $\text{Cos}_{\text{EK}}(\omega)D\omega = 1$. The asserted global chain rule fails already at $x = \omega$. $\square$

Status. **[C]**, and it is an *incompatibility of two structures*, not a contradiction in either established construction. A derivation of numbers and a chosen global periodic extension are different objects, and neither construction claims that they combine with all familiar identities without qualification.

Corollary 16.3 (**[C]** The canonical global phase is constant in ordinary time). *For every ordinary real t , the product ωt is purely infinite or zero, so*

$$\text{Exp}_{\text{EK}}(i\omega t) = 1 \quad (t \in \mathbb{R}). \quad (16.4)$$

As a function of an ordinary real time parameter the right-hand side is constant. It is therefore not a coefficientwise real-time solution of $\psi' = i\omega\psi$ with $\psi(0) = 1$.

Status. **[A]** A derivative defined using arbitrarily small *surreal* increments is a different limit operation and does not repair this real-time substitution. The lesson is to choose a dynamical function calculus first, rather than infer one from the existence of a phase homomorphism. Equation (16.4) should be included as an explicit test in any proposed phase interface: a construction that silently turns $\text{Exp}_{\text{EK}}(i\omega t)$ into a real-time infinite-frequency oscillator is mixing incompatible conventions.

Proposition 16.4 (**[C]** Infinite periods of a global phase homomorphism). *Suppose $E : (\mathbf{No}, +) \rightarrow \{u \in \mathbf{No}[i] : u\bar{u} = 1\}$ is a group homomorphism extending the finite-angle rule (16.1). Then E has an infinite period.*

Proof. Every unit-modulus u is limited, so its complex standard part is e^{ia} for some ordinary real angle a , and $ue^{-ia} = 1 + \eta$ with η infinitesimal. The formal logarithm of $1 + \eta$ is defined, and $(1 + \eta)\overline{(1 + \eta)} = 1$ shows that logarithm is purely imaginary; hence $u = \text{cis}(a + \delta)$ for a limited angle. Given any infinite X , choose such a limited θ representing $E(X)$; then $E(X - \theta) = 1$ and $X - \theta$ is infinite. $\square$

Remark 16.5 (Theorem 16.4 and Exp_{EK} agree). These two results are often mistaken for a conflict, so the merge records the resolution explicitly. Theorem 16.4 says any global phase homomorphism extending the finite-angle rule *must* have an infinite period; Exp_{EK} has kernel $2\pi i \mathbf{Oz}$, which contains infinite omnific integers. The construction is therefore an *instance* of the proposition, not a counterexample to it. What fails is only the ordinary assertion that all periods are ordinary integer multiples of 2π : finite-angle agreement plus the homomorphism law do not reproduce it. Neither result invalidates a chosen global normalization, and whether such a construction represents a physical high-frequency limit requires a link to the phase *function*, its state and its derivative, not merely to its scalar value.

16.3 Why the naive Taylor series also fails

Remark 16.6 ([C] Non-summability of an infinite-angle Taylor family). For an infinite monomial A with $v(A) < 0$, the terms of $\sum_{n \geq 0} (iA)^n / n!$ have exponents $n v(A)$ descending without a least element, so the family is not strongly Hahn summable. In particular the formal series for $e^{i/\varepsilon}$ has exponents unbounded below and is an inadmissible Hahn Taylor substitution. Even for an ordinary nonzero real angle the direct Taylor family fails at exponent 0, all terms landing there; its ordinary complex sum must be taken in $\mathbb{C}$ first, as in (16.1).

[A] An appropriate oscillatory differential algebra, a transferred internal function space, or a sectorial analytic construction might handle an infinite-frequency limit. Such a theory may use surcomplex coefficients, but it must specify which phases are functions, what derivatives act on them, and how measured finite-time motion is recovered. A global extension of sine alone is not that theory.

16.4 The phase obstruction belongs to a companion report

The sharp differential-algebraic statement about oscillation over $\mathbf{No}[i]$ is not reproved here. It is proved, in three independent ways, in the companion merged report *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers*, held in `docs/surcomplex/differential-equations/` of this repository, whose oscillator corollary establishes that for $\lambda \in \mathbb{C}$ the equation $\partial y = \lambda y$ has a nonzero solution in $\mathbf{No}[i]$ if and only if $\lambda \in \mathbb{R}$ — so $\partial y = iy$ and $\partial^2 y + y = 0$ have only the zero solution. Theorem 16.2 and Theorem 16.3 are the *physical* reading of that obstruction, and they land squarely on that report’s phase conventions. This report therefore adopts its vocabulary without exception.

Convention 16.7 (The three phase symbols, imported). Three different objects in this literature are called “phase”, and one of them is not an angle at all. The companion report fixes three names and three symbols; this report uses them and no others.

The finite angle θ , living in $\mathcal{O}$.

The argument of cis ; the *whole* angle of a unit, limited by theorem. No value of a circular function at an infinite angle is ever used.

The infinitesimal phase residue η , living in $\mathfrak{m}$.

Every unit factors uniquely as $u = cE(i\eta)$ with c an ordinary unit complex number; c carries all the ordinary phase and η all the variable intrinsic phase. It is the infinitesimal *part* of θ , not a synonym for it: $\theta = \text{st}(\theta) + \eta$ and $c = e^{i\text{st}(\theta)}$.

The accumulated phase B , an arbitrary surreal.

A primitive of the imaginary coefficient. It is *not* the angle of any surcomplex number; it need not be limited, and when it is infinite it lies outside $\mathcal{O}$ entirely. Its purely infinite part is the obstruction, and the solvability criterion is exactly the finiteness of B .

Warning 16.8 (The sentence not to write). A merged sentence such as “the phase of the solution is $\log \omega$ ” is meaningless, and the companion report deliberately prints it beside the true one. The correct statement is: *the accumulated phase of the coefficient i/ω is $\log \omega$, which is infinite, so there is no solution*. Likewise “every surcomplex number has a limited phase” is a theorem about θ ; “the variable phase of a unit is infinitesimal” is a theorem about η ; and neither implies anything about B . This report never prints the false form, and it does not re-derive the obstruction that the companion report owns.

17 Quantum fields, divergent expansions, and renormalization

17.1 The jump from finite matrices to fields is substantial

[I] Quantum field theory on a curved spacetime needs local field algebras, states, distributional correlation functions, a suitable spectrum condition and renormalized composite observables; in a Hilbert-space presentation it also needs infinite-dimensional completion and operator domains. Hollands and Wald give a precise formulation of these issues for ordinary globally hyperbolic spacetimes [32].

[A] Replacing the coefficient field supplies none of a normed-space completion, countable additivity, a spectral measure, or the hypotheses of a unitary-evolution theorem. A chosen set-sized topology or a coefficientwise construction may be useful, but it is additional input.

17.2 Formal perturbative coefficients are a realistic application

[I] Formal parameters are already a powerful organizing principle in mathematical physics; deformation quantization is a well-developed example in which associative products are built as formal series rather than convergent numerical sums [37].

[A] A Hahn or surreal realization can place several such scales in one ordered framework and add fractional powers, logarithms or exponential sectors where a justified differential structure is available; a scale-resolving field can represent quantities of the schematic form

$$\sum_{\alpha, k, n} c_{\alpha k n} \varepsilon^\alpha [\log(1/\varepsilon)]^k \exp(-nA/\varepsilon), \quad (17.1)$$

provided the chosen transseries construction admits the displayed support and operations. The display is schematic: *an arbitrary triple-indexed family is not automatically admissible*. In suitable classes it keeps algebraic, logarithmic and beyond-all-orders sectors separate, which can help formulate matching conditions, compare competing terms, and retain exponentially small information a power expansion discards. Much of this work can nevertheless be done in a smaller Hahn or transseries field; the full proper class is usually an ambient unifying language, not a computational necessity.

[A] This improves the handling of orders of magnitude. It does not make an arbitrary formal expansion into a unique actual state or scattering amplitude.

[A] For one precisely delimited class the admissibility question has an exact answer elsewhere in the collection. For the Wick diagram families of a finite polynomial interaction with a symmetric covariance over K_Γ , Γ divisible, the companion report held in `docs/surcomplex/wick-summability-certificates/` proves that strong Hahn summability is equivalent to finitely many strict valuation inequalities (its Theorem 6.1, `wick:thm:main`), and exhibits a mixed quartic family that fails although each interaction alone passes. That is a theorem about formal Gaussian coefficient data. It does not concern the display above and is not a statement about any physical expansion.

17.3 A divergent series, an exact element, and hidden data

Computation 17.1 (**[E]** The factorial series is an exact Hahn element). The series

$$\sum_{n \geq 0} (-1)^n n! \varepsilon^n \quad \text{and} \quad \hat{F}(\varepsilon) = \sum_{n \geq 0} n! \varepsilon^{n+1} \quad (17.2)$$

are well-defined Hahn elements for positive-valuation ε , although the associated ordinary power series have zero radius of convergence: for nonzero real z the ratio of successive term magnitudes in $\hat{F}$ is $(n+1)|z|$. Exact denotation at ε does not give a real-valued sum at z ; the formal series and an ordinary function with that asymptotic expansion are different objects.

Computation 17.2 (**[E]** The beyond-all-orders ambiguity, made explicit). Consider the dimensionless equation

$$y'(x) + y(x) = \frac{1}{x}, \quad x \rightarrow +\infty. \quad (17.3)$$

Substituting $\sum a_n x^{-n-1}$ gives $a_0 = 1$ and $a_n = n a_{n-1}$, so $a_n = n!$ and the formal inverse-power solution is $\hat{y}(x) = \sum_{n \geq 0} n! x^{-n-1}$. For the truncation y_N ,

$$y'_N + y_N - \frac{1}{x} = -\frac{(N+1)!}{x^{N+2}}, \quad (17.4)$$

checked exactly for $N = 1, \dots, 8$. At $x = \omega$ the series is Hahn summable, and under a compatible strong derivation with $D\omega = 1$ its differentiated terms cancel in the same way. But the homogeneous equation has solutions Ce^{-x} , so

$$y(x) = y_{\text{part}}(x) + Ce^{-x} \quad (17.5)$$

shares the same inverse-power expansion for every ordinary constant C ; in the enlarged exponential field $Ce^{-\omega}$ is a distinct beyond-all-orders scale. The exponentially small homogeneous ambiguity is itself a checked identity.

Status. **[A]** The coefficient sequence does not specify the complete solution: C must come from boundary or initial data, or from a prescribed summation and matching condition. A richer field can *retain* that information once supplied; it cannot infer it from data that do not determine it. For positive ordinary $x \rightarrow 0$, two functions differing by $Ce^{-1/x}$ have the same power asymptotic expansion to every order, and arithmetic alone selects neither C , nor a contour prescription, nor boundary conditions. This is a miniature of the difficulty faced when an asymptotic black-hole solution is mistaken for complete dynamics.

Computation 17.3 ([E] Borel transform and lateral summation). With the Borel convention sending $n!z^{n+1}$ to ξ^n , the transform of $\widehat{F}$ is

$$\mathcal{B}\widehat{F}(\xi) = \frac{1}{1-\xi}, \quad (17.6)$$

and the lateral Laplace integrals bypassing the pole on opposite sides,

$$F_{\pm}(z) = \int_0^{\infty e^{\pm i0}} \frac{e^{-\xi/z}}{1-\xi} d\xi, \quad (17.7)$$

differ by a quantity of magnitude and phase proportional to $2\pi i e^{-1/z}$, by the residue of $e^{-\xi/z}/(1-\xi)$ at $\xi = 1$; the sign depends on the orientation convention. The residue is symbolically checked. The two functions have the same formal power asymptotics away from the Stokes direction and differ by information invisible at every algebraic order.

Remark 17.4 ([C] A genuine infinitesimal needs a larger workspace). For a genuine positive infinitesimal ε , the element $e^{-1/\varepsilon}$ is smaller than every ε^N , and it is *not* an element of the rank-one field $\mathbb{R}((\varepsilon^{\mathbb{Q}}))$ with that prescribed monomial scale, since a nonzero element there has a least rational exponent. A larger workspace or a named transseries subfield is required. Extending the arithmetic and choosing how it represents an actual function are two separate steps.

[A] A transseries must therefore include a suitable exponential sector *and* data selecting its coefficient. Naming the Hahn element has selected no boundary condition, no lateral summation and no physical state. [I] This is exactly why Stokes data and realization matter in resurgent physics [36]. [A] The extension and summation choices in the established surreal integration theories are precisely additional structure, not consequences of an unadorned infinite sum [16, 12].

17.4 A conditional asymptotic-to-surreal pipeline

[A] A rigorous application can proceed through a restricted pipeline: start from a specified family of real or complex solutions and a dimensionless parameter; derive its admissible asymptotic expansion, with uniformity on a named ordinary domain; embed that expansion in a differential Hahn or transseries field whose derivation represents the selected variable; and prove that the observables extracted from the representation agree with the original solution up to a stated error. The output is a formal or an analytic certificate in the sense of Section 13.4. For singular ODE reductions, existing surreal integration results suggest a feasible bridge where their hypotheses apply [16]; for the unrestricted nonlinear Einstein system that bridge would require substantially more work. A precise restricted model is preferable to an assertion of universal transfer.

17.5 Unlimited values do not define products of distributions

Computation 17.5 ([E] A Gaussian regulator and its square). For an ordinary positive real regulator ε put $\delta_{\varepsilon}(x) = (\varepsilon\sqrt{\pi})^{-1} \exp(-x^2/\varepsilon^2)$. Gaussian integration gives

$$\int_{\mathbb{R}} \delta_{\varepsilon}(x) dx = 1, \quad \int_{\mathbb{R}} \delta_{\varepsilon}(x)^2 dx = \frac{1}{\varepsilon\sqrt{2\pi}}. \quad (17.8)$$

Both integrals are symbolically checked. The first family tends to the Dirac distribution; the second has a divergent integral.

Status. [A] Naming the right-hand side of the second identity an unlimited surreal value constructs no canonical distribution δ^2 and proves no regulator independence. Note also what was used: the integral in this example was the *ordinary* integral for each real regulator, and no unannounced surreal integration rule appeared.

Computation 17.6 ([E] Pointwise infinitesimals do not produce a distribution). In ordinary distribution theory

$$\lim_{\eta \downarrow 0} \frac{1}{x + i\eta} = \text{PV} \frac{1}{x} - i\pi\delta(x), \quad (17.9)$$

which follows from $\frac{1}{x+i\eta} = \frac{x}{x^2+\eta^2} - i\frac{\eta}{x^2+\eta^2}$: against a smooth compactly supported test function the imaginary kernel integrates to π in the limit, by $x = \eta y$ and $\int_{\mathbb{R}} (1+y^2)^{-1} dy = \pi$, while the odd real kernel tends to the principal value after locally subtracting the test function's value at zero. This is an *integral* statement, not a pointwise identity of functions. Setting η instead to a fixed positive infinitesimal, pointwise standard part gives $1/x$ at each nonzero ordinary x , and at $x = 0$ the value is unlimited so standard part is unavailable. *The $-i\pi\delta(x)$ contribution cannot be recovered by pointwise standard part at all.*

Status. [A] A test-function pairing, with an explicitly justified coefficientwise, asymptotic or nonstandard integration construction, is required. Writing a surcomplex $i\varepsilon$ has supplied no distributional prescription. The general lesson recurs throughout field theory: information can concentrate in a shrinking region whose pointwise ordinary shadow vanishes away from that region, so a global integration rule must resolve the region *before* taking the observable shadow.

[I] Generalized-function approaches to nonlinear equations do exist, including frameworks used in general relativity, and they retain carefully specified regularization data and notions of equivalence or association [35]. **[A]** They should not be confused with scalar field extensions. Surreal coefficients could be incorporated into such a framework, but the mathematical work lies in defining its operations and their relation to physical observables. Substituting an unlimited scalar for a delta distribution does not retain its action on test functions, and a black-hole proposal involving weak stress tensors must address that explicitly whatever field it chooses.

17.6 Removing negative powers is not a renormalization theory

Computation 17.7 ([E] Finite part is neither multiplicative nor reparametrization invariant). On the ring of limited elements, standard part is multiplicative. On all Laurent series, the operation FP extracting the exponent-zero coefficient is not:

$$\text{FP}(\varepsilon) = \text{FP}(\varepsilon^{-1}) = 0, \quad \text{FP}(\varepsilon \cdot \varepsilon^{-1}) = \text{FP}(1) = 1. \quad (17.10)$$

Nor is a bare finite-part prescription invariant under a harmless-looking change of regulator coordinate. With $\theta = \varepsilon/(1 + a\varepsilon)$ for $a \in \mathbb{R}$,

$$\varepsilon^{-1} = \theta^{-1} - a, \quad \text{FP}_\varepsilon(\varepsilon^{-1}) = 0, \quad \text{FP}_\theta(\varepsilon^{-1}) = -a. \quad (17.11)$$

Equivalently, in the other manuscripts' parametrization: $F(\varepsilon) = \varepsilon^{-1} + a$ has exponent-zero coefficient a , but under the equally legitimate formal change $\varepsilon = \eta + c\eta^2$,

$$F(\eta + c\eta^2) = \eta^{-1} + (a - c) + c^2\eta - \dots, \quad (17.12)$$

so the constant coefficient becomes $a - c$. Both reparametrization shifts are symbolically checked.

Status. So “keep the finite part” of an unlimited expression is scheme-dependent. **[A]** This does *not* contradict the uniqueness of standard part in Theorem 4.4, because F was never in $\mathcal{O}_F$. Physical renormalization must specify observables, allowed counterterms and normalization conditions; field enlargement does not fix that freedom, and effective-field-theory reasoning remains necessary for identifying which interactions and parameters have physical significance [24]. A coefficient-extraction rule must be shown to respect the relevant coordinate and gauge symmetries, and its observable consequences must not depend on arbitrary regulator choices — otherwise a supposedly resolved curvature is a scheme-dependent number rather than a physical prediction.

[A] Surreal arithmetic can still help here, and the help is precisely specifiable: divergent powers and logarithms can be retained symbolically, counterterm cancellation can be tested exactly, and residual prescription dependence can be *exposed* rather than hidden. That is an improvement in representation and verification. It is not a proof that ultraviolet divergences have been physically removed.

17.7 Hawking radiation and information require state evolution

[I] The Hawking effect can be formulated using ordinary quantum fields on suitable classical black-hole backgrounds, with the usual care about states and renormalized observables [32]. **[A]** A black-hole information proposal requires an account of the full evolution and of the observables accessible to relevant observers. Making an interior curvature or an intermediate amplitude surcomplex neither supplies that evolution nor proves its unitarity. A successful asymptotic calculation could certainly assist — controlling a near-horizon mode expansion, a late-time tail, or a nonperturbative correction — but to claim a physical information-resolution mechanism one would still need to identify a state map, prove the relevant consistency properties, and relate the result to finite measurement outcomes. Those are the missing physical statements, not missing names for large numbers.

17.8 A realistic wave-physics benchmark

[I] Black-hole quasinormal-mode and scattering problems already use complex analysis, asymptotic series and boundary conditions; Hatsuda relates quasinormal frequencies to high-order WKB expansions and shows, in the cases studied, how Borel summation recovers the numerical frequencies [34]. **[A]** That is a concrete and *testable* benchmark for a surreal-compatible implementation: preserve the perturbative and nonperturbative sectors, enforce the physical boundary conditions, and reproduce an established complex frequency calculation. Success would demonstrate calculational utility. It would not resolve an interior spacetime singularity, because the wave problem and its boundary conditions are separate from a replacement of the central geometry. The benchmark is attractive precisely because the claim is limited and can fail.

Part V

Context, interpretation, and what to do next

18 Existing proposals and the status of the idea

18.1 A direct surreal-singularity proposal deserves a direct test

[I] Nieto’s *Some mathematical and physical remarks on surreal numbers* explicitly discusses gravitational singularities and proposes that the availability of infinite surreal values removes the usual singularity problem; a later paper connects surreals to dualities, matroids, qubits and twistors [41, 42]. In the Schwarzschild discussion the argument passes from a larger scalar domain to a claim that the singularity disappears, with no corresponding construction of a regular differential spacetime or a geodesic continuation. That such work exists establishes that the suggestion has precedents, and it is not a question that can be dismissed as unasked or meaningless.

Assessment 18.1 ([A] The missing step, located). The computations of Sections 7 and 9 identify precisely where the inference fails, in four places.

- (1) At a nonzero infinitesimal radius the metric is evaluable, but by Theorem 7.5 and Theorem 7.8 the invariant curvature is still unlimited in the real-boundedness sense, and by Theorem 7.7 the $r = 0$ pole persists under every radial reparametrization of the stated form; its order is multiplied by q and its leading coefficient is rescaled by $w(0)^{-6}$.
- (2) The exact identity $r^6 \mathcal{K} = 48m^2$ still has *no* field-valued solution at $r = 0$, by Theorem 7.12, in *any* characteristic-zero field and not merely in $\mathbb{R}$.
- (3) The classical endpoint is still reached at finite — indeed infinitesimal, hence limited — proper time, by Theorem 9.2, and the tidal equations (9.4) retain their singular coefficients.
- (4) No continuation law follows from any of the above, so none of (Q1)–(Q4) of Section 3.3 is answered.

Therefore the proposed inference is not a singularity-resolution theorem. The useful intuition in it is the *resolution of scales*, not a demonstrated removal of the endpoint.

Status. [A] This criticism concerns that *inference*. It is not an assertion that no future theory using surreal variables could make nontrivial physical predictions, and it is not an impossibility theorem. A complete alternative might replace the metric, the state space, the observables or the dynamical law — but those replacements must be constructed.

[A] The literature reviewed by the three manuscripts does not establish an empirically tested theory in which scalar replacement alone resolves a black-hole singularity. That is a bounded assessment of the cited work. It is not a proof that no future non-Archimedean theory can do so, and the established mathematical results on surreal differential fields and integration are genuine and must not be conflated with speculative gravitational interpretations. Each manuscript records that its literature search was targeted rather than systematic: absence of a predictive model in the inspected sources is not proof that no other proposal has ever been written.

18.2 Comparison with other non-Archimedean and singular methods

[A] The choice of tool should be driven by the operation required, not by the size of the number system, and a research program should explain why surreals are preferable to an existing smaller tool for a particular task. Universality alone is neither an efficiency theorem nor a prediction. Table 3 merges the three manuscripts' comparison tables. Its last column records a *missing implication*, not a blanket impossibility, and the table is not a ranking of research programs.

Table 3: What each framework is naturally for, and what does not follow automatically. The third column is a missing implication, never a blanket impossibility.

Framework	Principal useful structure	What does not follow automatically
Real/complex asymptotics	Limits, error bounds, analytic continuation, well-developed PDE and operator methods	Existence of a quantum-gravity completion of a classical singularity
Formal power and Puiseux series	Finite coefficient algorithms; ramified scaling; clear order-by-order equations	A convergent or uniquely selected real solution
Hahn fields	Exact ordered powers, multiple scales, support-certified arithmetic, finite algebra and valuation balances	Function domains, derivatives, integration, realization of formal series; arbitrary smooth function theory or physical interpretation
Transseries and resurgence	Logarithmic and exponential asymptotics, beyond-all-orders sectors, asymptotic differential algebra, selected reconstruction methods	Admissible support, summation, Stokes and boundary data, error estimates; universal applicability to nonlinear spacetime PDEs
Full sur-real/surcomplex scalars	An exceptionally broad ordered and exponential ambient system, algebraic closure, compatible differential structures for many scales, distinguished normal forms	An ordinary continuum topology, a canonical global oscillatory calculus, a probability theory, a measurement rule, or singularity resolution; set/class discipline and a chosen analytic framework remain to be supplied
Hyperreal and nonstandard models	Infinitesimal encodings with a specified transfer and internal-set framework, with ordinary shadows	New predictions merely from reformulating an old model; internal models, saturation, measure and dynamics hypotheses must be supplied
p -adic models	Non-Archimedean norm geometry and explicitly constructed alternative field theories	New geometric and physical interpretations; no inherited real ordering or Lorentzian causality

Framework	Principal useful structure	What does not follow automatically
Distributions and generalized-function geometry	Weak limits, singular sources, boundary values, a specified notion of weak field and regularized nonlinear operation	Products or nonlinear equations may need extra structure; uniqueness or regulator independence without additional hypotheses
Regular-core or other modified dynamics	New field equations or effective sources, possibly a regular core	Derivation, empirical validity, global stability, geodesic completeness, unitarity, or observable consistency from local regularity alone

[A] Four specific cautions belong with the table.

First, if a calculation is unchanged when performed in a small Hahn field, its success does not demonstrate that the full proper class of surreals is physically necessary. A positive result should identify whether the essential contribution is the larger field, a support theorem, a new extension operator, or simply a more transparent notation for an existing asymptotic argument. For an individual expansion with rational powers a Puiseux field may suffice; for several ordered scales a Hahn field may be the right working object. Surreals become attractive as an ambient framework connecting many such structures, or supporting a particularly useful normal form — not because they are larger.

Second, an ordered non-Archimedean field can sit inside a surreal ambient field without thereby identifying the accompanying structures. A hyperreal formulation usually specifies a nonstandard enlargement, internal functions and sets, and a transfer principle for that enlarged structure; those are more than a field embedding [9, 39].

Third, p -adic physics is not surreal physics in different notation: $\mathbb{Q}_p$ carries no compatible field order, whereas order is central to the real-surreal modulus and the causal-sign constructions used throughout this report. **[I]** Gubser and collaborators' p -adic AdS/CFT model is instructive precisely because it chooses a p -adic boundary and a tree-like bulk explicitly and computes correlation functions [40]: a concrete alternative geometric model, neither an ordered extension of the real spacetime metric nor evidence about surreal numbers and Schwarzschild.

Fourth, a complex contour can detour around a pole, but an analytic continuation along that contour is not necessarily a real Lorentzian trajectory. Ordinary $\mathbb{C}$ already permits such detours; changing to $\mathbf{No}[i]$ specifies no continuation, no boundary condition and no causal history. To turn such a construction into a real physical continuation one must recover a real Lorentzian slice, specify the reality conditions, choose the branch and matching data, and show that the resulting evolution satisfies the intended equations. Analytic continuation in a parameter is not the same operation as extending an infalling observer's causal worldline.

[A] For this repository, a set-sized Hahn or transseries layer is the lowest-risk first implementation: it uses the existing summability and finite-algebra work while avoiding an unnecessary commitment to a full surreal-valued event manifold. A later theory can explain when embedding those calculations into **No** supplies a substantive additional benefit.

19 Measurement, units, recoverability, and empirical novelty

19.1 Which structure is supposed to be physical?

[A] A scalar replacement must say whether it changes the set of events, the values of fields on an unchanged spacetime, the codomain of measurement results, or only an approximation calculus. These lead to different theories. Limited surreal field values on a real manifold can have an ordinary shadow while retaining formal infinitesimal corrections; a manifold modelled directly on a non-Archimedean field instead requires a new account of paths, integration and initial data. *There is no single object called “the surreal spacetime” determined by the field axioms.*

[A] The simplicity relation and the birthday of a surreal number are extra mathematical structure, not automatically a physical age. A rule identifying an ordinal birthday with time would have to explain how ordinary time translation, changes of units and coordinate covariance act on it. No such identification is needed for the coefficient-based program developed here, and none is made.

19.2 What an ordinary apparatus is supposed to report

[A] There are at least two readings of a surreal-valued observable. It can encode the asymptotic behaviour of a family of ordinary observables, in which case there is a parameter-to-physical-system map and an approximation problem. Or it can be taken literally as the value in a new ontology, in which case an apparatus rule must map it to records and probabilities.

The standard-part map is a simple apparatus rule for limited values. It is *not forced by field algebra*; it is natural when infinitesimals are meant to represent deviations below every real resolution, and it is an additional modelling assumption. It is undefined on an unlimited curvature such as (7.6), and extending the codomain to include $+\infty$ still provides no ordinary finite detector output. Theorems 12.3 and 15.2 show how conservative this rule is: regular limited data reduce to the usual theories in the stated models.

[C] More generally, any expression built from limited inputs by addition, multiplication, conjugation and inverses with nonzero standard part commutes with reduction. So a finite algebraic readout protocol cannot distinguish two inputs whose difference is infinitesimal under those conditions. **[A]** That is a mathematical statement, not an assertion that every imaginable physical apparatus is such a protocol.

19.3 Ways to leave the conservative regime

[A] A new prediction could arise through a singular rescaling, an unbounded duration, a nonstandard observable rule, or an explicit change of dynamics. Each is a meaningful research possibility, and each changes more than the bare choice of scalar field — Theorem 15.4 is the cleanest illustration of the second, and Theorem 17.7 of the third. If an ordinary finite experiment with bounded couplings is claimed to see such an effect, the model should derive that result without silently inserting an infinite resource.

19.4 A new scalar ontology may have no new finite predictions

[A] Theorem 4.4 and Theorem 15.2 show how a large numerical extension can reduce to ordinary predictions for a specified class of observations. That is not a defect: many valuable mathematical reformulations are empirically equivalent. It does mean that an ontological claim — “the true

amplitude contains an infinitesimal term” — needs an operational consequence before it becomes a testable physical distinction.

[A] Invariance of predictions is the decisive test. A physically meaningful result should survive changes of admissible coordinates, equivalent asymptotic parametrizations, choices of embedding into a larger Hahn workspace, and regulator prescriptions subject to the same physical normalization conditions. That requirement is strictly stronger than formal equality inside one chosen representation, and (17.11) demonstrates how an apparently finite prediction can shift under a reparametrization when the observable rule is underspecified. Conversely the invariant curvature identities and Theorem 5.4 exhibit quantities that can be controlled intrinsically. If a prediction changes when an arbitrary infinitesimal is replaced by another with no accompanying change of physical inputs, the model has not yet separated physics from its regulator; if the choice is intended to be physical, it must be promoted to a law or a measurable parameter rather than hidden in notation.

[A] No empirical preference for a literal surreal scalar ontology follows from anything in this report. The immediate case for these methods rests on mathematical utility. An experimentally distinct theory would have to exhibit a definite finite prediction, show why ordinary effective descriptions do not already contain it, and connect its parameters to a reproducible physical procedure.

19.5 The relevant notion of resolution must be stated

[A] A theory might allow unlimited intrinsic quantities while maintaining finite detector responses. Another might replace a spacetime endpoint by a transition between quantum states. A third might define a weak geometric extension. These are logically possible strategies and nothing here rejects them by definition. Each carries a distinct burden of proof: the first must *derive* detector responses rather than postulate a standard part of a nonexistent limited value; the second needs a dynamical state map and an interpretation of time; the third needs a regularity class and an evolution or matching principle. Examples of what “resolved” might mean include extension of every relevant causal geodesic, bounded detector responses, existence and uniqueness of a weak evolution, and a specified quantum propagation criterion — and these conditions are *not equivalent*. In all cases the physical gain would be the successful construction and its consequences, never the permission to write an unlimited scalar.

20 Relation to the companion reports in this repository

[A] This report is the only member of its collection whose subject is empirical. Nothing in the neighbouring reports contains general relativity, quantum mechanics or field theory, so the report carries the whole burden of its own reading contract and establishes the theorem-versus-interpretation discipline of Section 2 for itself. Two companion reports are nevertheless load-bearing, and in both cases the correct relation is to *inherit and cite*, not to restate.

Foundations for Surreal and Surcomplex Mathematics. In docs/foundations-and-computation/foundations/. That report owns the workspace-localization theorem and the size obstructions any candidate foundation must respect. Section 4.2 cites its localization theorem for the workspace discipline used throughout, and defers to its size obstructions for anything bearing on proposal (P3), a genuinely surreal-valued spacetime. This report does not re-argue set-theoretic

legality, and there is no content duplication: the physics manuscripts ask a different question of the same facts, namely which of (P1)–(P3) a given calculation actually needs. Readers evaluating a proposed surreal-valued event manifold should begin there and not here.

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers. In docs/surcomplex/differential-equations/. That report owns the oscillation obstruction, the finite-primitive criterion and the three phase symbols. Section 16.4 cites its oscillator corollary rather than re-deriving the obstruction; Theorem 16.7 adopts its three phase symbols verbatim, and Theorem 16.8 reproduces its warning about the sentence not to write. Theorem 16.2 and Theorem 16.3 are the physical reading of that obstruction and land on that report’s phase conventions; they strengthen nothing there and contradict nothing there.

The surcomplex analysis and trigonometry reports. Sections 5.1 and 5.3 use the analysis report’s distinction between Hahn summation and ordinary convergence, its analytic function classes, and the common-domain warning example (3.1); Section 16.1 uses the trigonometry report’s inclusion of the Ehrlich–Kaplan integer-part global normalization together with its explicit refusal to select a scalar derivation [3, 4]. Neither is treated as a verified physics foundation, and the trigonometric and complex-algebraic material must not be read as a finished unrestricted phase calculus for quantum evolution.

The surquaternionic family (docs/surquaternions/). Named only to prevent a transfer. That material is noncommutative, so conjugation, modulus multiplicativity and valuation need separate treatment there, and nothing in this report — every result of which uses commutativity somewhere — may be lifted into it without reproof. The last row of Table 2 records the one specific hazard: the quaternionic radial exponential is not an additive-to-multiplicative homomorphism for noncommuting arguments, while Exp_{EK} of (16.3) is a homomorphism on the commutative surcomplex field. Both statements are true of different objects.

21 A staged research program

[A] Everything in this section is tier **[A]**: these are proposals, not descriptions of existing modules and not results. The central research challenge is to make a proposal precise enough that it can *fail* a test. A larger collection of named numbers is not yet such a proposal.

21.1 Recommended architecture

The most defensible starting point is

$$\boxed{\begin{array}{c} \text{ordinary spacetime} \\ + \text{controlled differential Hahn/transseries coefficients} \\ + \text{explicit real-shadow map} \end{array}} \quad (21.1)$$

This preserves the classical geometric domain while enriching the asymptotic fields. **[A]** It is called the most defensible starting point; it is *not* a derived necessity. A fully non-Archimedean spacetime can be studied separately, but should not be required before the coefficient approach has produced a demonstrable advantage.

A coefficient interface should record the value group, support certificates, coefficient domains, conjugation, the real form, and the derivatives under which the space is closed; and it should distinguish a spacetime derivative from a scalar derivation and from a formal-parameter derivative *at the type or interface level*, per Table 2. It should store the coefficient domain, exponent group, derivative convention and any summation prescription alongside each series; reject a product or substitution whose support condition fails; and distinguish an exact Hahn equality from an asymptotic equality of functions and from equality after standard part. The repository's existing finite algebra, Neumann support, rational-jet and standard-part layers are useful prerequisites, and their stated current scope must be respected.

21.2 A minimal model specification

A candidate theory should specify: its scalar workspace and foundational size; its spaces of states, events and fields; its topology and derivatives; its rules for infinite sums and integration; its field equations and admissible initial data; its causal or quantum evolution law; its gauge and unit conventions; and its operational readout. Recovery of ordinary physics must be a theorem or a controlled approximation *in the chosen model*, never an appeal to the inclusion $\mathbb{R} \subset \mathbf{No}$. For a singularity claim the specification must additionally state which of the meanings in Section 19.5 is intended.

21.3 Stage A: formalize a conservative local theory

Target the model of Section 12, not unrestricted quantum gravity. Define a common-domain algebra of globally supported coefficient functions; prove support preservation under products and coefficient derivatives; construct matrix inversion around a nonsingular ordinary leading matrix; and make standard part commute with these operations *by theorem*, not by convention. Then define tensors, the Levi-Civita connection, curvature and the Einstein tensor, and prove Theorem 12.2, Theorem 12.3, the finite Bianchi identities, Theorem 5.4, and compatibility with ordinary coordinate transformations, recording the exact coefficient-ring assumptions. Proposed module names such as `Physics/CoefficientFields`, `Physics/FormalMetric` and `Physics/StandardPartGR` are *proposed additions*, not claims about the inspected tree.

The failure conditions matter as much as the successes, and they should be *reported* rather than papered over: a zero leading determinant; a non-well-ordered global support; a coefficient family lacking a common domain; an unsupported substitution. A system that assigns a symbolic infinity and proceeds has failed.

21.4 Stage B: verify exact backgrounds and observer diagnostics

Use Schwarzschild as a mandatory control case: verify the cancellation of the horizon-coordinate singularity (Theorem 7.1), the vacuum Ricci tensor away from zero, the Kretschmann invariant (Theorem 7.3), the radial proper-time scaling and the Jacobi exponents (Theorems 9.1 and 9.4), in *both* static and ingoing coordinates, and the Kasner family (Theorem 11.1) to test multiple real exponents and vanishing leading coefficients. Add the regular-core ansatz to distinguish changed dynamics from scalar extension.

The benchmark is passed only if the method identifies the horizon as removable and $r = 0$ as obstructed under the unchanged invariant identity. A system that declares both regular merely because the denominators have become surreal symbols has failed. Success means more than

producing a large expression: the tool should retain units, verify agreement of invariant quantities between charts, certify which denominators are units, and report when a requested evaluation meets a zero denominator or an unsupported expansion. In particular it must distinguish g_{rr} at the Schwarzschild horizon from (7.6) at the central curvature pole. The supplied programs check the concrete identities used in this article; they are not a proof-assistant verification of the general framework.

21.5 Stage C: certified dominant balances, matching, and one matter problem

Implement the support test and the exact finite remainder of Theorems 14.5 and 14.6. The desired output is a proof-carrying classification of outer, transition and inner scales together with the exact rational model the expansions came from; extend the test to several small parameters and to fractional-power ansatzes. A further benchmark is a reduced gravitational or matter ODE with a controlled singular expansion: compute its coefficients, prove the support pattern inductively, and compare with a real solution through a remainder theorem or a valid summation construction. *Formal recurrence alone is not an existence theorem*, and in applying a surreal integration result one must verify its nonresonance, function-class and branch hypotheses explicitly.

A defensible matter experiment is a scalar field on a fixed spherically symmetric background with a specified asymptotic region and initial or characteristic data: derive a consistent expansion for the linear equation, compare it with ordinary asymptotic estimates, and only then attempt backreaction in the Einstein–massless-scalar system

$$R_{ab} = 8\pi \partial_a \phi \partial_b \phi, \quad \square_g \phi = 0, \quad (21.2)$$

in geometrized units with zero cosmological constant, these being the equations of the usual minimally coupled massless scalar action, which also fix the normalization of ϕ . In double-null spherical coordinates, with

$$ds^2 = -\Omega^2(u, v) du dv + r^2(u, v) d\Omega_2^2, \quad g_{uv} = -\Omega^2/2, \quad (21.3)$$

one null constraint and the geometric mass aspect are

$$\partial_u \left(\frac{\partial_u r}{\Omega^2} \right) = -\frac{4\pi r}{\Omega^2} (\partial_u \phi)^2, \quad 1 - \frac{2M}{r} = -\frac{4 \partial_u r \partial_v r}{\Omega^2}, \quad (21.4)$$

with the analogous equation in v . **[E]** The normalization in (21.4) is checked through $R_{uu} = -2r^{-1}(r_{uu} - 2\Omega_u r_u/\Omega)$ together with (21.2), and the areal-radius gradient norm is checked as well. **[A]** These give explicit constraints and invariants to test; they are not a promise that arbitrary characteristic data admit a Hahn solution. For a candidate expansion, publish the coefficient domain, the scale ordering, the equations solved at each order, the residual valuation, and real error information where available — and check both evolution equations *and* constraints, since a proof that one reduced ODE is satisfied does not establish a solution of the coupled field equations.

21.6 Stage D: prove a realization bridge

The most useful single result would connect a formally certified solution in a named asymptotic class to an ordinary solution with a controlled remainder. In an ODE reduction this might use an existing extension or summation theorem after verifying its hypotheses; in a PDE setting it would need appropriate analytic estimates and a common domain, with the selection of boundary

or Stokes data explicit. A substantive target is a selected Fuchsian or quiescent Einstein–matter construction formulated in a differential Hahn setting, with a proof that its shadow and remainder bounds reproduce the known classical solution, including the constraints, gauge conditions, and compatibility of evolution with the constraints. Only then should one ask whether a larger support class yields a genuinely stronger result.

The concrete obstruction of Section 13.3 — position-dependent exponents — is exactly the kind of thing a prototype should expose. And the success criterion is not a longer formal expansion; it is an expansion whose interpretation can be checked against a real family or against an independently defined generalized dynamics. A failure of realization can itself reveal missing sectors or inconsistent assumptions.

21.7 Stage E: only then, test a singularity-resolution hypothesis

Only after the conservative benchmarks should one propose a modified action, a constitutive law, a quantum effective equation, or a nonclassical observable algebra. State which classical assumptions are changed. Then test regularity in a specified sense, continuation of physical probes, preservation or replacement of constraints, stability, and ordinary-limit agreement.

For a regular-core hypothesis, specify whether the core scale is a fixed positive real, an asymptotic parameter, or a literal infinitesimal, and check the distinction of Theorem 14.4. An alternative target is a controlled inner-layer model with a finite physical scale whose source comes from a stated action or effective equations rather than being chosen solely to cancel a curvature pole; examine causal structure, regularity, energy conditions, stability under perturbations, and matching to the exterior. A finite central scalar is one benchmark among several, not the complete result. For a generalized spacetime hypothesis, define its topology, curves and notion of maximal extension *before* claiming completeness. For a quantum hypothesis, define the state space and its observable map before claiming unitary evolution through the singular regime. For a compatible quantum dynamics, begin with finite systems whose Hamiltonian, real-time evolution and phase calculus are specified together; prove reduction to the ordinary Schrödinger equation under common-domain limited-coefficient hypotheses; treat large-phase and long-time scalings as separate asymptotic regimes; and include the explicit test (16.4) in any proposed phase interface.

The decisive deliverable would be a model with a definite observable consequence, or a stronger mathematical existence theorem — not an expression in which the old divergent quantity has been assigned an infinite name. The program permits both positive and negative results and states criteria for recognizing each.

21.8 Four concrete projects, and what would count as an advance

Project A: certified multiscale infall.

Develop the crossover problem (10.3) in a controlled Puiseux or Hahn setting, prove uniform remainder estimates on matching domains, and compare with high-precision real integration. Success is a theorem connecting the symbolic sectors to actual geodesics, not a list of formal dominant terms.

Project B: a regular-core singular perturbation problem.

Use (14.1) or another explicitly specified model to track the outer Schwarzschild and inner-core expansions, their overlap, and invariant curvature bounds as functions of the regulator. Success

distinguishes regularity at fixed regulator from uniform control as the regulator vanishes, and identifies the required stress tensor.

Project C: a wave and resurgence benchmark.

Select one black-hole wave equation with known boundary conditions and well-tested frequencies, construct its transseries representation, prove the needed support and summation statements, and recover the physical solution. Success is reproducible accuracy or a new theorem about the asymptotic structure, not a numerical coincidence from an unspecified infinite sum.

Project D: a genuine non-Archimedean modification.

Only after the preceding interfaces are understood, choose an explicit new dynamical or operational postulate. Prove a well-posedness result in a reduced model, show what replaces the classical endpoint, and determine whether the ordinary observable prediction differs from the corresponding real model. Success would be a substantive physical proposal; *no such postulate is forced by surreal arithmetic alone.*

[A] A mathematically convincing advance would be a new controlled solution class, a sharper singular-asymptotic estimate, a support-based algorithm that detects invalid expansions, a rigorous matching theorem that is materially easier or stronger in the new language, an extension theorem for a relevant gravitational solution class compatible with gauge transformations and controlled remainders, or a spectral result for a precisely defined surcomplex wave operator. Such results would justify surreal methods *even with no new physical prediction.*

A physically convincing advance would additionally identify a well-defined finite observable, show how it is computed, prove recovery of the appropriate classical or quantum regime, and explain why the result is independent of unphysical choices. For a singularity-resolution claim it would specify what continues through the formerly singular regime and prove the relevant continuation or dynamical theorem. This report offers testable mathematical targets. It offers no completed quantum-gravity model.

22 Final assessment

[A] As a language for near-singularity behaviour: yes, with substantial potential. Surreal and surcomplex methods record distinct unlimited and infinitesimal scales exactly. Fractional powers, logarithms, exponentials and suitably supported series expose structure that a single divergent limit suppresses: the infall exponents (9.5)–(9.6), the competing angular-momentum scales of Theorem 10.4, the logarithmic weights of Theorem 11.3, the exponentially small sector of Theorem 17.2, and the inner/outer criteria of Theorem 14.5 are all concrete gains. The established differential and integration theories provide a serious foundation for selected extensions beyond elementary examples, and the repository’s separation of algebra, summability, domains and topology is an asset rather than a technical inconvenience.

[A] As an automatic cure for black-hole singularities: no. This is the report’s conclusion and it is negative. The Schwarzschild invariant remains exactly $48m^2/r^6$ for $m \neq 0$. Under a formal radial change $r = s^q w(s)$ with integer $q \geq 1$ and $w \in \mathbb{R}[[s]]$, $w(0) \neq 0$, the pole order becomes exactly $6q$ and the leading coefficient becomes $48m^2 w(0)^{-6}$. The finite checks used $w = 1 + s$ and $q = 1, 2, 3$, so their leading coefficient remained $48m^2$. The general proof is Theorem 7.7: no radial reparametrization of this stated form removes the pole. The proper-time invariant remains $64/(27u^4)$; the proper time from rest at R to $r = 0$ is $(\pi/2)\sqrt{R^3/2M}$, finite — and for infinitesimal

R it is infinitesimal, hence still finite, so the infall terminates either way. The tidal component $2M(2M - r)/r^4$ diverges, and the tidal equations retain their singular coefficients. The identity $r^6\mathcal{K} = 48m^2$ has no field-valued solution at $r = 0$ in any characteristic-zero field. A regular-core model changes the metric and its effective source, and making its core scale infinitesimal does not make its central curvature finite in real units. What is true is that surreal numbers make the hierarchy near a singularity *exactly representable* while every invariant obstruction survives. The sentence to refuse, in any form, is that surreal numbers resolve the black-hole singularity.

[A] As a fundamental replacement for all real and complex physics: a possible research direction, not an established physical theory. Finite algebra transfers well: Theorem 15.1 and Theorem 15.2 are real results. Topology, differentiation, integration, infinite-dimensional quantum theory and empirical interpretation require explicit choices that cannot be inferred from the field alone. In the full fine topology, ordinary real-time continuous curves into $\mathbf{No}[i]$ are constant; coefficientwise dynamics or a different topology avoids that, but must be declared. There are already rich positive mathematical constructions, and their mutual compatibility must be *proved* rather than assumed — Theorem 16.2 and Theorem 16.3 illustrate that requirement sharply, and they do so about the most canonical construction available.

[A] The best-supported next step. Use surreal ideas to strengthen *the analysis of physical models*, especially multiscale and singular perturbation problems, while retaining a clear ordinary observational limit: retain ordinary spacetime and operational observables, use a carefully chosen formal or Hahn coefficient structure, connect it to actual solutions by proved realization and matching results, and formalize the algebraic interfaces. Such a program could produce valuable mathematics — reusable even if surreals never become fundamental physical scalars — and might eventually help a physical theory of singular regimes. A more radical theory remains possible, but its singularity-resolving content must lie in explicit new dynamics or a rigorously defined replacement of the classical state space. The larger numerical universe supplies expressive resources. Dynamics and evidence must supply the physics.

Part VI

Appendices

A Derivation of the curvature formulas

A.1 The general static spherical metric

For

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2,$$

work first in a region $f > 0$ and choose the orthonormal coframe

$$e^0 = \sqrt{f} dt, \quad e^1 = f^{-1/2} dr, \quad e^2 = r d\theta, \quad e^3 = r \sin \theta d\phi.$$

With the curvature convention

$$R^a_{bcd} = \partial_c \Gamma^a_{db} - \partial_d \Gamma^a_{cb} + \Gamma^a_{ce} \Gamma^e_{db} - \Gamma^a_{de} \Gamma^e_{cb}, \quad R_{bd} = R^a_{bad}, \quad (\text{A.1})$$

the independent orthonormal components, up to the Riemann symmetries, may be taken as

$$\begin{aligned} R_{0101} &= \frac{1}{2} f'', & R_{0202} &= R_{0303} = \frac{f'}{2r}, \\ R_{1212} &= R_{1313} = -\frac{f'}{2r}, & R_{2323} &= \frac{1-f}{r^2}. \end{aligned}$$

A simultaneous overall sign change from another Riemann convention does not change $\mathcal{K}$. Each listed squared component has multiplicity four in the full contraction: four entries of magnitude $f'/(2r)$, one of magnitude $f''/2$, and one of magnitude $(1-f)/r^2$. Therefore

$$\mathcal{K} = 4 \left[\frac{(f'')^2}{4} + 4 \frac{(f')^2}{4r^2} + \frac{(1-f)^2}{r^4} \right],$$

which is (7.3); contracting before squaring gives (7.4).

[A] These coordinate expressions are rational differential identities in f and r . They continue to regions $f < 0$ where the metric is nondegenerate, and to a regular horizon when a nonsingular chart such as (7.2) is used; the static orthonormal coframe itself is not asserted to exist with that real square root inside the horizon. Substitution into a Hahn field preserves the resulting rational identities wherever the denominators remain nonzero, and no transfer theorem for arbitrary real functions is invoked. Each of the three companion programs performs the coordinate calculation from (A.1) without relying on a static orthonormal frame across the horizon.

For $f = 1 - 2m/r$ the three contributions are $16m^2/r^6$ each, giving (7.5); for $f = 1 - r^2/\ell^2 + O(r^5)$ they tend to $4/\ell^4$, $16/\ell^4$ and $4/\ell^4$, giving (14.3).

A.2 Kasner and the tidal coefficients

For the Kasner metric (11.1) the analogous components have magnitudes $p_j(p_j - 1)/\tau^2$ for the time-space planes and $p_j p_k/\tau^2$ for the spatial planes; counting the same multiplicity four gives (11.2), and the vacuum constraints in (11.1) follow by contracting these components.

For Schwarzschild, the time–radial component yields the radial stretching coefficient $2m/r^3$ and the transverse components yield $-m/r^3$ in the geodesic-deviation equation. A radial boost preserves these principal tidal coefficients, which is why (9.4) holds in a radially freely falling frame and not only in the static frame.

A.3 Dimensional conventions

All curvature powers are stated with $[\mathcal{K}] = \text{length}^{-4}$. The proper-time variable in (9.2) is a length; if $\Delta\tau_{\text{sec}}$ is measured in seconds, replace u by $c\Delta\tau_{\text{sec}}$, so that for the $E = 1$ radial infall

$$\mathcal{K} = \frac{64}{27 c^4 (\Delta\tau_{\text{sec}})^4}.$$

This convention prevents dimensional constants from being mistaken for new asymptotic exponents.

Warning A.1 ([A] $\mathcal{K}$ is not a universal regularity criterion). The Kretschmann scalar is a convenient diagnostic for the Schwarzschild and regular-core examples, where it is positive. It is *not* a universal positive norm of curvature in Lorentzian geometry. More general spacetimes can require frame components or other invariants to detect a singular regime, and Kretschmann finiteness alone constrains no higher derivatives. *No conclusion in this report equates a bounded Kretschmann scalar with general spacetime regularity.*

B The verification programs: what they check and what they do not

B.1 The three suites

The three source programs are preserved unmodified in `code/` and their recorded outputs in `data/`. All three use exact symbolic arithmetic with a computer algebra system, compare residuals exactly against zero — never against a numerical tolerance — and raise on any nonzero residual, so a failure aborts rather than being reported as a pass.

Table 4: The three verification suites, as recorded and as re-run.

Member	Checks	What it covers
12	45	Independent coordinate Riemann, Ricci and mixed Einstein reconstruction for the general static spherical metric, matching (7.3) and (7.4) (checks 01–03); Schwarzschild $\mathcal{K}$, vanishing Scal and G^t_t , horizon curvature, the dimensionless $48\epsilon^{-6}$; the ingoing radial–time block determinant; the $E = 1$ first integral; the proper-time coefficients $64/27$, $4/9$, $-2/9$; both indicial root sets; the Kasner constraints, coefficient $64/27$ and flat case; regular-core central curvature $24/\ell^4$ and Scal = $12/\ell^2$ by limit, ρ and $\rho(0)$, the Einstein source identity, the expansion through r^5 , the infinitesimal-core $24\epsilon^{-4}$; the outer, inner and transition-scale rational rewrites; eight exact geometric remainders $N = 1, \dots, 8$; the boundary-layer $b''(0) = -2$ and derivative scaling; the Planck threshold; the two-scale exponent; both Gaussian regulator integrals; two-level Born normalization and the rare-outcome limit.
13	50	The generic static-spherical Riemann contraction; all sixteen Schwarzschild Ricci components; $\mathcal{K}$; horizon curvature $3/(4m^4)$; the ingoing Eddington–Finkelstein radial determinant; the infinitesimal-radius identity; the radial proper-time cubic relation and the $E = 1$ geodesic equation; the proper-time curvature and both tidal coefficients; all four Jacobi powers $-1/3$, $4/3$, $1/3$, $2/3$; the Kasner linear and quadratic constraints, the $64/27$ coefficient and <i>all three</i> flat exceptional cases; six Hayward identities; the curvature power-counting radius; the two-scale classification; the regulator finite-part reparametrization; the Borel pole residue; two weak-tide primitives; and the double-null R_{uu} normalization and areal-radius gradient norm.
14	37	The Kretschmann invariant from a full four-index Riemann loop and the mixed Einstein component; Schwarzschild $\mathcal{K}$ and its Ricci components; Eddington–Finkelstein determinant; horizon curvature; the $E = 1$ first integral; $\mathcal{K} = 64/(27s^4)$; the radial tidal coefficient and all four Jacobi exponents; the nonradial $2/5$ leading balance; the <i>exact crossover integrand denominator factor</i> of (10.3); Hayward core curvature and density and the $\ell \rightarrow 0$ Schwarzschild limit at $r > 0$; the finite-part shift $a \mapsto a - c$; factorial ODE residuals at truncations $N = 1, \dots, 8$; and the exponentially small homogeneous ambiguity.

All three recorded runs pass. They were re-run for this merge, on *copies*, and all three pass again on a newer interpreter than the one recorded: 45, 50 and 37 checks, 132 in total, every suite exit 0. *None of the three writes any file beside itself*; the build helpers shipped with one member do create a build directory and copy a PDF, which is why the suites were run on copies as a matter of discipline.

B.2 What the programs establish, and what they do not

Warning B.1 (The scope disclaimers are part of the result). Each program prints its own scope disclaimer, and those disclaimers are reproduced here because they are exactly the honest limitations a merge must carry.

- Member 12: *algebraic identities only; no claim of a verified surreal field, general PDE existence, geodesic completion, or a physical singularity cure.* The program does not implement a surreal number field at all: substitutions such as $r = m\varepsilon$ use an ordinary positive symbol.
- Member 13: *general mathematical proofs, asymptotic realization, and physical claims were not machine-verified.* It does not verify set-theoretic foundations, global support lemmas, the standard-part reduction theorem, analytic existence, or the interpretation of quantum probabilities, and it runs no black-hole evolution numerically. Its role is to catch concrete algebraic mistakes in the examples, not to confer machine-checked status on the general proofs.
- Member 14: *finite symbolic identities only; no general analytic or physical validation.* It does not verify Theorem 5.1, Theorem 12.2 or Theorem 15.1 — those have ordinary text proofs — does not construct the full surreal field, does not prove convergence of any transseries, does not validate a physical model, and does not simulate across a physical singularity.

In the tier language of Theorem 2.1: *the programs certify tier [E] and nothing else.* No statement in tier [C] and no statement in tier [A] is machine-checked anywhere in this report, and no Lean formalization of this article is included or claimed.

B.3 Independent verification performed for this merge

Because the central conclusion is negative, and because a merge is most likely to damage exactly that, the load-bearing facts were re-derived from the metric rather than read from the sources. Specifically: the Christoffel symbols and the Riemann tensor of (7.1) were built from scratch and contracted, giving $\mathcal{K} = 48M^2/r^6$ exactly, hence $p = 6$ in Theorem 7.7; substituting $r = s^q(1+s)$ gives a pole of order exactly $6q$ with leading coefficient $48M^2$ for $q = 1, 2, 3$ and $M \neq 0$, the normalized case $w(0) = 1$ of the general coefficient $48M^2w(0)^{-6}$; the proper-time quadrature (9.3) evaluates in closed form to $(\pi/2)\sqrt{R^3/2M}$; and the tidal component $R^r_{trt} = 2M(2M - r)/r^4$ diverges as $r \rightarrow 0$. All four agree with the manuscripts. The negative conclusion of Section 22 therefore rests on independently reproduced computations and not on a source’s summary.

C Notation and assumptions at a glance

Symbol or phrase	Meaning in this report
[E], [C], [A], [I]	Exact symbolic identity; conditional theorem proved in the text; assessment; imported from the literature. Theorem 2.1.
No, No[i]	The full proper-class surreal field and its complexification. <i>Not</i> an ordinary set-sized metric space.
F_Γ, K_Γ	Set-sized real and complex Hahn fields with well-ordered support and small monomials t^γ for $\gamma > 0$.

Symbol or phrase	Meaning in this report
v	Least exponent of a nonzero Hahn series; $v(0) = +\infty$ as a bookkeeping symbol, never an adjoined field element.
$\mathcal{O}_F, \mathfrak{m}_F$	The limited elements (bounded by an ordinary real bound) and the infinitesimals (smaller in magnitude than every positive ordinary real).
limited / infinitesimal / unlimited	Theorem 4.1; used in place of “finite/infinite” to block the slide from “is a field element” to “is a finite physical magnitude”.
st	Real or complex standard part, defined <i>only</i> on limited elements.
strong Hahn summability	Well-ordered union of supports <i>and</i> finitely many contributions per exponent (Theorem 4.2). A support condition, not convergence.
ε	A positive dimensionless infinitesimal when working in a Hahn field; a positive <i>real</i> regulator where a real family is explicitly under discussion. The two uses are always flagged.
h, π_0	Formal parameter fixed by the coordinate derivatives, and reduction modulo h in Framework F1.
$\mathcal{A}(U), \mathcal{A}_\Gamma(U)$	$C^\infty(U, \mathbb{R})[[h]]$ and the Hahn coefficient-function algebra with global well-ordered support; Frameworks F1 and F2 of Section 12.1.
$\partial_\mu, d/d\varepsilon, D$	Coefficientwise ordinary spacetime derivative; parameter derivative; a scalar derivation on No . <i>Never interchangeable</i> ; see Table 2.
θ, η, B	Finite angle; infinitesimal phase residue; accumulated phase. Imported verbatim from the differential-equations report (Theorem 16.7).
Oz , Exp_{EK} , E , cis	Canonical initial integer part of No ; the integer-part-based global surcomplex exponential; the infinitesimal Hahn exponential; the finite-angle phase.
m, ℓ, ℓ_P, L	Geometrized mass GM/c^2 ; regular-core length scale; Planck length; an ordinary reference length.
τ, u, s, S	Proper time in length units; remaining proper time $u = \tau_* - \tau$ (written s where a source’s notation is followed); and the dimensionless remaining clock of Theorem 10.2.
$\mathcal{K}$	The Kretschmann scalar $R_{abcd}R^{abcd}$; <i>not</i> the coefficient field K .
FP	Extraction of the exponent-zero coefficient on <i>all</i> Laurent series, as opposed to st on $\mathcal{O}$. Non-multiplicative and prescription-dependent (Theorem 17.7).
formal solution / formal certificate	An identity coefficient by coefficient, with support admissibility and residual vanishing through a valuation threshold. It asserts <i>no</i> convergence, summability to a real solution, or physical validity (Section 13.4).

D Consolidated register of what is not claimed

The three manuscripts state 36 explicit non-claims between them. All of them survive here, consolidated into 31 numbered items and not one dropped. This register is part of the result.

D.1 Provenance and status

- N1. Every member self-labels as prepared with AI assistance, not refereed, and makes no research-priority claim for any proposition, computation or benchmark. The proved results are supplied as explicit mathematical deductions for an assessment, and this merge adds no priority claim of its own. The physical equations used in the examples are standard models, not newly discovered black-hole solutions.
- N2. No Lean formalization of any member, or of this report, is included or claimed. No member ran the repository’s Lean build, and no declaration-level audit was performed; neither was one performed for this merge. A theorem in an unrefereed report and a successfully checked generic Lean declaration have different evidential status.
- N3. Repository inspection was targeted: between them the members read the root README, the documentation catalogue, the surcomplex analysis README (one member cites the analysis source file) and the trigonometry README, at one pinned commit. Not every report in the collection was read; no file in the repository was changed by any member or by this merge.
- N4. The literature search was targeted, not systematic. Absence of a predictive model in the inspected sources is not proof that no other proposal has ever been written, and the survey does not establish that no future non-Archimedean theory could resolve a singularity — only that the reviewed sources do not.
- N5. The Berarducci–Mantova derivation, the Aschenbrenner–van den Dries–van der Hoeven universal H -field results, van den Dries–Ehrlich surreal exponentiation, the Ehrlich–Kaplan exponential and trigonometric constructions, and the Costin–Ehrlich integration theory are all *imported* with their stated hypotheses and function-class restrictions intact. None is reproved and none is strengthened. The Ehrlich–Kaplan erratum concerns in-text bibliography reference numbering, not a withdrawal of the global phase construction.
- N6. Penrose incompleteness, the null Raychaudhuri focusing bound, the Dafermos and Dafermos–Luk interior results, Donoghue’s effective-field-theory corrections, Hollands–Wald, Steinbauer–Vickers, Unruh–Schützhold, Horowitz–Marolf, Hatsuda, Hayward, Carballo-Rubio et al., Andersson–Rendall, Damour–Henneaux–Nicolai, Benci–Horsten–Wenmackers, Loeb, Gubser et al. and the Nieto papers are cited, not proved here, and source hypotheses and regularity qualifications are retained. No current universal result about all black-hole interiors is claimed.
- N7. One member’s package README lists a checksum file that is not present in the delivered directory. The discrepancy is recorded here rather than silently repaired, and nothing in this report depends on it.

D.2 What the programs do not verify

- N8. The three suites check *finite symbolic identities only*: $45 + 50 + 37 = 132$ checks. They do not construct or verify a surreal field — indeed they implement no surreal arithmetic, using

ordinary positive symbols for infinitesimal substitutions — and they do not verify Hahn summability in general, set-theoretic foundations, global support lemmas, the standard-part or formal Einstein reduction theorems, the discreteness proposition, the finite spectral theorem, analytic existence, PDE existence, geodesic completion, convergence of any transseries, quantum probability interpretation, or any proposed physical interpretation.

- N9. They run no black-hole evolution numerically and do not simulate across a physical singularity. Their role is to catch algebraic transcription errors in the worked examples.
- N10. Re-running them establishes reproducibility on a newer interpreter and nothing further. One member's README notes that a given Python version is the intended interface but only the recorded environment was tested, and one member's build records one minor underfull-box LaTeX diagnostic.

D.3 Scope of the mathematical results

- N11. Theorem 5.1 is proved only for *set-sized* subsets. It explicitly does not claim that the proper class $\mathbf{No}[z]$ is a discrete space, since no radius can be chosen below a proper class of positive distances; it does not apply to the intrinsic valuation topology of a set-sized Hahn field; and it does not rule out coarser topologies or coefficientwise smooth structures.
- N12. Theorem 7.7 and Theorem 7.8 are statements about Hahn-field valuation and about order in an ordered extension respectively. Neither is a statement about real limits, and neither rules out a modified metric or a rescaled observable.
- N13. Theorem 7.12 is conditional on preserving the same invariant identity at the endpoint. It is not a theorem excluding a modified metric, a different equation, nonclassical observables, a quantum replacement of the endpoint, or a geometry with no zero-area-radius event.
- N14. Theorem 12.2 and Theorem 12.3 assume a nondegenerate smooth leading metric on one common domain, with strictly positive order or positive global support for the perturbation. Singular perturbations are explicitly outside them; Theorem 13.1 is offered as evidence that the hypotheses cannot be dropped. Both are conditional and neither is machine-verified.
- N15. Theorem 12.4 is a conditional obstruction and not a universal no-go. Five escape routes are enumerated in its status note: a degenerate or nonexistent shadow metric, negative-order fields, singular boundary layers, new equations, and nonclassical observables or a different notion of spacetime.
- N16. Theorem 14.5 is a criterion on Hahn-field valuation data and not on real convergence. It detects misuse of an expansion; it invents no solution and selects no metric.
- N17. Theorem 15.2 is conditional on adopting real standard parts as measured probabilities. It excludes conditioning on standard-probability-zero events, and excludes infinite and hyperfinite experiments. It is not a theorem that non-Archimedean models are experimentally indistinguishable in general, and it does not cover non-standard-part probability interpretations.
- N18. Infinite-dimensional quantum theory, functional integration, measures and unbounded self-adjoint operators are *not* supplied by algebraic closure. A proper-class path sum is not licensed by Hahn summability.
- N19. Theorem 16.2, Theorem 16.3 and Theorem 16.4 concern one specific commutative construction and one specific normalized derivation. They do not assert that global surcomplex exponentials

cannot be defined — one is constructed in (16.3) — and the Ehrlich–Kaplan constructions must not be described as nonexistent.

- N20. The realization bridge of Section 21.6 is open, and one concrete obstruction is flagged: differentiating $t^{p(x)}$ produces $p'(x) \log t$, so the constant-exponent coefficient algebra is not closed under position-dependent exponents.
- N21. The transseries display (17.1) is schematic. An arbitrary triple-indexed family is not automatically admissible, and the schematic $e^{\kappa v} v^{-p}$ form is not a universal theorem about all Kerr interiors. For finite polynomial Gaussian models over K_{Γ} the diagramwise admissible domain is determined exactly in `docs/surcomplex/wick-summability-certificates/`; that result concerns formal coefficient data and does not extend to the display.

D.4 Scope of the physical statements

- N22. **A physical resolution of black-hole singularities is not supplied by scalar extension and is not claimed anywhere in this report.** This report is a mathematical assessment and a research proposal, not a quantum-gravity resolution of black-hole singularities, and not a claim of a new empirically validated theory, a new black-hole solution, or a new prediction.
- N23. All negative conclusions are scoped as *missing implications*, not impossibility theorems. A future surreal-variable theory could still make nontrivial predictions; a quantum bounce is not excluded; and non-Archimedean probability interpretations other than standard-part are not covered.
- N24. r_Q of (8.1) is dimensional power counting, not a derived universal threshold, not a sharply defined transition radius, and not a derived minimum radius. Additional scales or state-dependent effects can invalidate a simple curvature criterion sooner. The ε^α , ε^β and ε^a , ε^b assignments describe a *family of models* and do *not* assert that the measured Planck length is literally infinitesimal; the Planck length is read throughout as an ordinary small positive real.
- N25. The regular-core effective stress-energy $T = G/8\pi$ is a consistent effective description with no microphysical Lagrangian and no stability proof. Finite central curvature is not global geodesic completeness, unique evolution, or stable evaporation, and the model’s global viability, stability and quantum derivation are left open.
- N26. The regular-core ansatz is *not* advertised as C^∞ at the centre: the r^5 term of (14.2) becomes $|x|^5$ along a Cartesian line through the centre. “Regular” there means finite curvature plus C^2 -level regularity, nothing more.
- N27. Kretschmann finiteness alone constrains no higher derivatives, and $\mathcal{K}$ is not a universal positive norm of Lorentzian curvature (Theorem A.1).
- N28. Generic oscillatory Kasner interiors may admit no single well-ordered expansion; Hardy-field and transseries technology targets eventually ordered nonoscillatory behaviour, so BKL-type interiors may need extra variables, sectors, or a different function category. A Schwarzschild Puiseux law licenses no inference about generic interiors, and charged and rotating interiors are explicitly not Schwarzschild copies.
- N29. Generic Jacobi rates in (9.5)–(9.6) do not apply to every Jacobi field; special initial conditions change them. The crossover limit of Theorem 10.3 is uniform only on bounded positive

R -intervals, and the large- R behaviour of the limiting function is a matched-asymptotic statement, not uniform convergence on an unbounded interval. The near-extremal bookkeeping of Theorem 11.4 is not a claim that extremality or a Cauchy horizon is physically harmless.

- N30. The conservative architecture (21.1) is called the most defensible starting point, not a derived necessity. Taking real standard parts as measured probabilities is a possible modelling choice, not a consequence of the field axioms. Declaring a dimensionless ratio infinitesimal is a stated convention; the report insists that “the radius is ω^{-1} ”, without a unit, is not well defined. The s -weighted tidal integral is a model criterion for singularity strength, and $\ell_{\text{P}}^4 \mathcal{K} \sim 1$ is a large-correction indicator; signature $(-, +, +, +)$ and $G = c = 1$ are conventions, and treating ε as a scalar constant with $\partial_\mu \varepsilon = 0$ is a choice.
- N31. Everything in Section 21 — the architecture, the five stages, the four benchmarks, the four projects, the proposed Lean module names, the five-item extension plan, the suggestions that a reduced gravitational ODE might fall in a resurgent class or that surreal methods might describe the incoming and outgoing regions of a bounce, the five-level hierarchy of success and the four-question test — is a *proposal* or conceptual framing. None of it is an implemented or experimentally validated theory, and none of it is a description of an existing module.

E Provenance: member by member

Table 6: Where each statement came from. “All three” means the shared spine, printed once.

Statement	Members	Note
Static-spherical $\mathcal{K}[f]$, Schwarzschild $\mathcal{K}$, horizon determinant, vacuum Ric, proper-time infall, Jacobi exponents, Kasner coefficient, Hayward central limits, r_Q , the two-scale exponent, the finite-part shift, the framework comparison, the research program	all three	Printed once. Each member ships its own independent symbolic re-derivation of the connection and curvature.
Theorem 4.4, Theorem 5.1, Theorem 15.1	all three	Same proofs; printed once, with the union of statements and caveats.
Theorem 15.2	all three	Union of three overlapping versions; the conditioning boundary (d) is from member 13, the projection and joint-probability clauses from member 12, the Born-weight clause from member 14.
Theorem 7.7, ramification invariance	12	Route 1. Unique to member 12.

Statement	Members	Note
Theorem 7.8	14	Route 2. Unique to member 14, and a different theorem; both are kept (Theorem 7.9).
Theorem 7.12	13, 14	Same one-line proof; printed once. Member 13 names the published proposal as its target.
Theorem 12.2 (Framework F1)	12, 14	Same theorem in two parametrizations; printed once.
Theorem 12.3, Theorem 12.4 (Framework F2)	13	Genuinely different hypotheses and a different proof ingredient (the Neumann support lemma); kept separately.
Theorem 13.1	12	The boundary-layer warning about differentiating standard parts.
Theorem 13.2	14	A purely real counterexample to the interchange.
Theorem 14.5, Theorem 14.6	12	The exact inner/outer Hahn support criteria, sharp at $\alpha = 2/3$.
Theorem 10.2, Theorem 10.3, Theorem 10.4	14	The small-angular-momentum crossover.
Section 8 as a headline	14	Member 12 makes the same argument inside its effective-field-theory section; member 13 in its ultraviolet section; member 14 promotes it, and this report follows member 14 on placement.
Section 16.1, Theorem 16.2, Theorem 16.3, Theorem 16.6	12	The global-phase section.
Theorem 16.4	13	Consistent with member 12's construction; Theorem 16.5 records why.
Theorem 5.4, Theorem 5.5, Theorem 11.3, Theorem 17.3, (21.3)–(21.4), Theorem 17.7 first half	13	Bounded frames, dominant balance, weak singularities, the Borel residue, the double-null identities, and non-multiplicativity of FP.
Theorem 17.2, Theorem 11.4, Section 17.8, Theorem 17.7 second half, Theorem 14.2(a)	14	Beyond-all-orders ambiguity, near-extremal bookkeeping, the wave benchmark, noncanonical subtraction, and the $ x ^5$ remark.
Theorem 17.5, Theorem 15.3, the units-and-falsifiability material of Sections 4.5 and 19.4	12	

Statement	Members	Note
Theorem 17.6, Theorem 15.4, Section 19 recoverability material, Section 21.3–Section 21.7 stage structure with reported failure conditions	13	
Section 21.8, Section 21.2, Section 3.3	14	
The claim tiers of Theorem 2.1 and Table 1	merge	Synthesized from the three members’ own claim ledgers, which they state separately; the tier tags and their use throughout are this report’s contribution to the merge and are not a new mathematical claim.

[A] No statement in this report rests on a member that does not prove what it is cited for. Where the three disagreed, they disagreed about *emphasis and placement* and not about content: the one substantive editorial conflict was whether the loss-of-control argument is a headline or a late aside, and Theorem 8.1 records its resolution.

References

- [1] V. Reshetnikov, *Surreal*, repository root README.md, pinned snapshot 39f2be6667ade51bca2b45daa47e289d69c09764. <https://github.com/VladimirReshetnikov/Surreal/blob/39f2be6667ade51bca2b45daa47e289d69c09764/README.md>. Accessed 21 September 2026. Scope and stated limitations of the implemented Lean prerequisites.
- [2] V. Reshetnikov, *The surreal and surcomplex reports*, repository documentation catalogue docs/README.md, same snapshot. <https://github.com/VladimirReshetnikov/Surreal/blob/39f2be6667ade51bca2b45daa47e289d69c09764/docs/README.md>. AI-assisted research documentation; includes the coefficient-category warnings and the status of the reports.
- [3] *Surcomplex Analysis: Germs, Coherent Hahn Sections, and Infinitesimal Zero Geometry*, research report in *Surreal*, docs/surcomplex/analysis/, same snapshot. Distinguishes Hahn summability from topological convergence, and separates germ, common-domain and formal coefficient classes. Not treated here as a verified physics foundation.
- [4] *Trigonometry on the Surcomplex Plane: Canonical Phases, Arbitrary-Scale Geometry, and Infinitesimal Degeneration*, research report in *Surreal*, docs/surcomplex/trigonometry/, same snapshot. Includes the Ehrlich–Kaplan integer-part global normalization and explicitly selects no scalar derivation.
- [5] *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers: the finite-primitive criterion, the phase obstruction, and support-certified coordinate systems*, merged research report in *Surreal*, docs/surcomplex/differential-equations/. Owns the oscillation obstruction and the three phase symbols adopted in Theorem 16.7.
- [6] *Foundations for Surreal and Surcomplex Mathematics: Sets, Classes, Universes, Type Theories, and a Pinned Lean Audit*, merged research report in *Surreal*, docs/foundations-and-computation/foundations/. Owns the workspace-localization theorem and the size obstructions cited in Section 4.2.
- [7] J. H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976.
- [8] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [9] P. Ehrlich, The absolute arithmetic continuum and the unification of all numbers great and small, *Bulletin of Symbolic Logic* **18** (2012), 1–45. <https://doi.org/10.2178/bsl.1327328438>.
- [10] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>.
- [11] A. Berarducci and V. Mantova, Surreal numbers, derivations and transseries, *Journal of the European Mathematical Society* **20** (2018), 339–390. <https://doi.org/10.4171/JEMS/769>; <https://arxiv.org/abs/1503.00315>.
- [12] A. Berarducci and V. Mantova, Transseries as germs of surreal functions, *Transactions of the American Mathematical Society* **371** (2019), 3549–3592. <https://doi.org/10.1090/tran/7428>; <https://arxiv.org/abs/1703.01995>.
- [13] M. Aschenbrenner, L. van den Dries and J. van der Hoeven, The surreal numbers as a universal H -field, <https://arxiv.org/abs/1512.02267>.
- [14] P. Ehrlich and E. Kaplan, Surreal ordered exponential fields, *Journal of Symbolic Logic* **86** (2021), 1066–1115. <https://doi.org/10.1017/jsl.2021.59>; <https://arxiv.org/abs/2002.07739>. In particular the section on trigonometric fields and surcomplex exponentiation.

- [15] P. Ehrlich and E. Kaplan, Surreal ordered exponential fields — Erratum, *Journal of Symbolic Logic* **87** (2022), 871. <https://doi.org/10.1017/jsl.2022.5>. Concerns in-text bibliography reference numbering.
- [16] O. Costin and P. Ehrlich, Integration on the surreals, *Advances in Mathematics* **452** (2024), article 109823. <https://arxiv.org/abs/2208.14331>, version 5 consulted. Used only with the stated function-class restrictions.
- [17] R. Penrose, Gravitational collapse and space-time singularities, *Physical Review Letters* **14** (1965), 57–59. <https://doi.org/10.1103/PhysRevLett.14.57>.
- [18] A. Raychaudhuri, Relativistic cosmology. I, *Physical Review* **98** (1955), 1123–1126. <https://doi.org/10.1103/PhysRev.98.1123>.
- [19] E. Witten, Light rays, singularities, and all that, *Reviews of Modern Physics* **92** (2020), 045004. <https://doi.org/10.1103/RevModPhys.92.045004>; <https://arxiv.org/abs/1901.03928>.
- [20] J. M. M. Senovilla, A critical appraisal of the singularity theorems, <https://arxiv.org/abs/2108.07296>; <https://doi.org/10.1098/rsta.2021.0174>.
- [21] K. Landsman, Penrose’s 1965 singularity theorem: from geodesic incompleteness to cosmic censorship, <https://arxiv.org/abs/2205.01680>, 2022.
- [22] M. Dafermos, The interior of charged black holes and the problem of uniqueness in general relativity, <https://arxiv.org/abs/gr-qc/0307013>.
- [23] M. Dafermos and J. Luk, The interior of dynamical vacuum black holes I: the C^0 -stability of the Kerr Cauchy horizon, *Annals of Mathematics* **202** (2025), 309–630. <https://doi.org/10.4007/annals.2025.202.2.1>; <https://arxiv.org/abs/1710.01722>. The use here is restricted to the hypotheses and interior conclusion of this work.
- [24] J. F. Donoghue, General relativity as an effective field theory: the leading quantum corrections, *Physical Review D* **50** (1994), 3874–3888. <https://doi.org/10.1103/PhysRevD.50.3874>; <https://arxiv.org/abs/gr-qc/9405057>.
- [25] J. F. Donoghue, Quantum general relativity and effective field theory, <https://arxiv.org/abs/2211.09902>. Review contribution to the *Handbook of Quantum Gravity*.
- [26] L. Andersson and A. D. Rendall, Quiescent cosmological singularities, *Communications in Mathematical Physics* **218** (2001), 479–511. <https://doi.org/10.1007/s002200100406>; <https://arxiv.org/abs/gr-qc/0001047>.
- [27] T. Damour, M. Henneaux and H. Nicolai, Cosmological billiards, *Classical and Quantum Gravity* **20** (2003), R145–R200. <https://arxiv.org/abs/hep-th/0212256>.
- [28] S. A. Hayward, Formation and evaporation of non-singular black holes, *Physical Review Letters* **96** (2006), 031103. <https://doi.org/10.1103/PhysRevLett.96.031103>; <https://arxiv.org/abs/gr-qc/0506126>.
- [29] R. Carballo-Rubio, F. Di Filippo, S. Liberati and M. Visser, Geodesically complete black holes, *Physical Review D* **101** (2020), 084047. <https://doi.org/10.1103/PhysRevD.101.084047>; <https://arxiv.org/abs/1911.11200>.
- [30] S. Bhattacharjee, S. Sarkar and A. Virmani, Internal structure of charged AdS black holes, <https://arxiv.org/abs/1604.03730>, 2016.
- [31] W. G. Unruh and R. Schützhold, On the universality of the Hawking effect, <https://arxiv.org/abs/gr-qc/0408009>.

- [32] S. Hollands and R. M. Wald, Quantum fields in curved spacetime, *Physics Reports* (2015).
<https://doi.org/10.1016/j.physrep.2015.02.001>; <https://arxiv.org/abs/1401.2026>.
- [33] G. T. Horowitz and D. Marolf, Quantum probes of spacetime singularities, *Physical Review D* **52** (1995), 5670–5675. <https://doi.org/10.1103/PhysRevD.52.5670>;
<https://arxiv.org/abs/gr-qc/9504028>.
- [34] Y. Hatsuda, Quasinormal modes of black holes and Borel summation, *Physical Review D* **101** (2020), 024008. <https://doi.org/10.1103/PhysRevD.101.024008>; <https://arxiv.org/abs/1906.07232>.
- [35] R. Steinbauer and J. A. Vickers, The use of generalised functions and distributions in general relativity, *Classical and Quantum Gravity* **23** (2006), R91–R114. <https://arxiv.org/abs/gr-qc/0603078>.
- [36] I. Aniceto, G. Başar and R. Schiappa, A primer on resurgent transseries and their asymptotics,
<https://arxiv.org/abs/1802.10441>; <https://doi.org/10.1016/j.physrep.2019.02.003>.
- [37] M. Kontsevich, Deformation quantization of Poisson manifolds, I,
<https://arxiv.org/abs/q-alg/9709040>.
- [38] V. Benci, L. Horsten and S. Wenmackers, Non-Archimedean probability, *Milan Journal of Mathematics* **81** (2013), 121–151. <https://arxiv.org/abs/1106.1524>.
- [39] P. A. Loeb, Conversion from nonstandard to standard measure spaces and applications in probability theory, *Transactions of the American Mathematical Society* **211** (1975), 113–122.
<https://www.jstor.org/stable/1997222>.
- [40] S. S. Gubser, J. Knaute, S. Parikh, A. Samberg and P. Witaszczyk, p -adic AdS/CFT, *Communications in Mathematical Physics* (2017). <https://doi.org/10.1007/s00220-016-2813-6>;
<https://arxiv.org/abs/1605.01061>.
- [41] J. A. Nieto, Some mathematical and physical remarks on surreal numbers, *Journal of Modern Physics* (2016). <https://doi.org/10.4236/jmp.2016.715188>; <https://arxiv.org/abs/1611.09699>. The black-hole discussion appears near the end of the 19-page preprint. Cited as a prior proposal, not as an assumed singularity-resolution theorem.
- [42] J. A. Nieto, Duality, matroids, qubits, twistors and surreal numbers,
<https://arxiv.org/abs/1810.04521>, 2018.